The 2024 Princeton-Combustion Institute Summer School on Combustion and the Environment:

Combustion Instability

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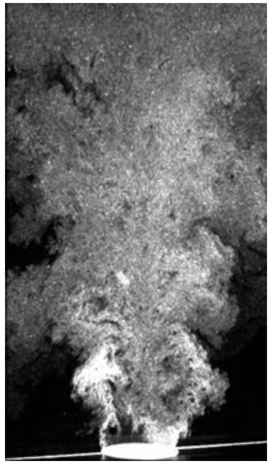
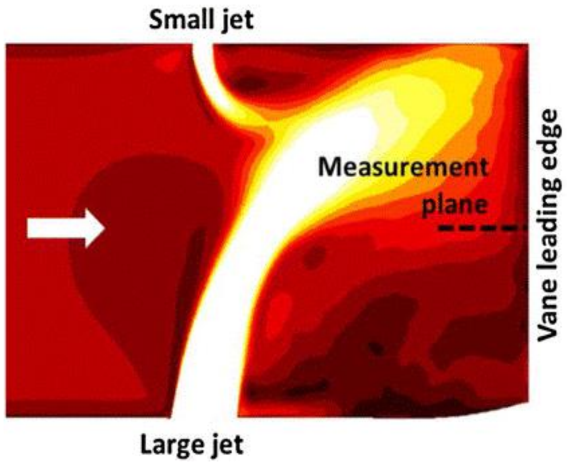
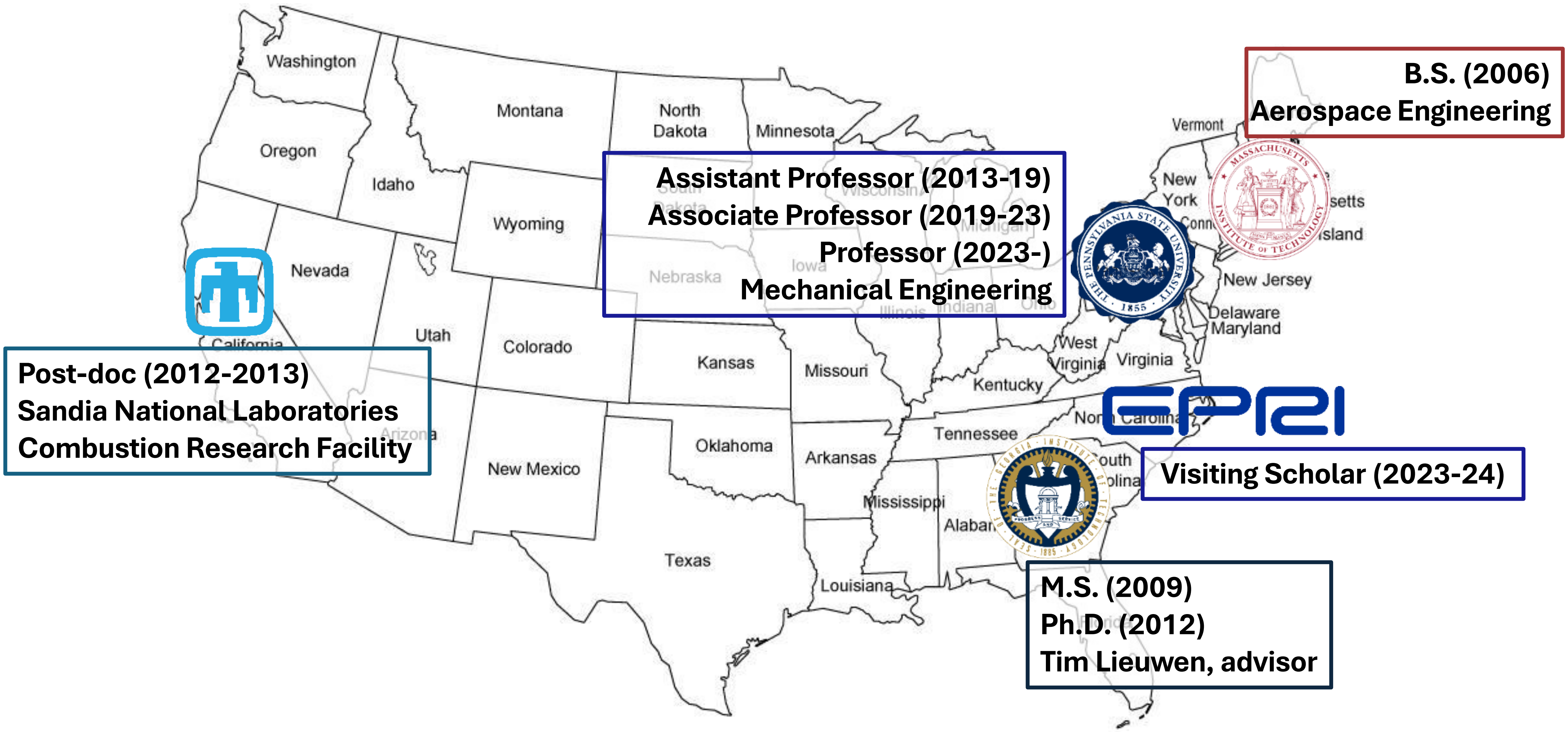
Pennsylvania State University
https://sites.psu.edu/rfdl/





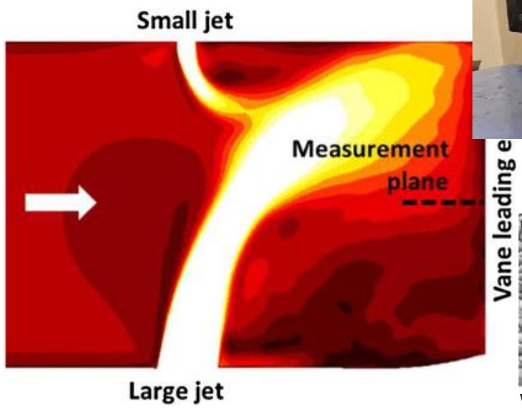
PennState

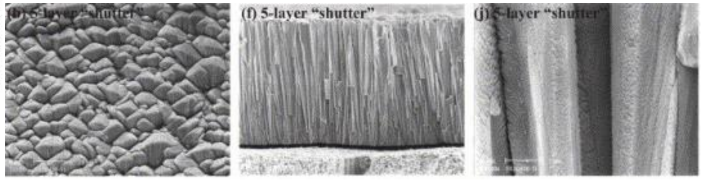

B.S. (2006)
Aerospace Engineering
Assistant Professor (2013-19)
Associate Professor (2019-23)
Professor (2023-)
Mechanical Engineering
Post-doc (2012-2013)
Sandia National Laboratories
Combustion Research Facility
Visiting Scholar (2023-24)
M.S. (2009)
Ph.D. (2012)
Tim Lieuwen, advisor

In the Reacting Flow Dynamics Lab, our research addresses these decarbonization challenges with three research areas

Combustor Operability

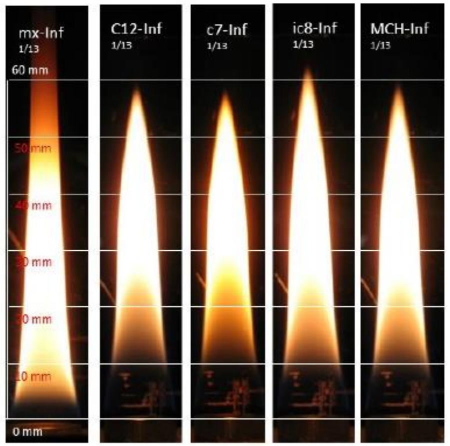
Materials and Durability

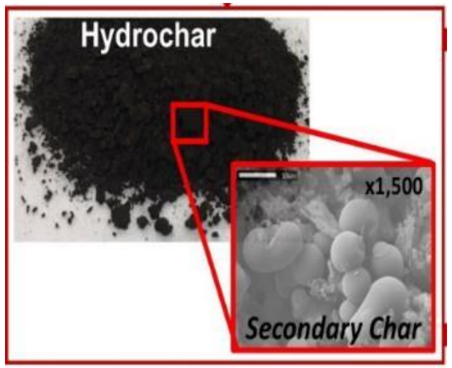




Wolfe et al. (2005). *Surface and coatings technology*, *190*(1), 132-149.

Alternative Fuels





Who are you?

Introduce yourself with three pieces of information:

— Name

— Institution

— Your thesis work in no more than two words (example: combustion instability)

Learning objectives of this course – at the end of the course, students should:

1. Identify the technologies in which combustion instability is an issue
2. Describe the basic thermoacoustic coupling process
3. Explain the kinematics of flame response to harmonic inputs
4. Assess a flame transfer function to identify the potential coupling processes
5. Use your understanding of flame response and thermoacoustic coupling to explain instability control techniques

Course schedule

Monday:
— 1400-1500: Introduction and combustion instability in technologies
— 1500-1530: Thermoacoustic feedback
— 1530-1545: Break
— 1545-1730: Flame kinematics (with crafts!) and flame response

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Combustion dynamics is one of the most expensive problems in several industries, including gas turbines, rockets, and boilers

— Damages expensive hardware

— Reduces operability

— Increases emissions (NO_x, CO)

Goy *et al.*, in *Combustion instabilities in gas turbine engines: operational experience, fundamental mechanisms, and modeling*, T. Lieuwen and V. Yang, Editors. 2005. p. 163-175.

Thermoacoustic combustion instability can occur when a flame is placed in a chamber with an acoustic mode

Photo credit: Jud McCranie

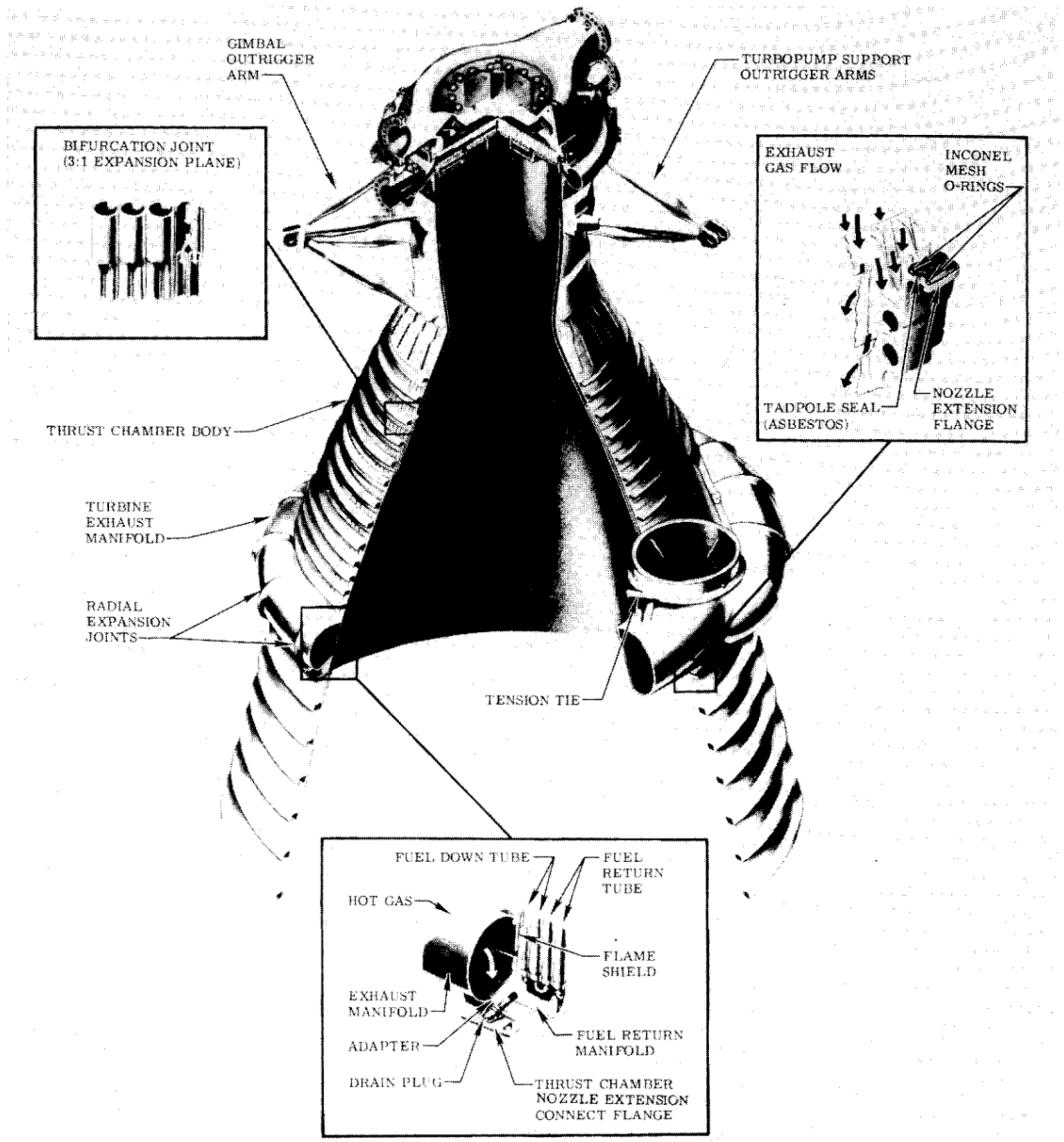
Photo credit: US Department of Energy

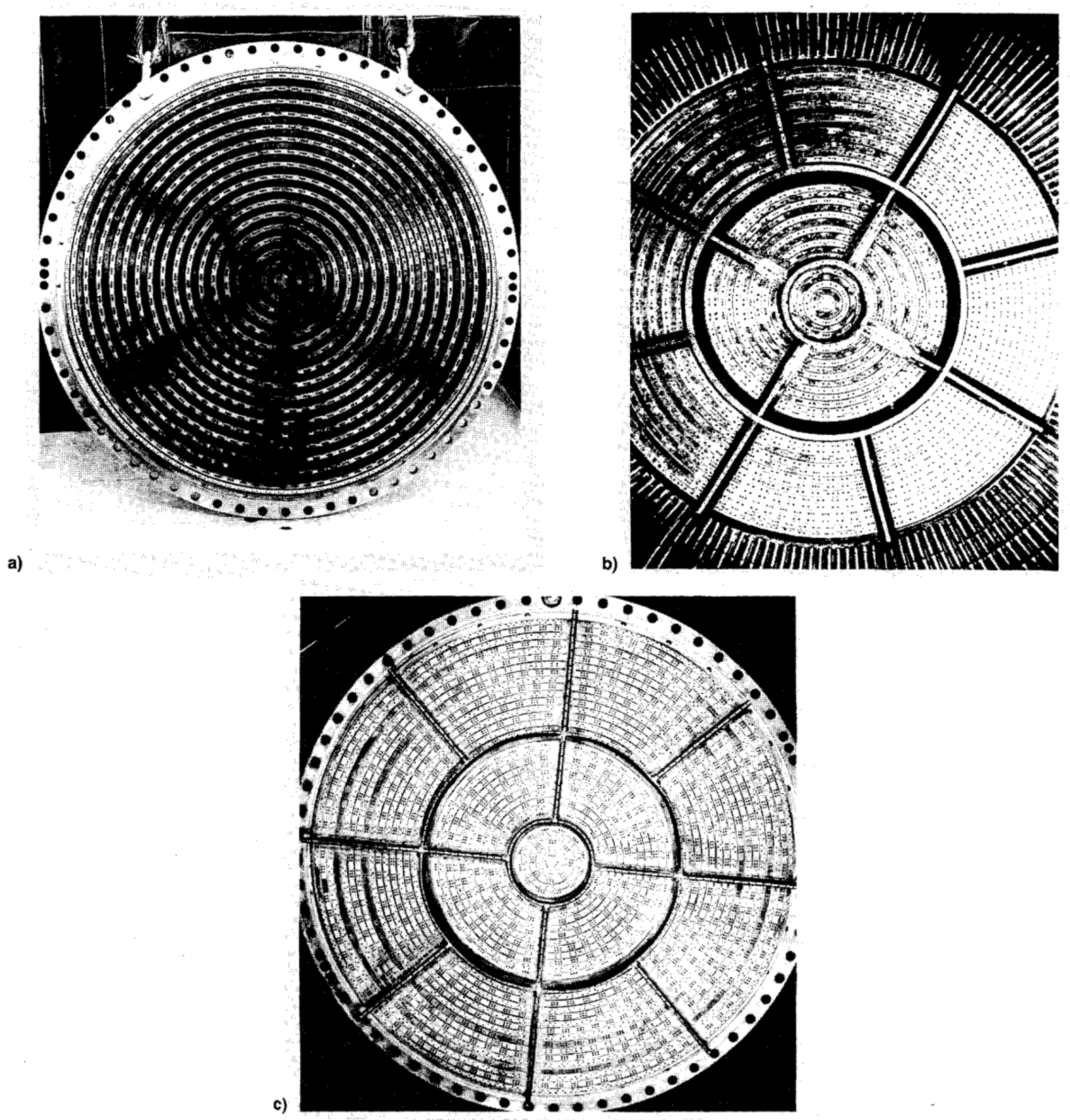
Photo credit: Sm faysal

Photo credit: Famartin

Example 1: F-1 rocket engine

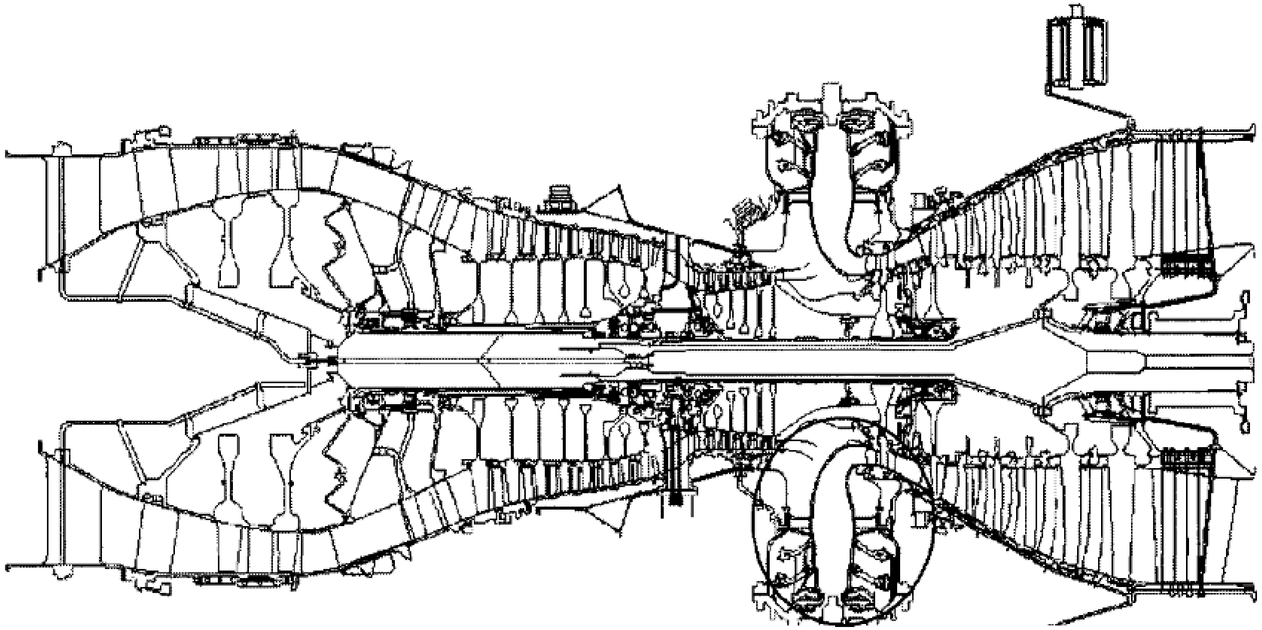
Oefelein, J. C., & Yang, V. (1993). Comprehensive review of liquid-propellant combustion instabilities in F-1 engines. *Journal of Propulsion and Power*, 9(5), 657-677.

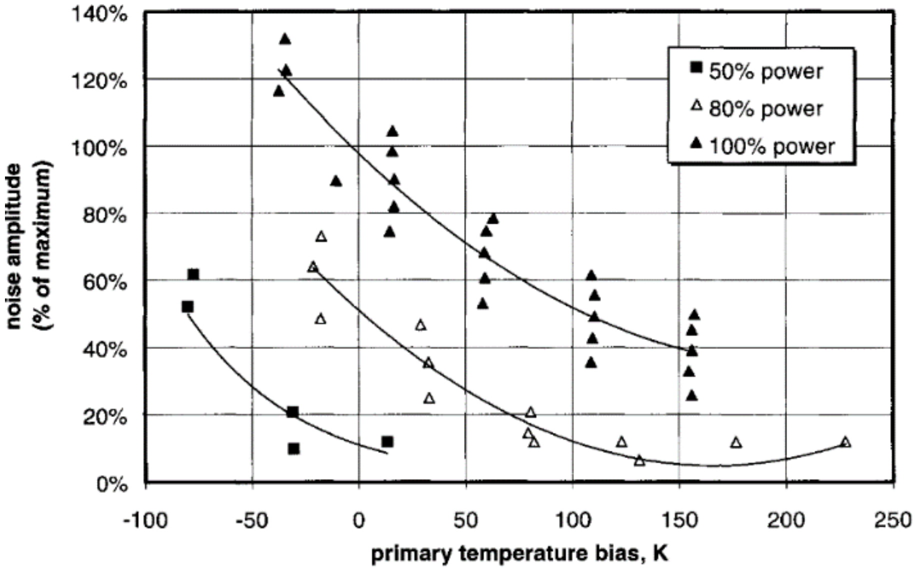


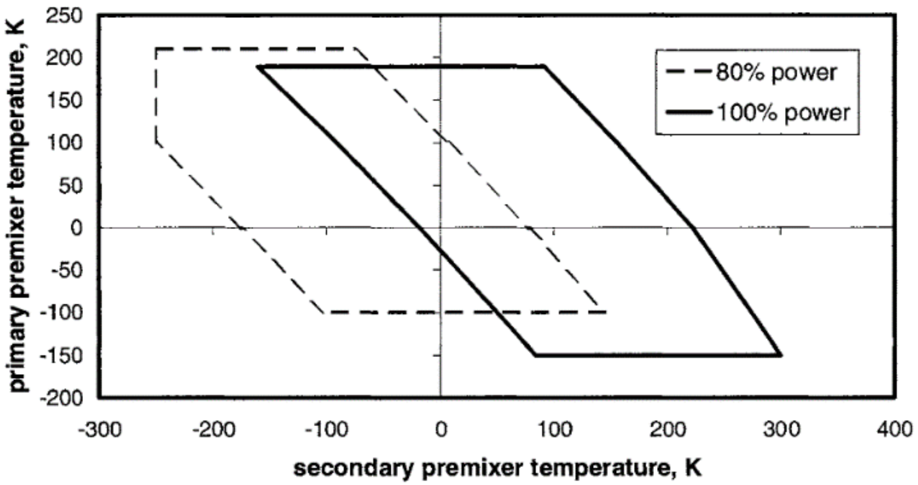


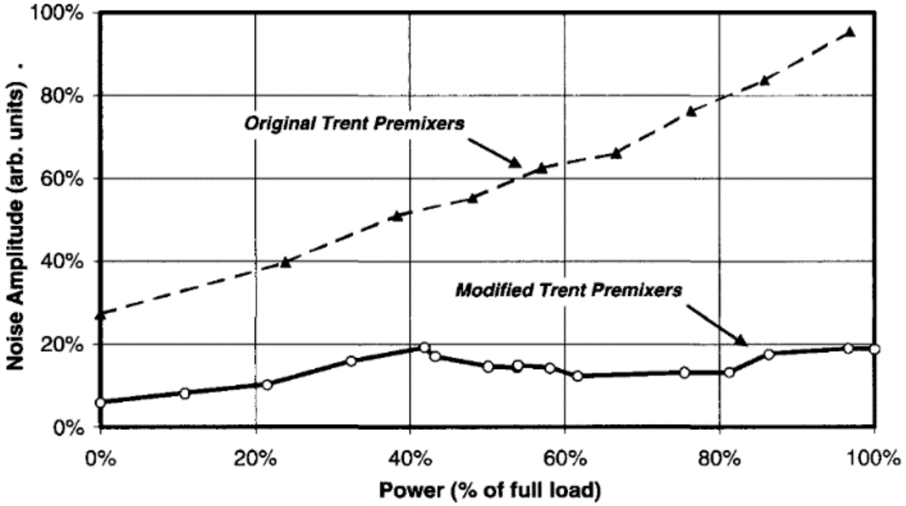
Example 2: Rolls-Royce Trent 60 Aeroderivative

Scarinci, T., Lieuwen, T. C., & Yang, V. (2005). Combustion instability and its passive control: Rolls-Royce aeroderivative engine experience. *Progress in Astronautics and Aeronautics*, *210*, 65.





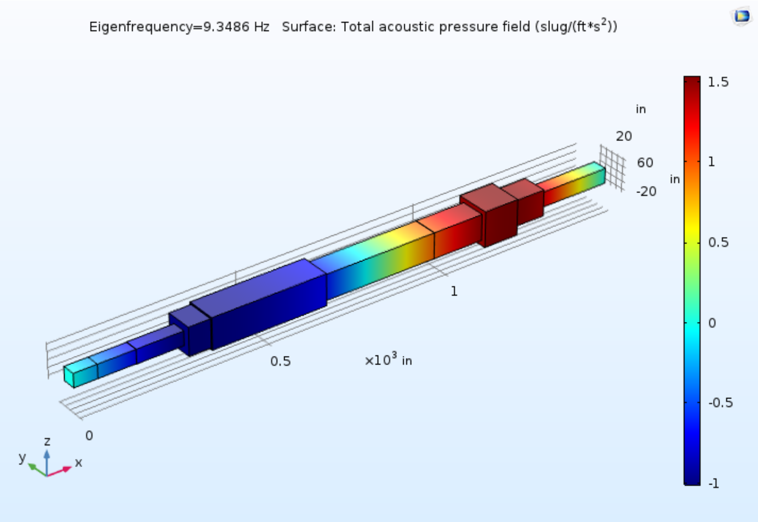


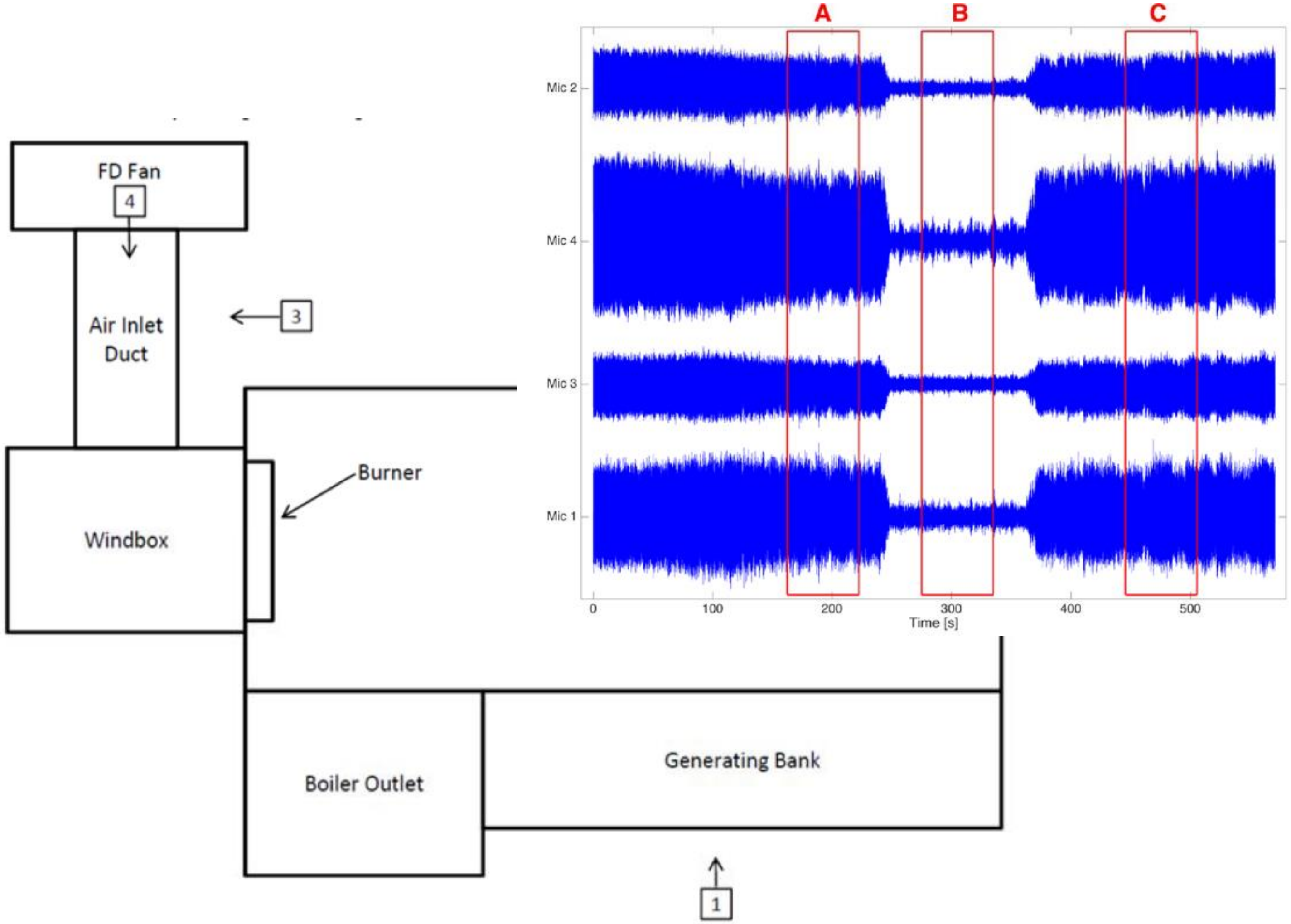


Example 3: Babcock and Wilcox package boiler

Rufener, S., Fuller, T. A., Flynn, T. J., Whelan, M. D., Finney, C., O'Connor, J. A., & Dare, T. P. (2023). Investigative approach to address thermoacoustic vibration in gas-fired heaters and boilers. Oak Ridge National Lab.(ORNL), Oak Ridge, TN (United States).

https://www.babcock.com/home/products/package-boilers





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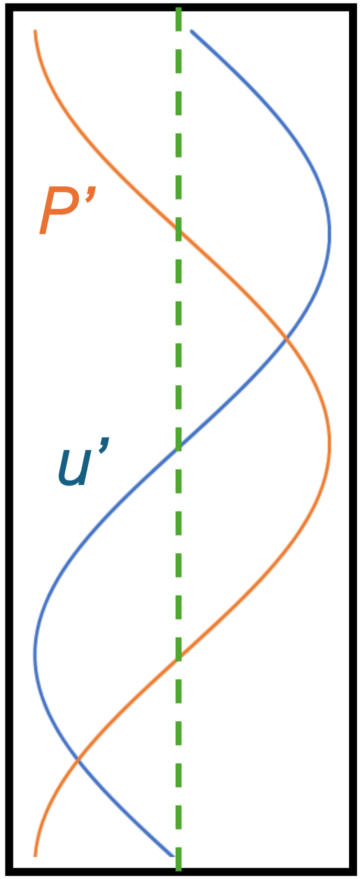
SIDEBAR

Acoustics time!

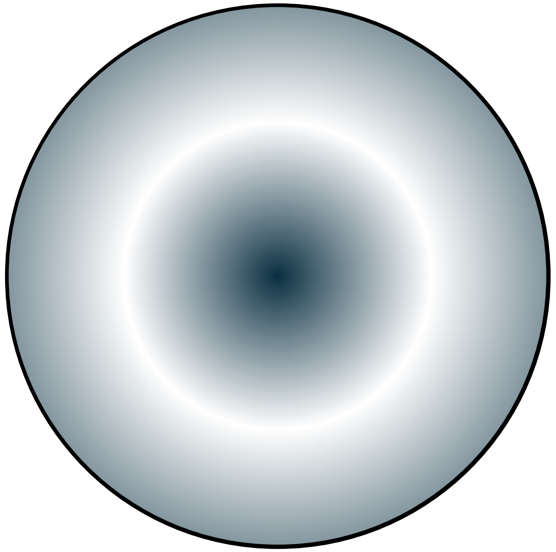
Acoustic wave propagation – to the board!

Acoustic modes in a cylindrical chamber

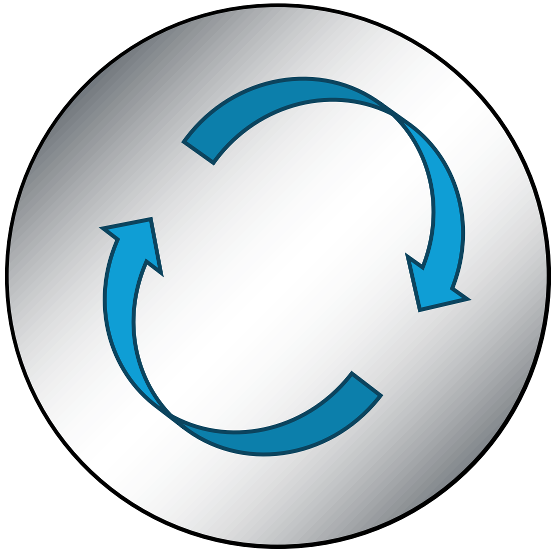
Longitudinal Mode



Radial Mode



Azimuthal Mode



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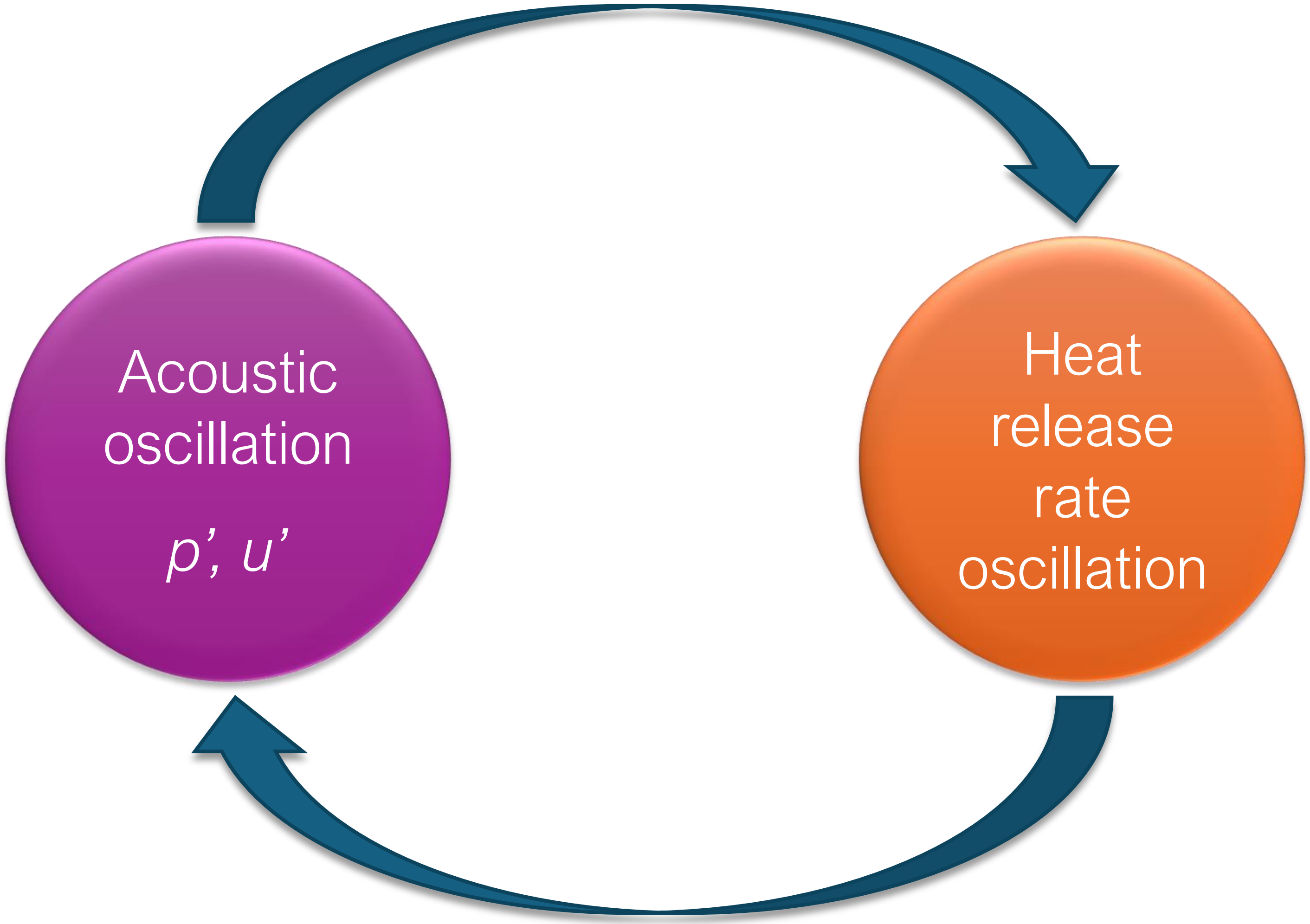
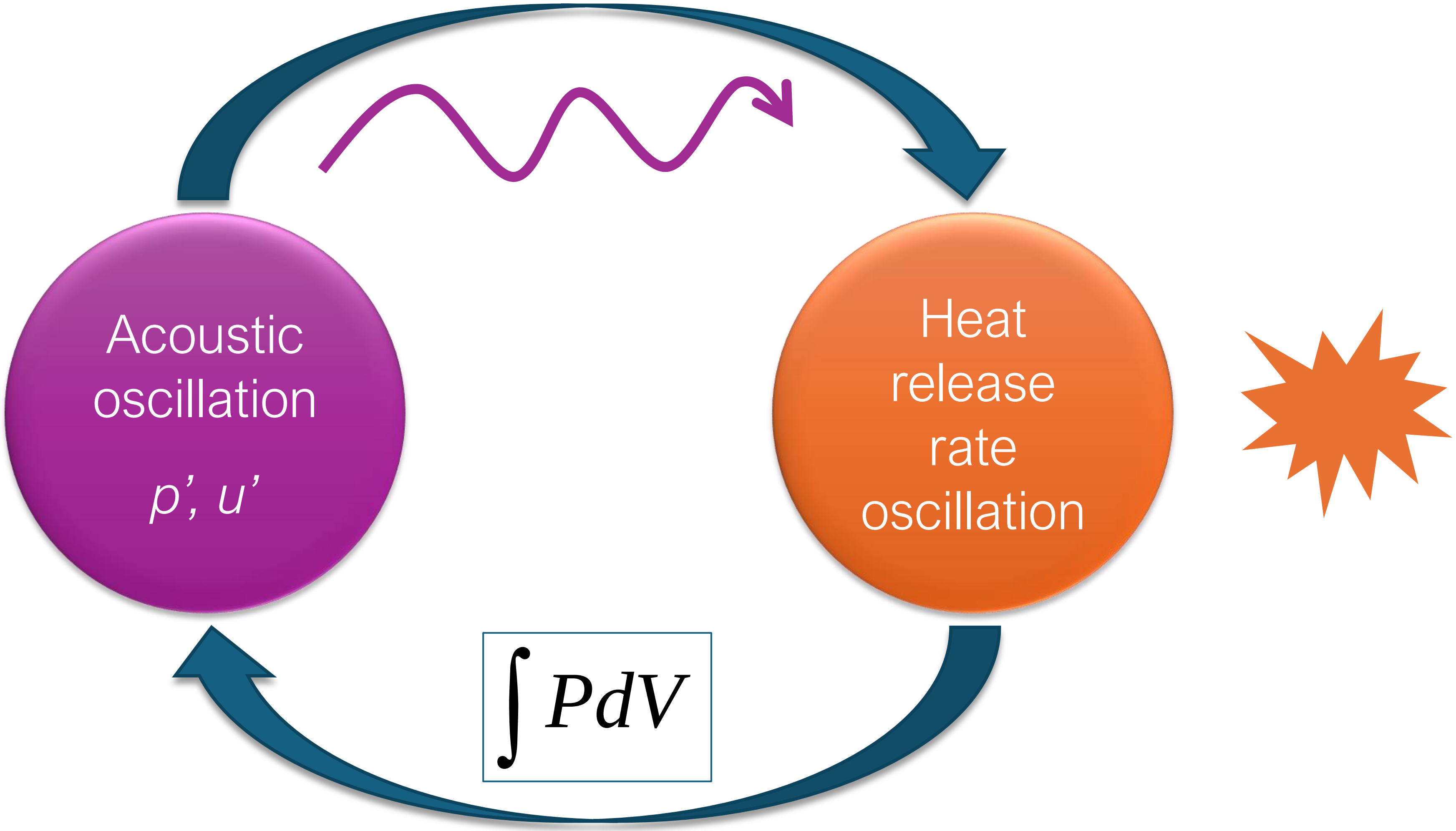
Back to thermoacoustics

Thermoacoustic instability is driven by a feedback process between acoustic oscillations and heat release rate oscillations of the flame



The energy transfer is facilitated by a compression/expansion work process



The Rayleigh criterion provides a way of determining the extent of coupling between the heat release rate oscillations and pressure

$$RI = \int_V \int_T p'(\vec{x},t)\,\dot{q}'(\vec{x},t)\,dt\,dV$$

- If RI > 0 → thermoacoustic feedback drives instability growth
- If RI < 0 → thermoacoustic feedback does not exist

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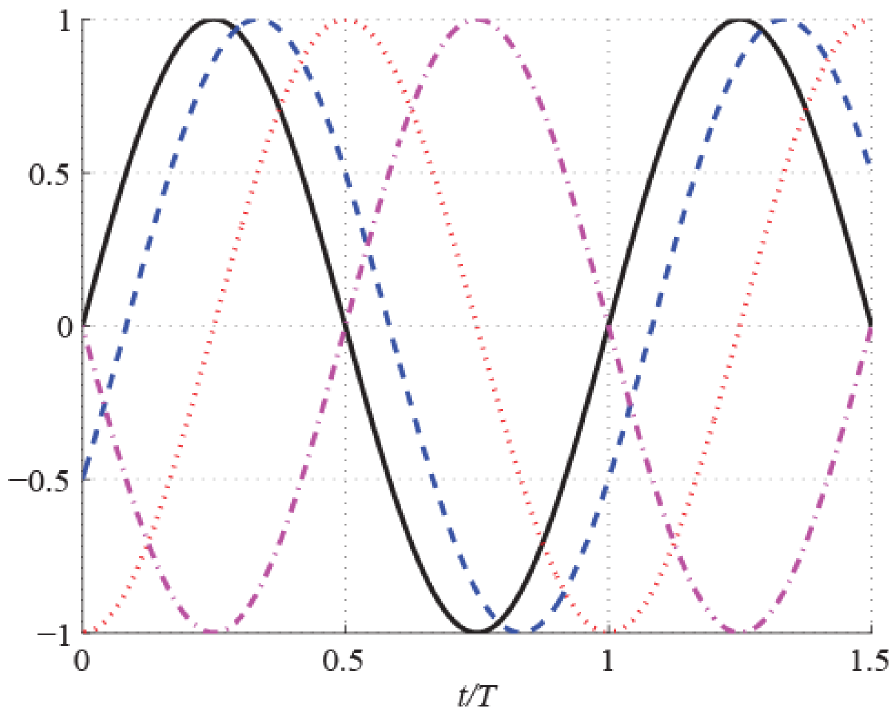
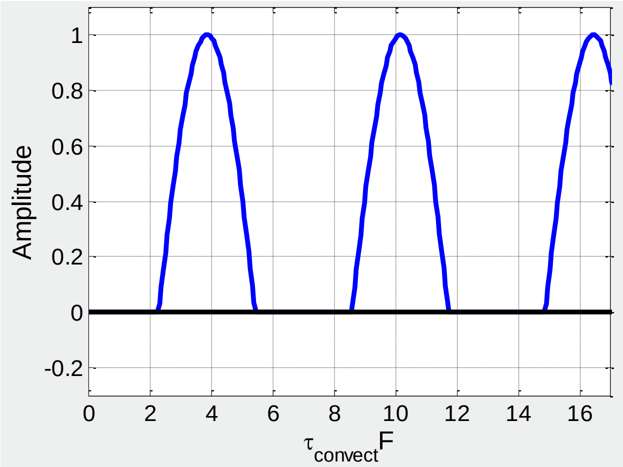
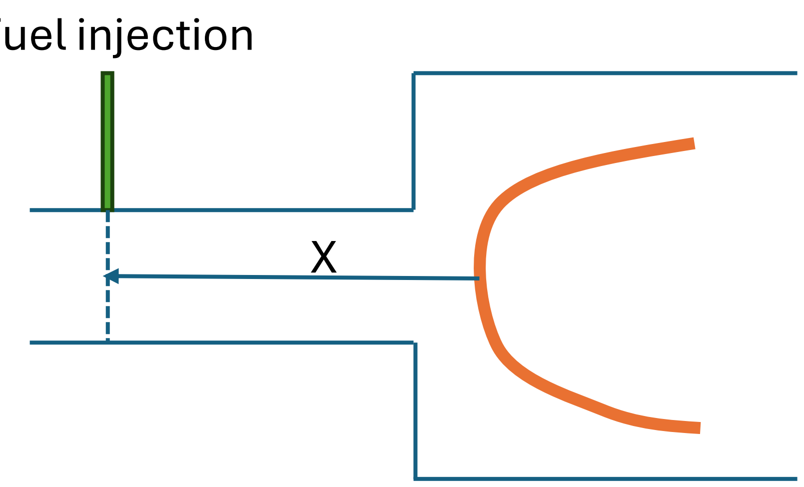


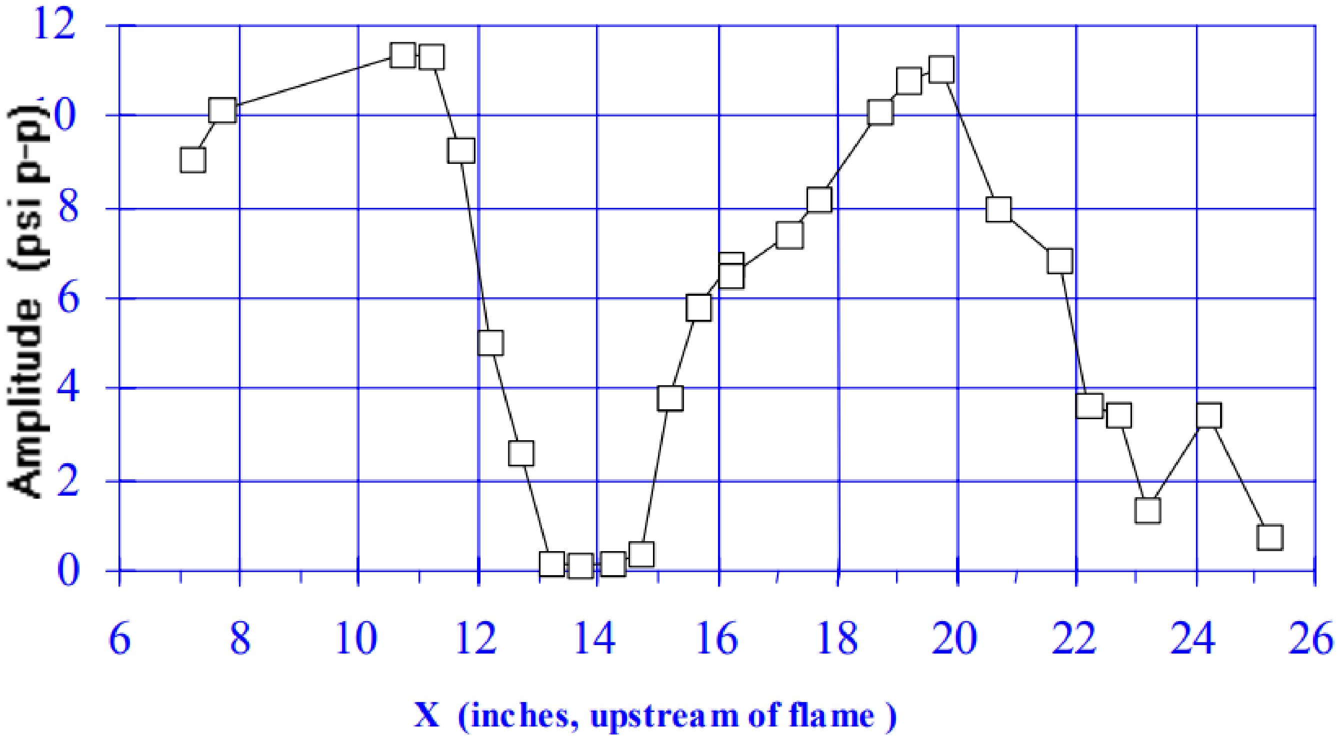
Phase relationship between pressure (solid lines) and three heat release rate oscillations at different phases:

- π/6 (dashed)
- π/4 (dotted)
- π/2 (dash-dot)



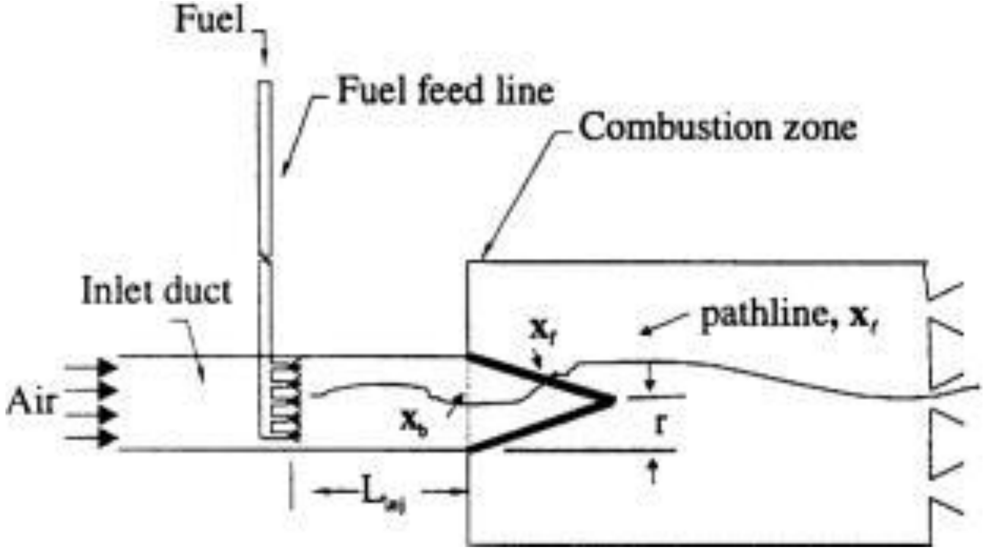
Phase effects example #1: impact of fuel injection point

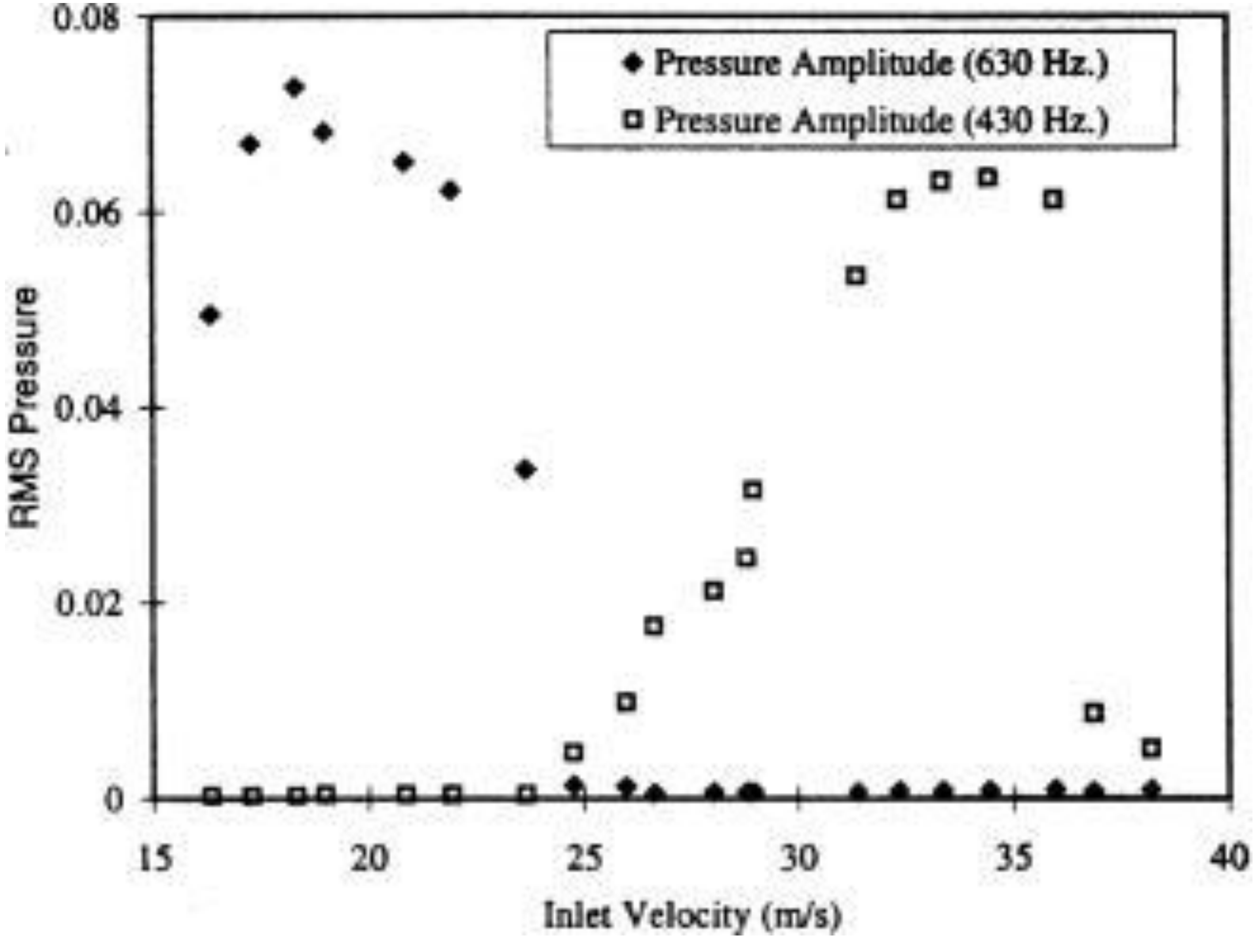




Lovett, JA, & Uznanski, KT. (2002) "Prediction of Combustion Dynamics in a Staged Premixed Combustor." *ASME Turbo Expo*, Amsterdam, The Netherlands.

Phase effects example #2: impact of bulk flow velocity



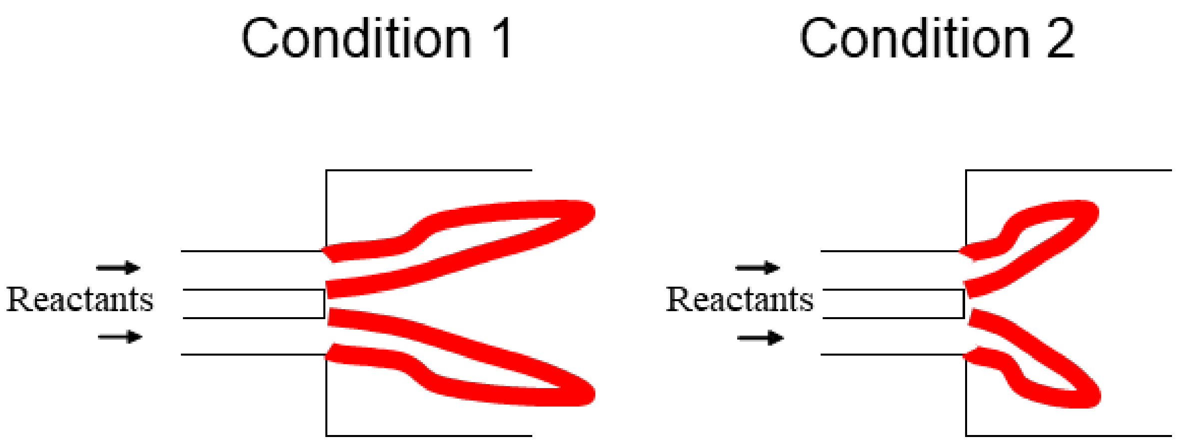


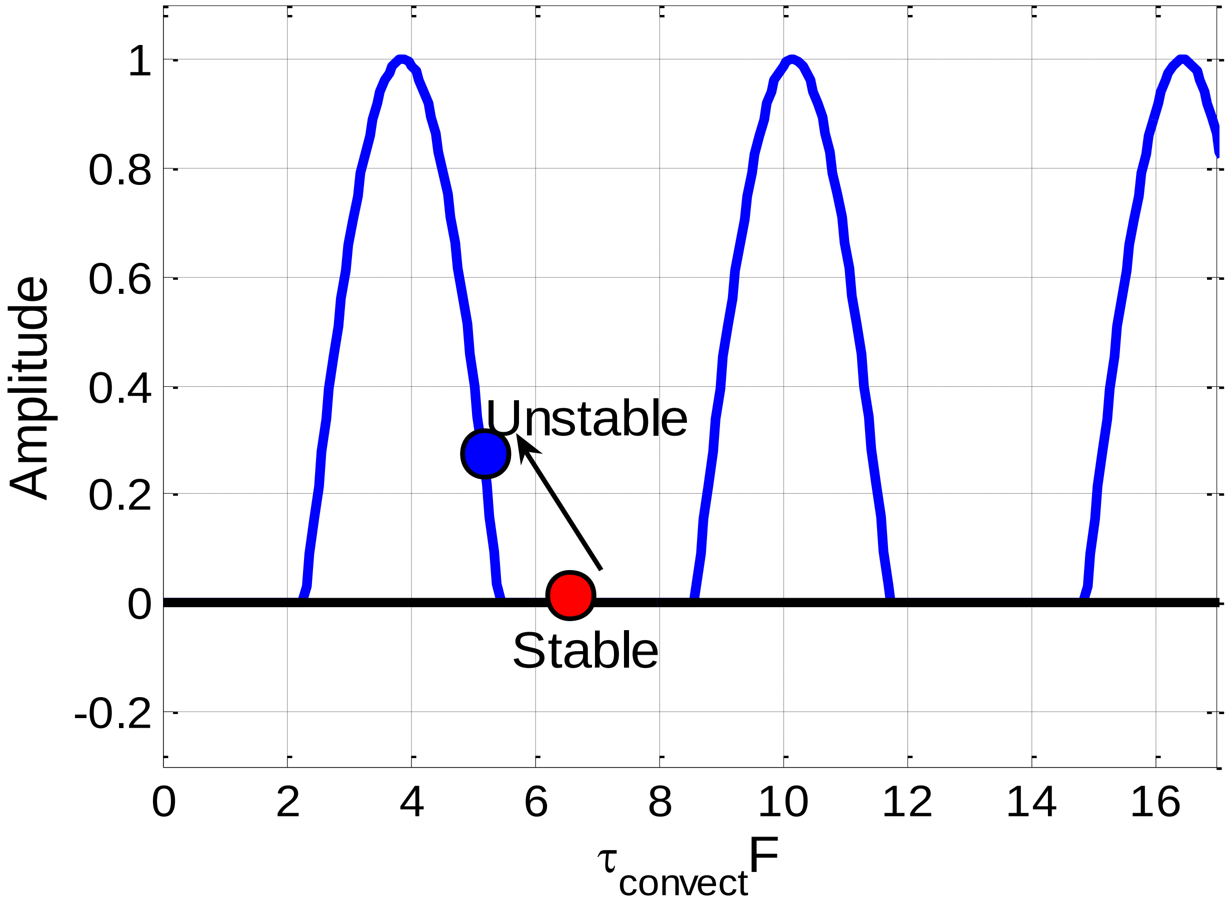
Lieuwen, T., Torres, H., Johnson, C., & Zinn, B. T. (2001). A mechanism of combustion instability in lean premixed gas turbine combustors. J. Eng. Gas Turbines Power, 123(1), 182-189.

Key Point:

Because instability is non-monotonic, changes to any number of operating parameters will never always suppress instability. It always depends.

Illustration of key point – flame speed variation by equivalence ratio, fuel composition, preheat temperature, etc.

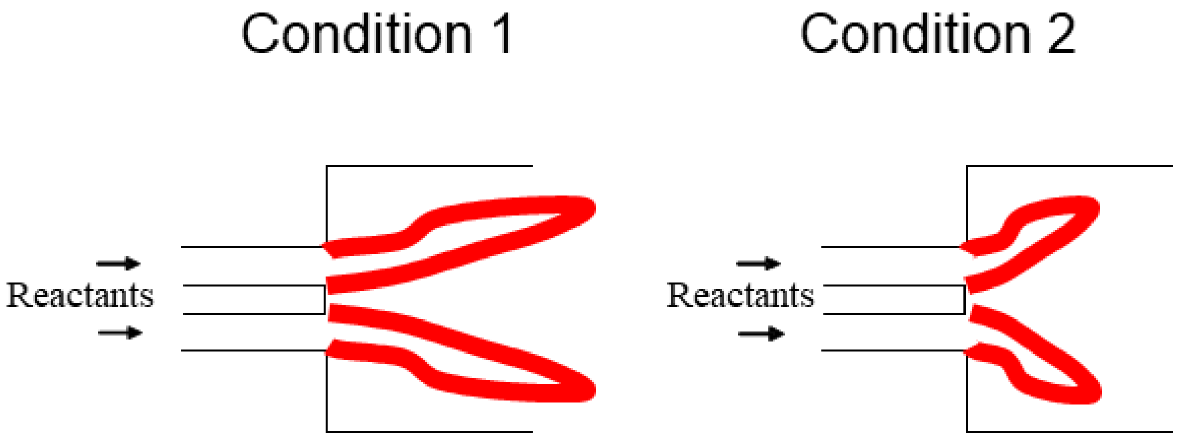


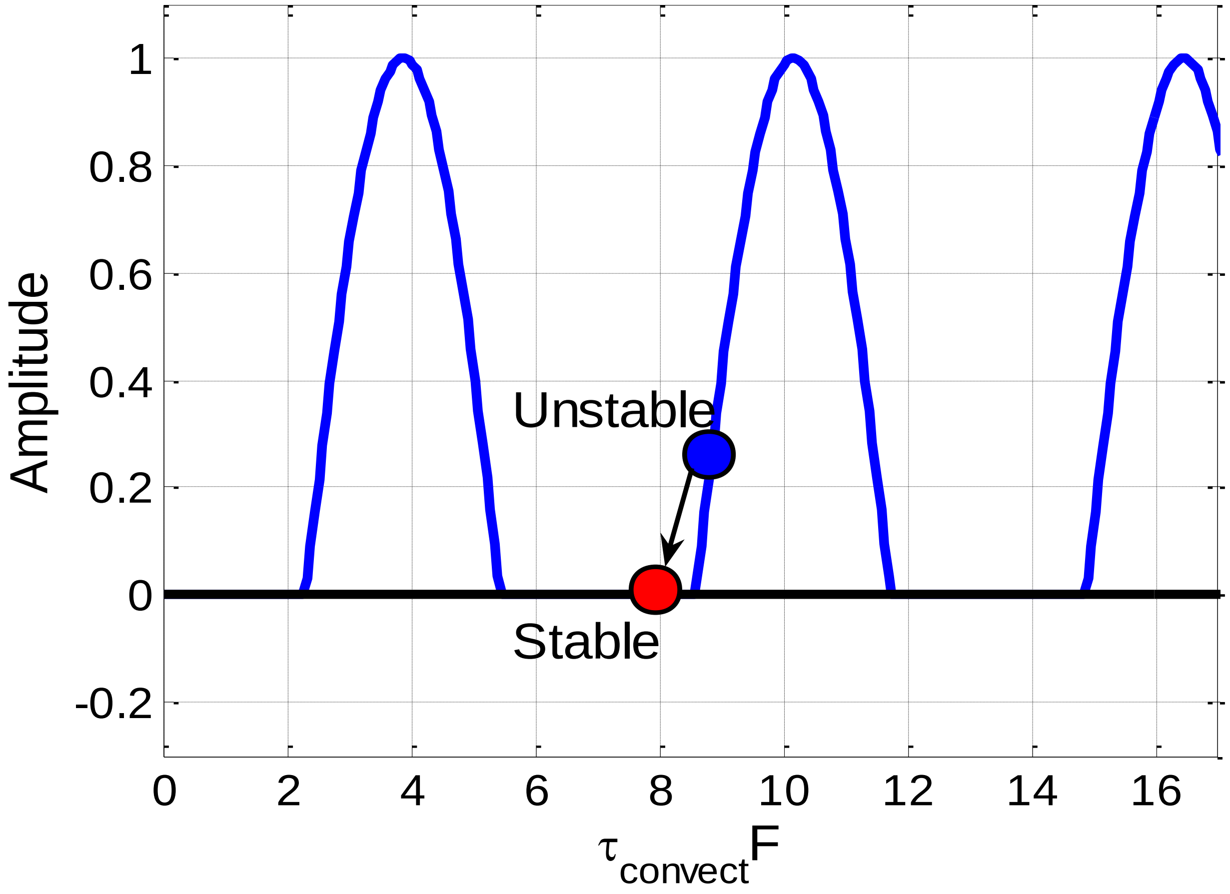


→ Making the flame shorter makes instability worse

Example courtesy of Tim Lieuwen, Georgia Tech

Illustration of key point – flame speed variation by equivalence ratio, fuel composition, preheat temperature, etc.

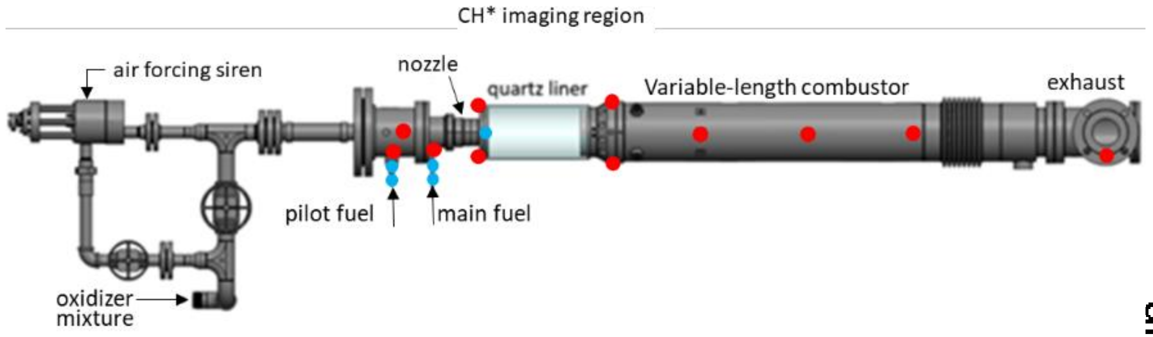




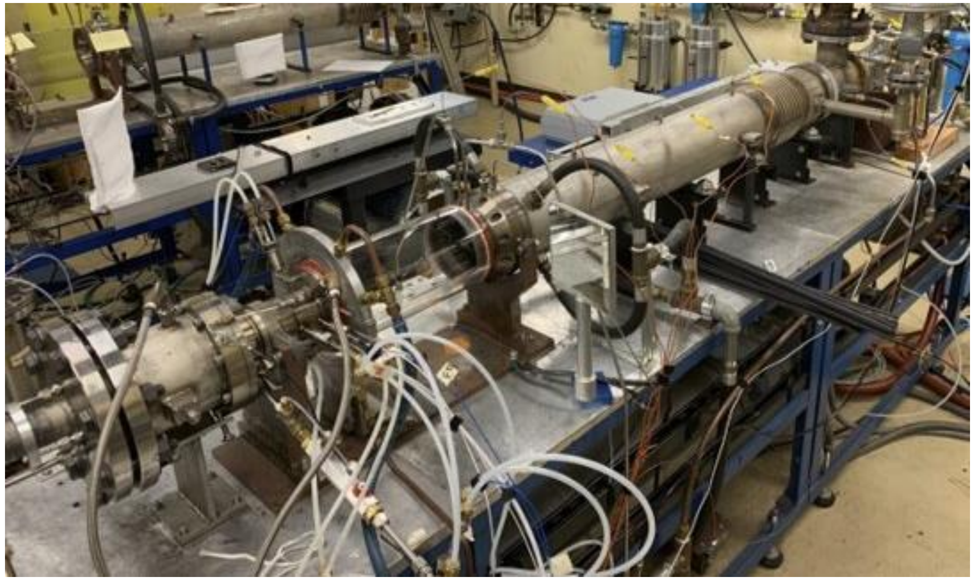
→ Making the flame shorter makes instability better

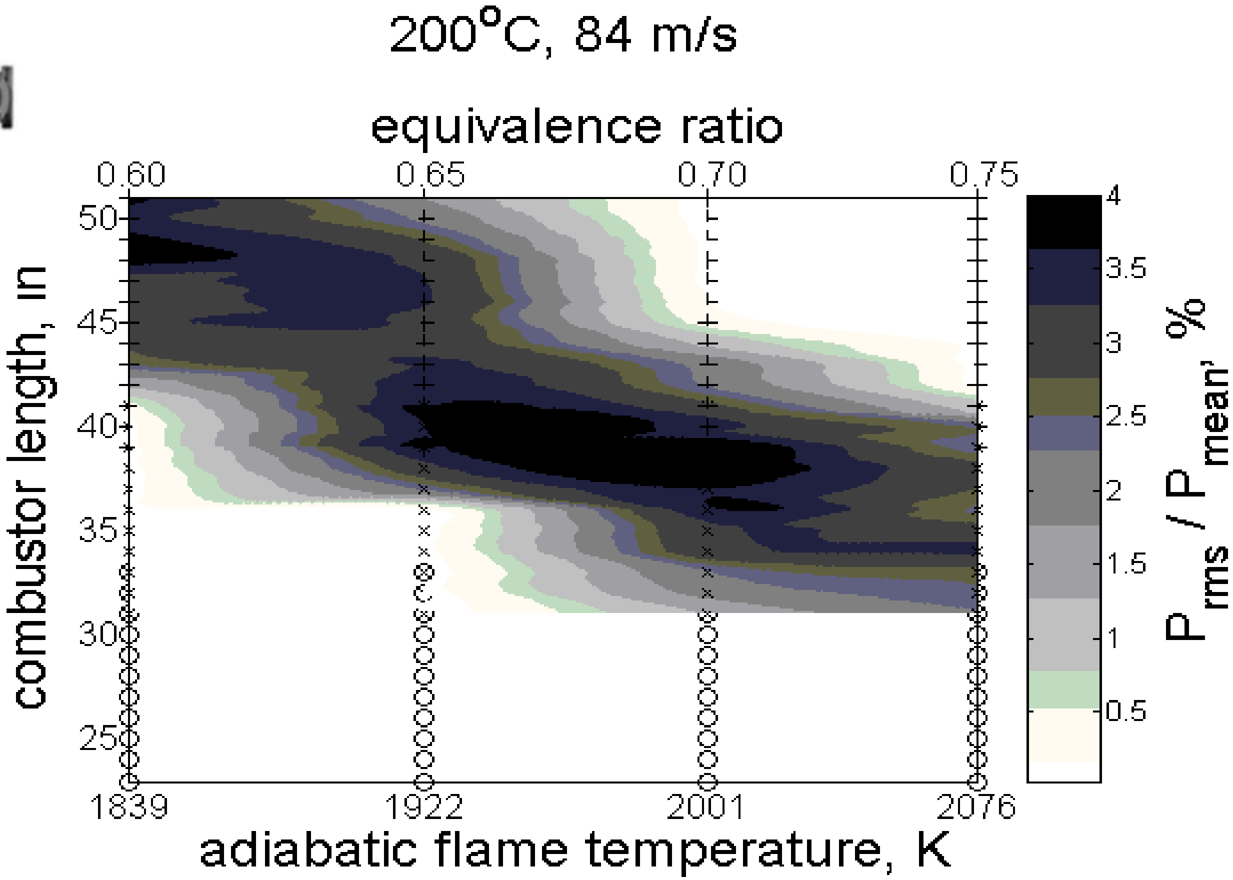
Example courtesy of Tim Lieuwen, Georgia Tech

As a result, you will always, somewhere, have the possibility of an instability



- Temperature measurements
- Dynamic pressure measurement
- Two-microphone method pair



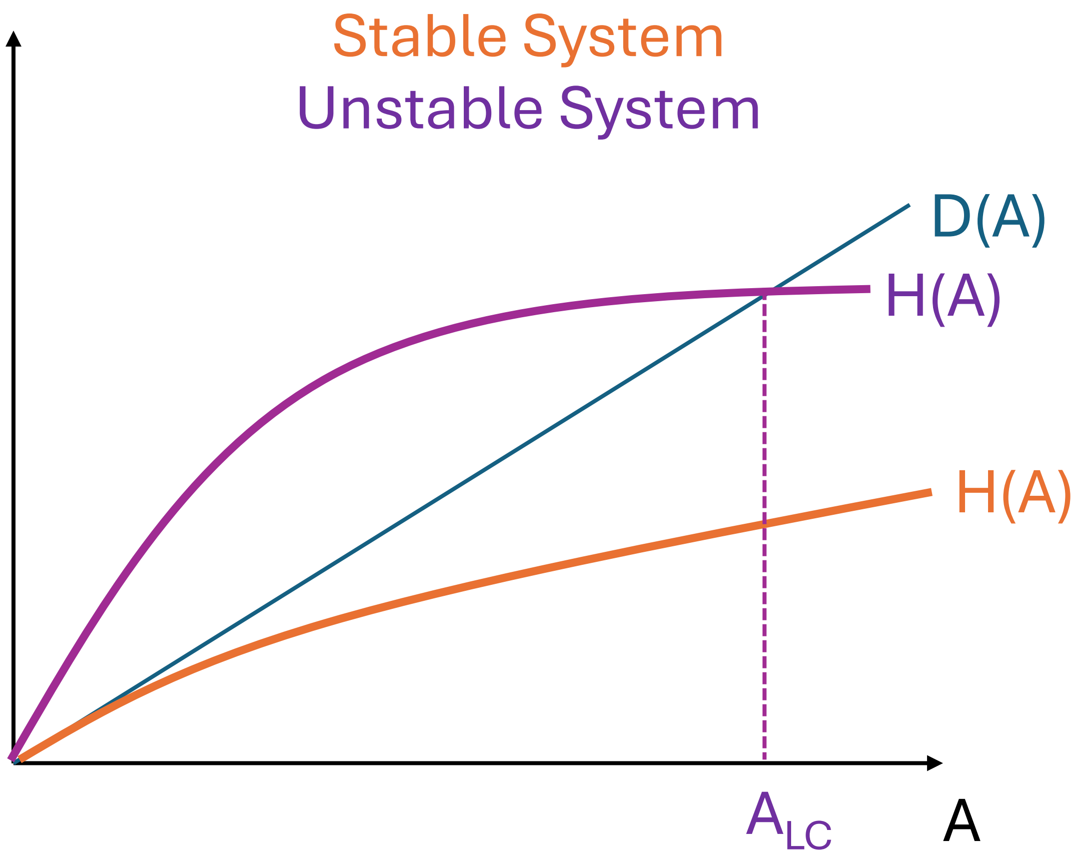


Data courtesy of D. Santavicca

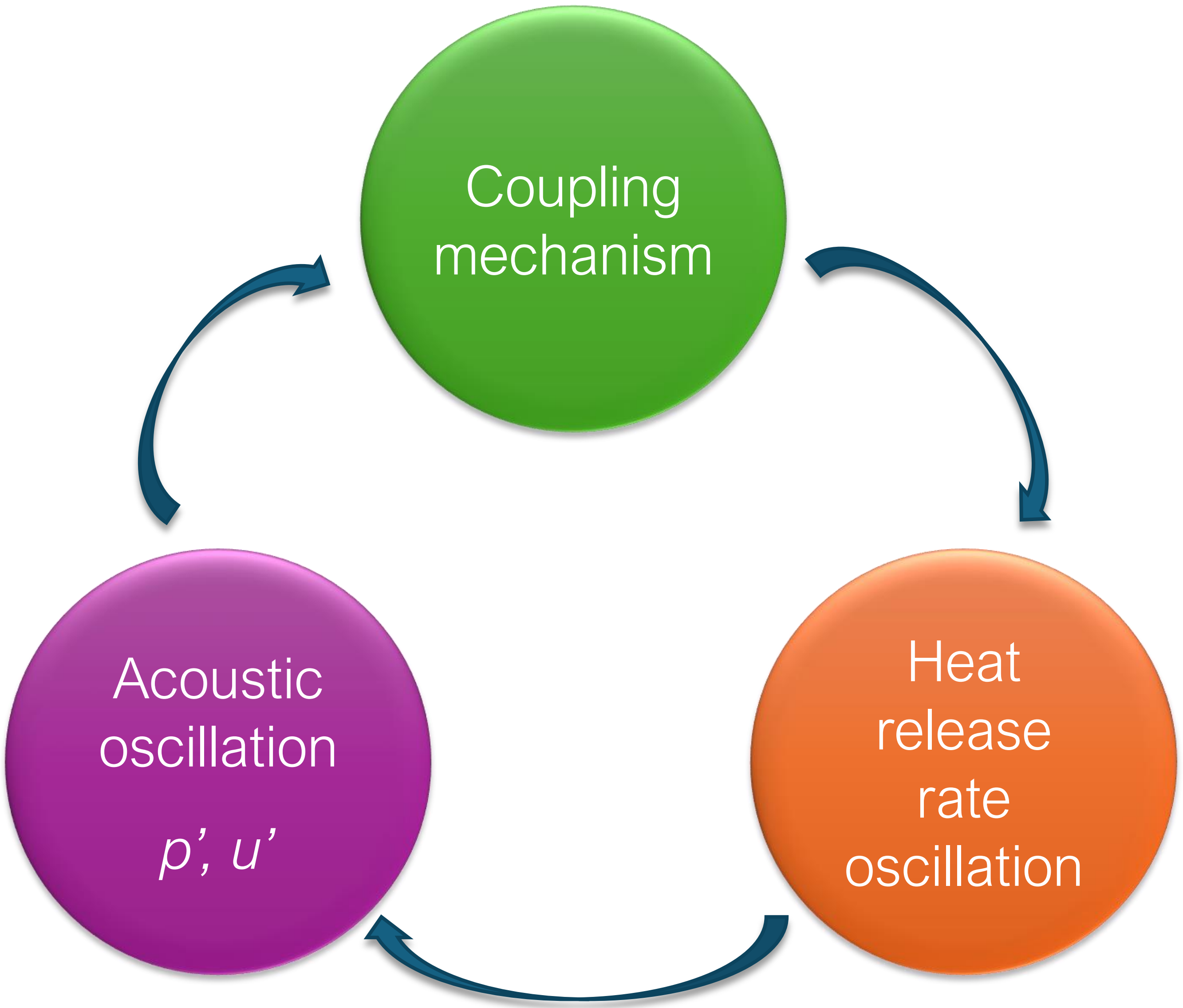
The amplitude of the instability is determined by the balance between damping and driving, producing a limit cycle oscillation

$$RI = \int_V \int_T p'(\vec{x},t)\dot{q}'(\vec{x},t)\,dt\,dV$$

- If RI > 0 → thermoacoustic feedback drives instability growth
- If RI > Damping → instability actually grows
- If RI < 0 → thermoacoustic feedback does not exist



Coupling mechanisms



Velocity coupling was identified in some of the earliest work on combustion instability as a critical mode for coupling

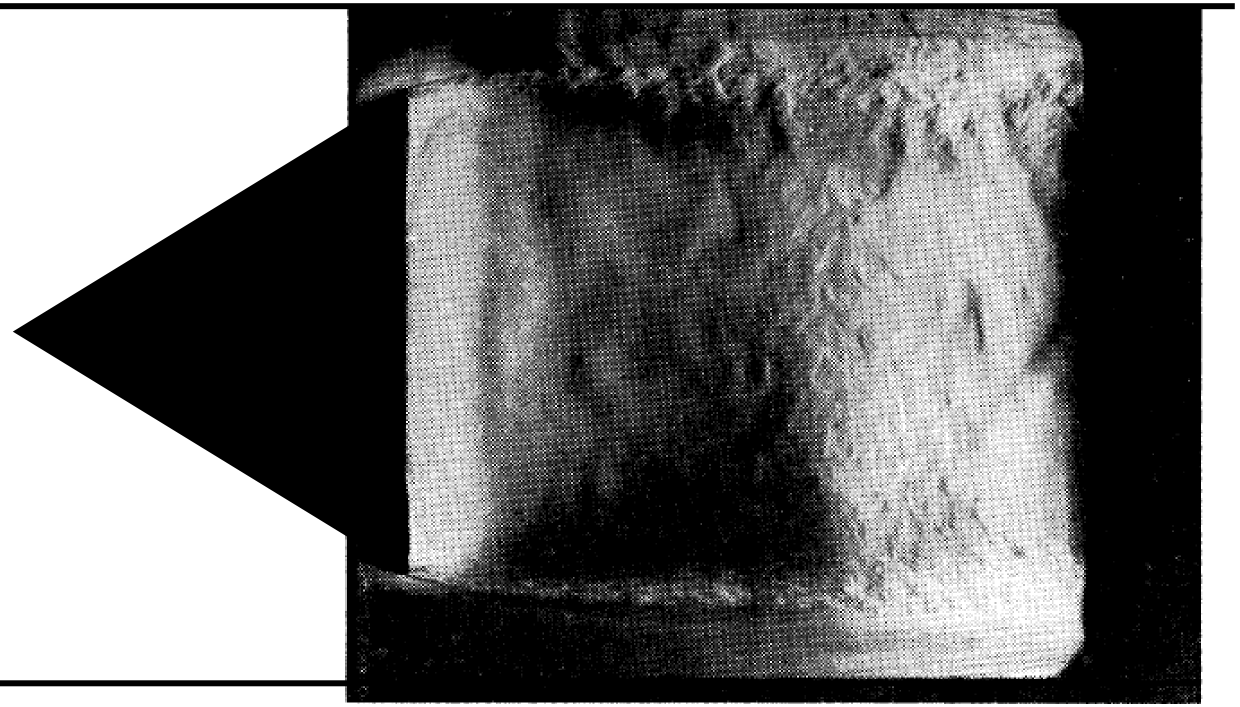


Fig. 12 Spark-schlieren photograph of smooth combustion

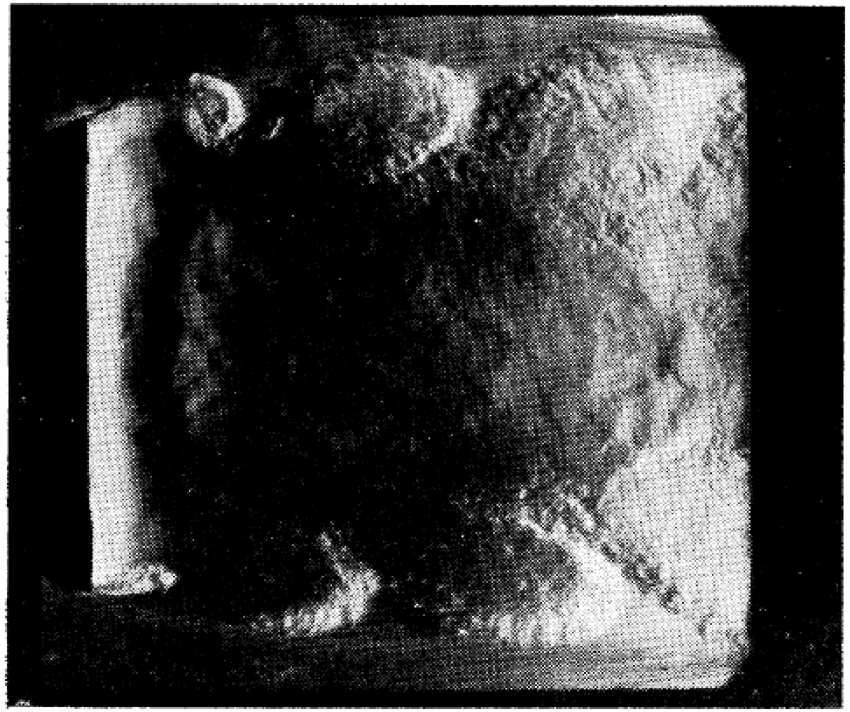
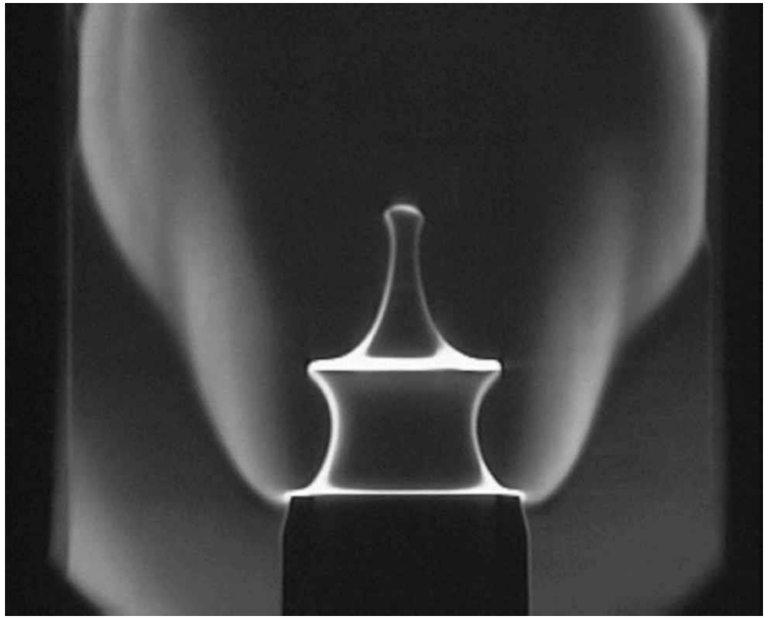


Fig. 13 Spark-schlieren photograph of screeching combustion

Rogers, D. E., & Marble, F. E. (1956). A mechanism for high-frequency oscillation in ramjet combustors and afterburners. Journal of Jet Propulsion, 26(6), 456-462.

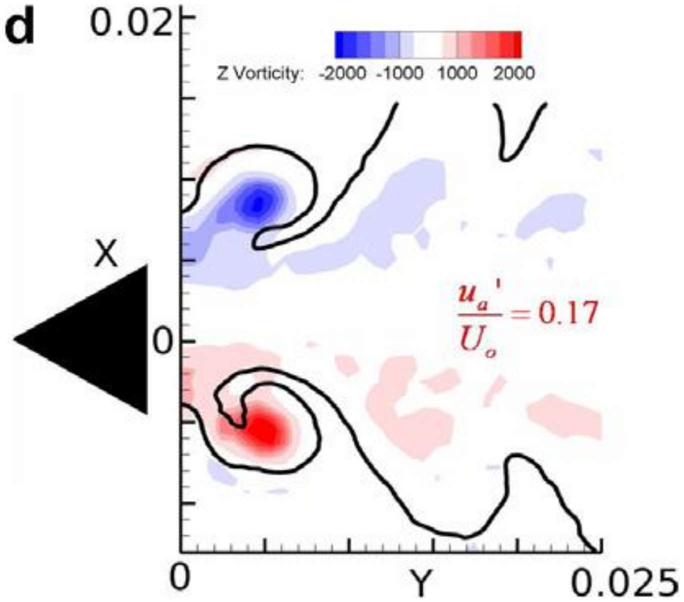
Velocity coupling can occur through multiple sources of velocity, although the most common two sources are acoustic and vortical

Acoustic



Source: S. Ducruix, D. Durox, S. Candel, Centre National de la Recherche Scientifique and Ecole CentralSupelec, Paris-Saclay

Vortical



Source: Shanbhogue, S., D.-H. Shin, S. Hemchandra, D. Plaks, T. Lieuwen, (2009) "Flame-sheet dynamics of bluff-body stabilized flames during longitudinal acoustic forcing" Proceedings of the Combustion Institute, **32**(2):1787-1794

SIDEBAR

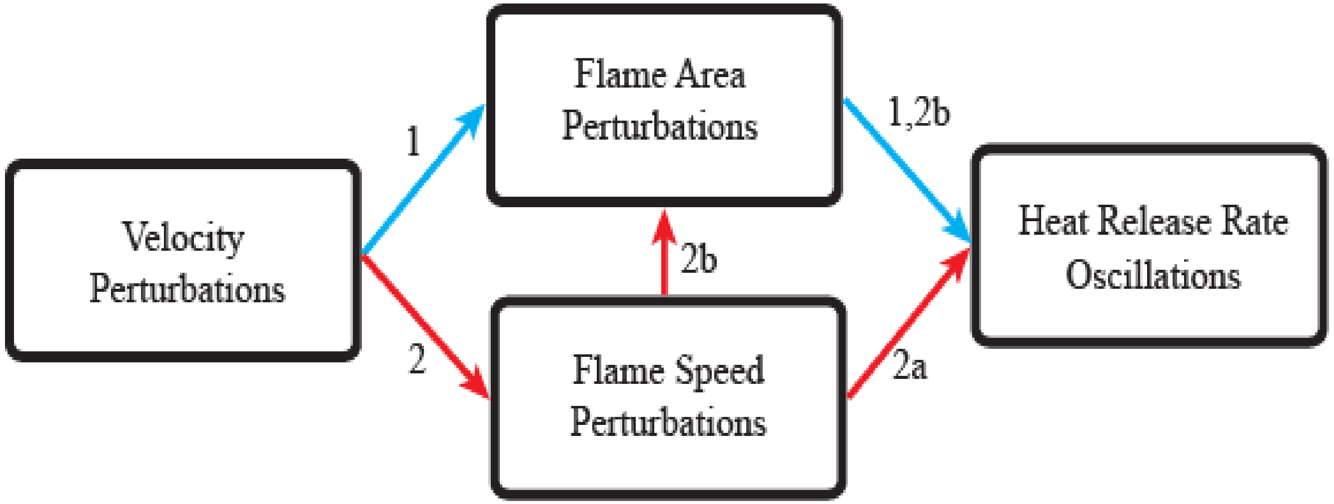
Disturbance decompositions – to the board!

SIDEBAR OVER

Back to velocity coupling

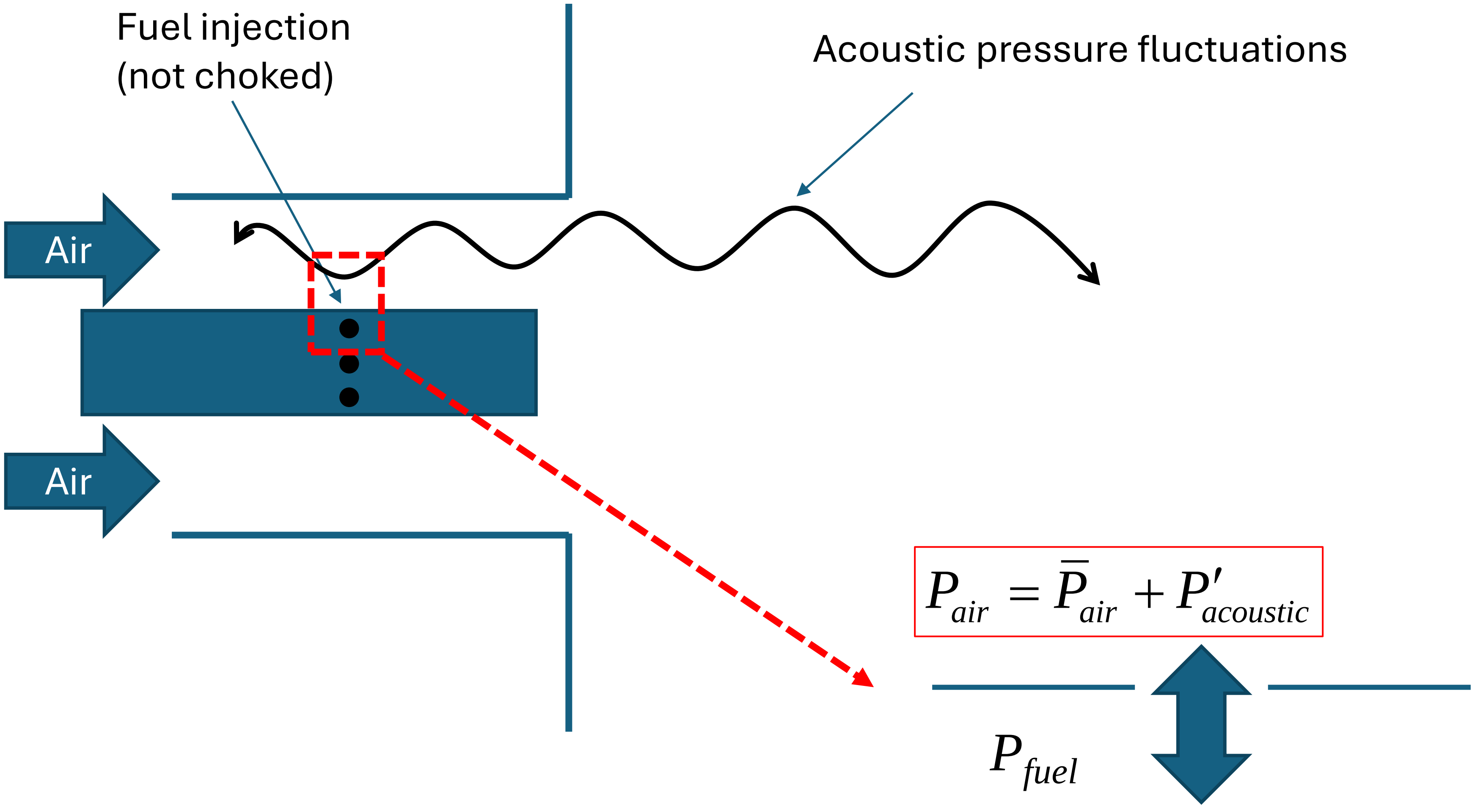
Velocity coupling works through a variety of mechanisms, though the area component is largely responsible

$$Q(t) = \int_{A_{flame}} \rho_u S_L h_R \, dA$$



Source: Shreekrishna, *Response mechanisms of attached premixed flames to harmonic forcing*, Aerospace Engineering, Georgia Institute of Technology (2011)

Equivalence ratio coupling occurs when fluctuations in the fuel and/or oxidizer drive local fluctuations in the equivalence ratio – sometimes called “mixture coupling”

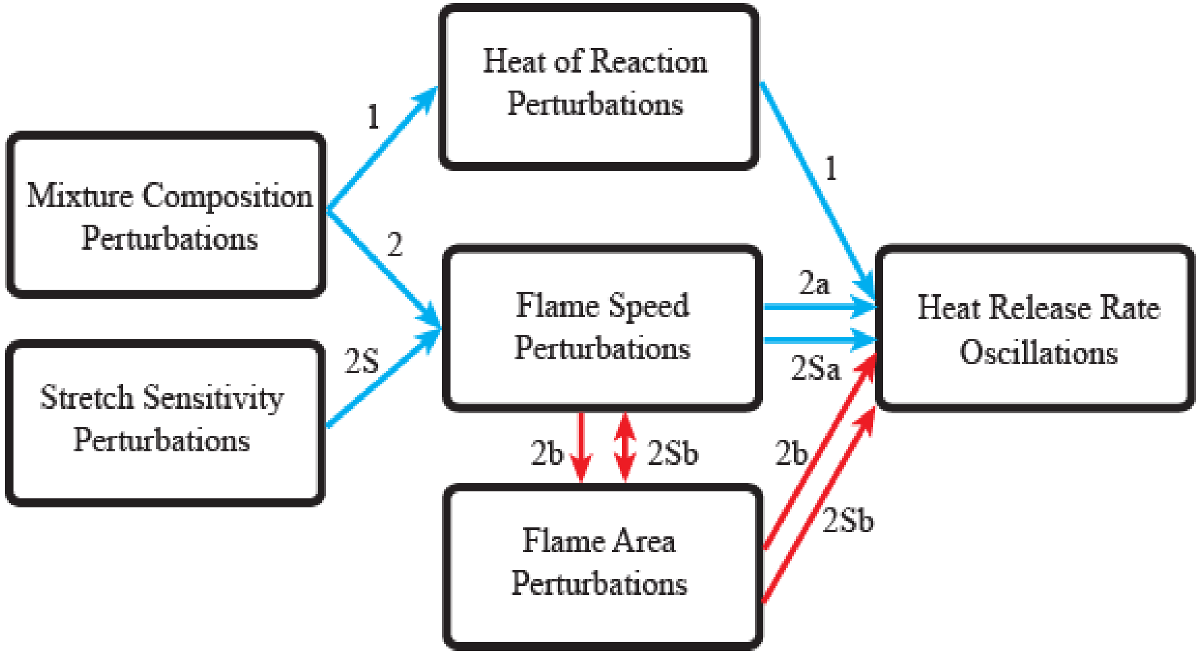


$$\boxed{P_{air} = \bar{P}_{air} + P'_{acoustic}}$$

$$\dot{m}_{fuel}(t) \sim c_v \sqrt{2\left(P_{fuel} - \boxed{P_{air}(t)}\right)}$$

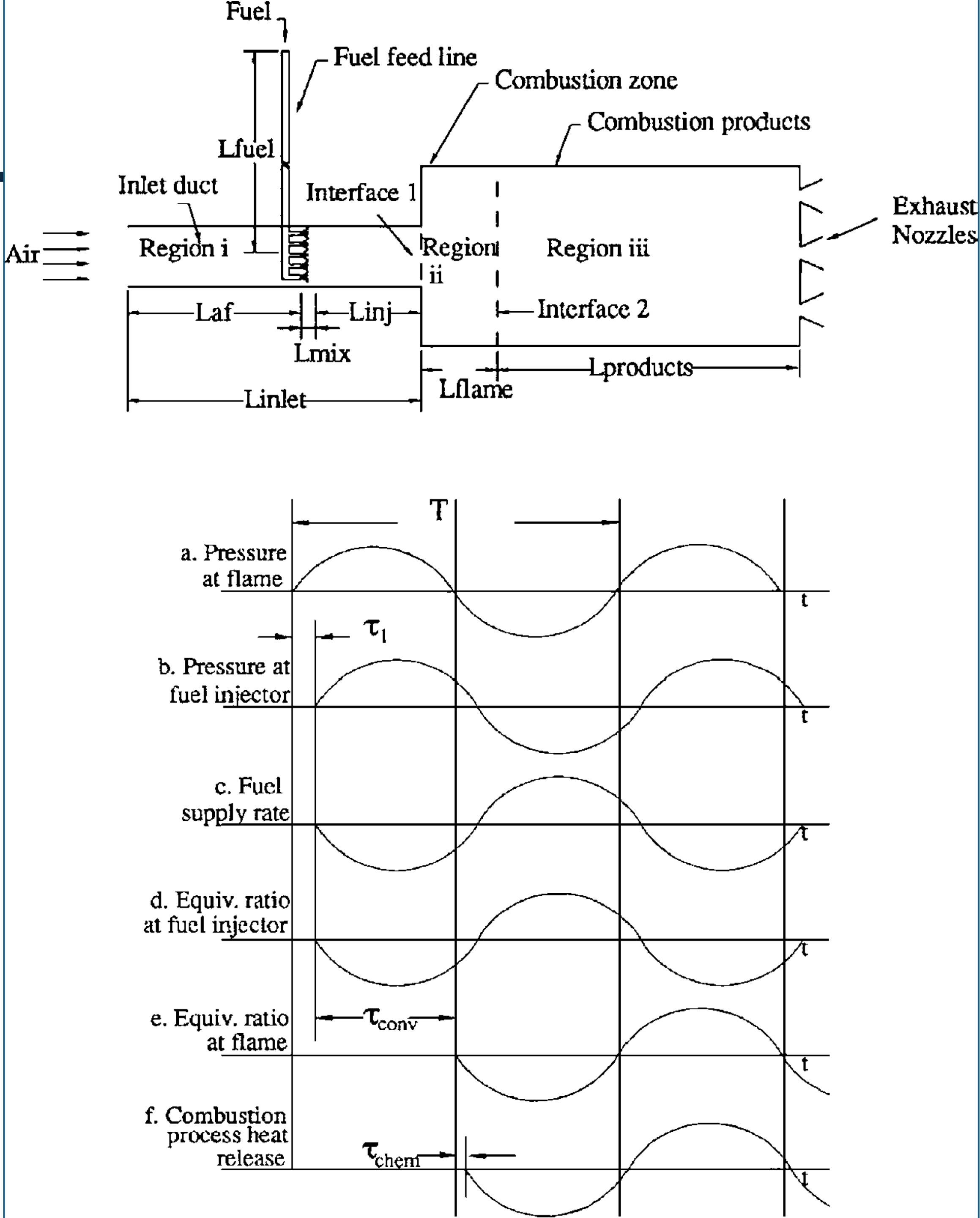
Equivalence ratio fluctuations drive heat release rate fluctuations through a complex web of pathways

$$Q(t) = \int_{A_{flame}} \rho_u S_L h_R \, dA$$



Source: Shreekrishna, *Response mechanisms of attached premixed flames to harmonic forcing*, Aerospace Engineering, Georgia Institute of Technology (2011)

Phase delays sum additively in realistic systems, resulting in interference



Ducruix, S., Schuller, T., Durox, D., & Candel, S. (2003). “Combustion dynamics and instabilities: Elementary coupling and driving mechanisms.” *Journal of Propulsion and Power*, 19(5), 722-734.

Key Point:

Keep track of your phase lags – they are the driver of instability.

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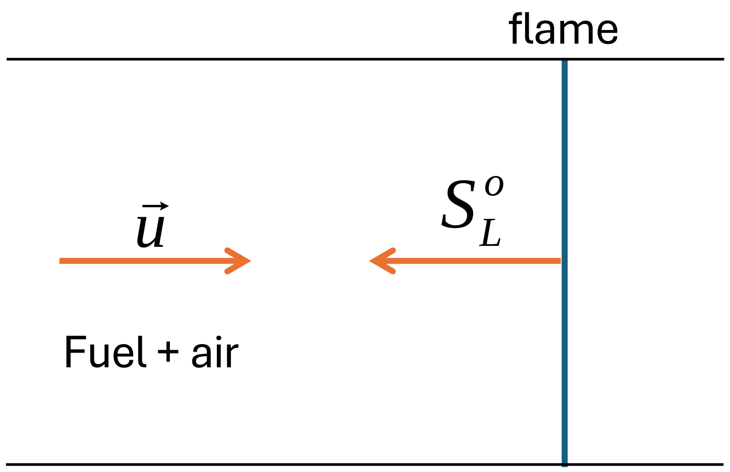
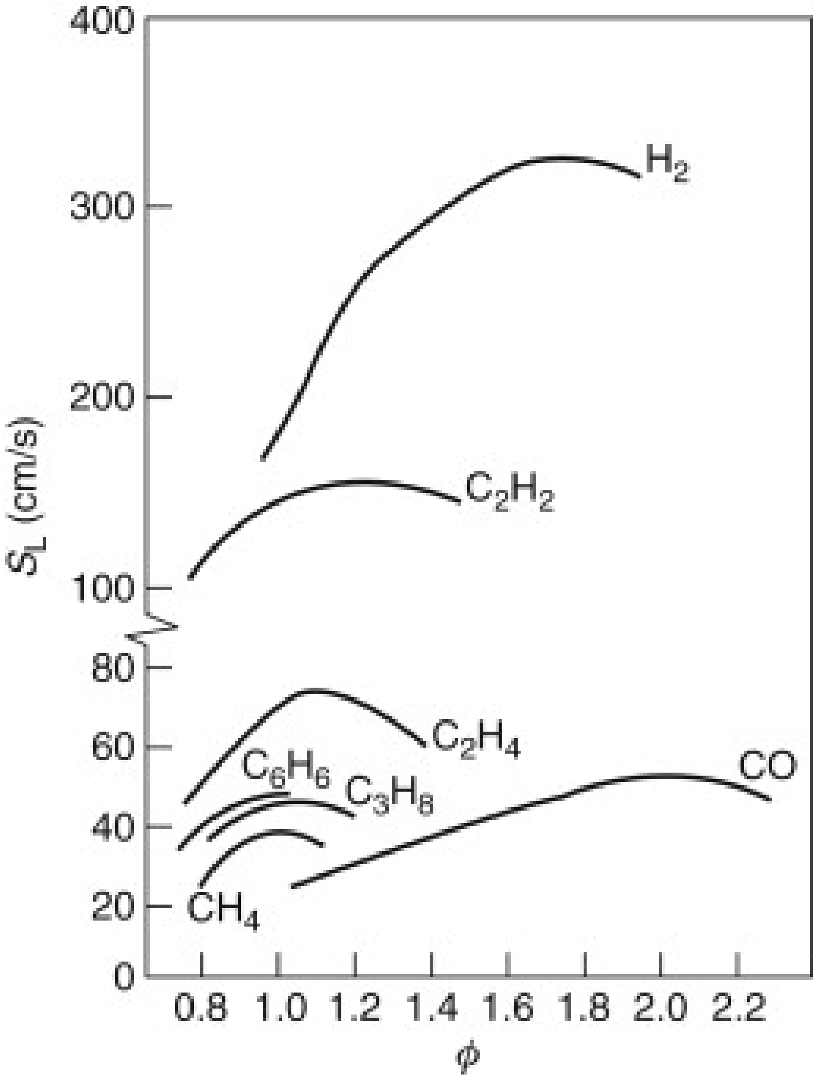
Flames respond to acoustic forcing in really beautiful ways and the physics behind the response is referred to as “flame kinematics”

Videos courtesy of D. Durox, EM2C Laboratory

Flames propagate normal to themselves at a speed S_L, the flame speed, which is not that fast for most mixtures

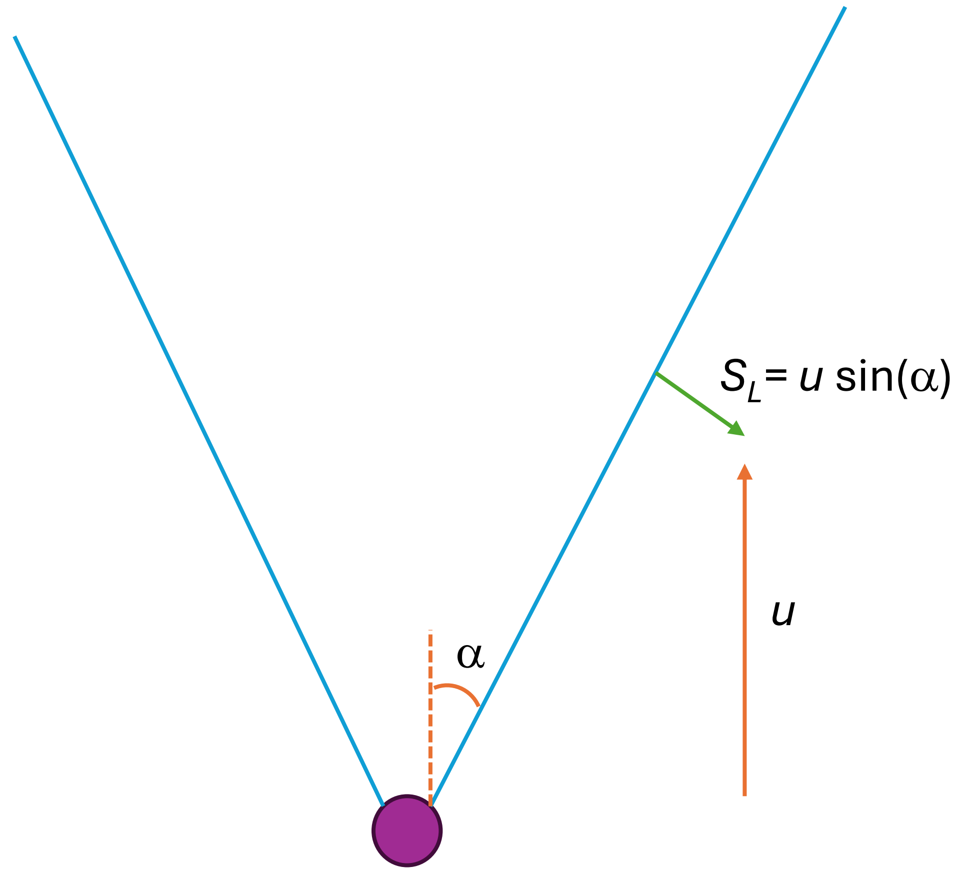


Keyword: Kinematic condition is where $S_L = u$

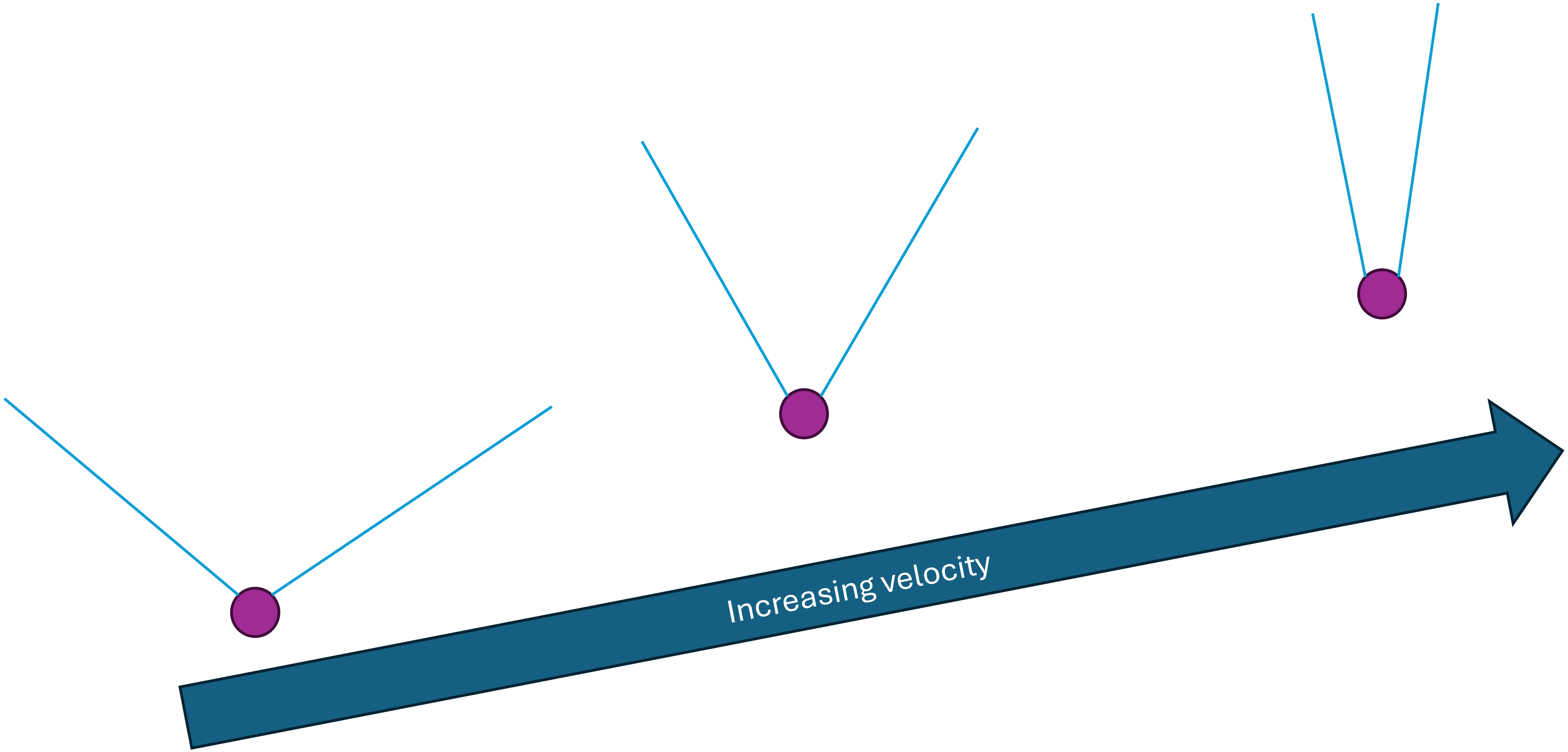


Glassman, I., Yetter, R. A., & Glumac, N. G. (2014). *Combustion*. Academic press.

Flame propagation in two dimensions follows the same basic concept, but now you have to do some trig to determine the flame angle



As you change the velocity of the flow, you change the angle at which the flame stabilizes relative to the flow where the kinematic condition is met



Now with crafts!

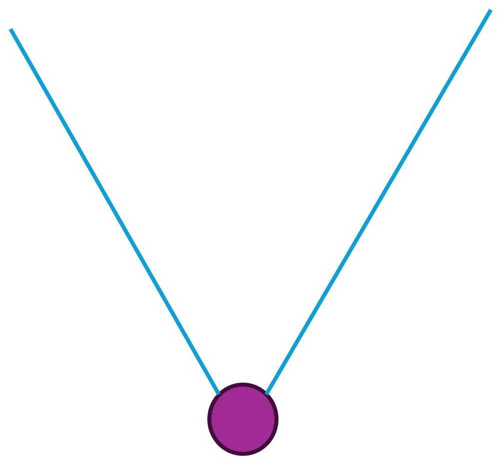
Let’s try this with our straws and pipe cleaners

Let’s explore what happens when the velocity varies very slowly

$$u(t) = \bar{u} + u' \cos(\omega t)$$

Would nominally result in this flame shape

Is a very low frequency



SIDEBAR

Convective wavelength – to the board!

SIDEBAR OVER

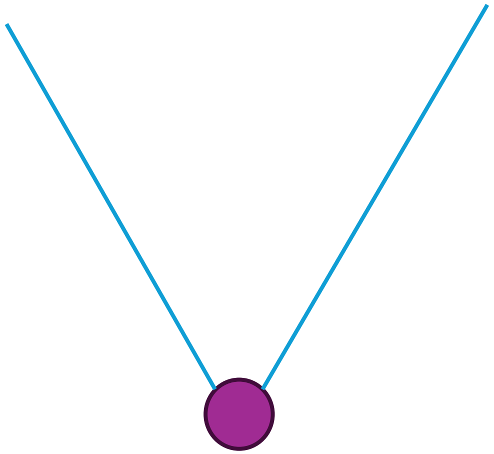
Back to flame response

Harmonic forcing – convective wavelength < Lf

$$u(t) = \bar{u} + u'\cos(\omega t)$$

Would nominally result
in this flame shape

Is no longer a small number…

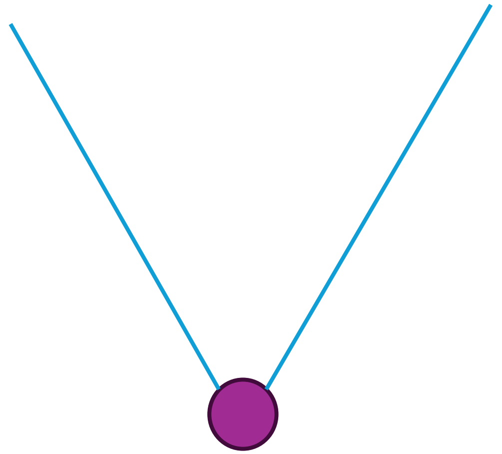


Harmonic forcing – convective wavelength << Lf

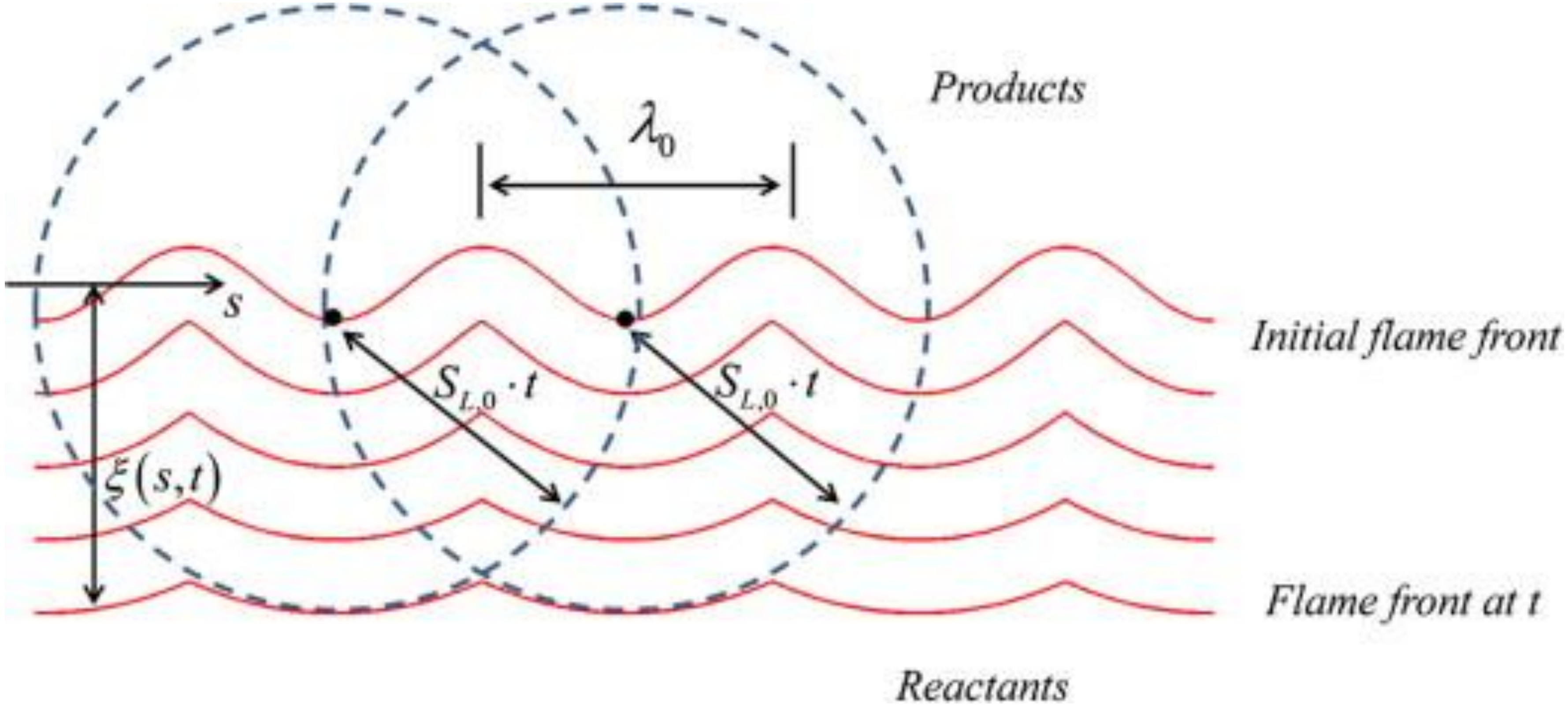
$$u(t) = \bar{u} + u' \cos(\omega t)$$

Would nominally result
in this flame shape

Is a large number…

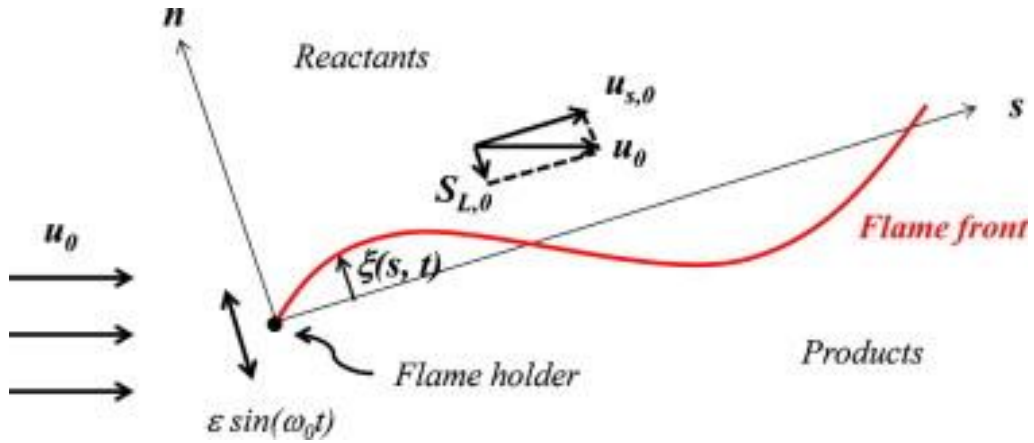


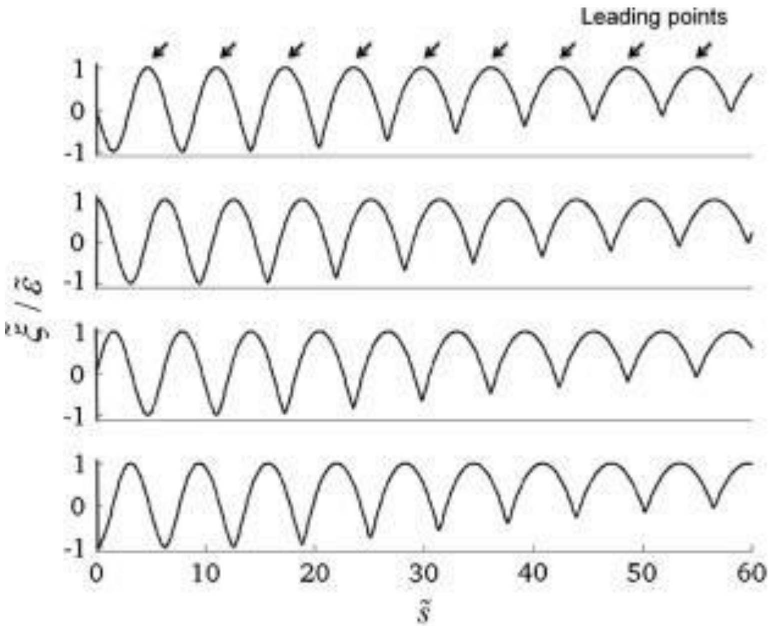
When flames propagate normal to themselves, they first cusp and then destroy the wrinkles imposed by an oscillating field

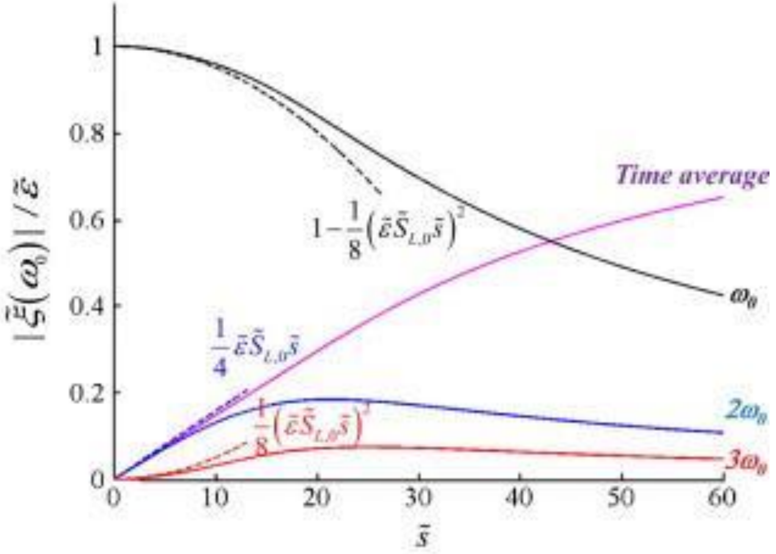


Shin, D. H., & Lieuwen, T. (2012). Flame wrinkle destruction processes in harmonically forced, laminar premixed flames. *Combustion and Flame*, *159*(11), 3312-3322.

The flame can only withstand a certain frequency of oscillation before kinematic restoration destroys wrinkles faster than they can cause heat release rate oscillation

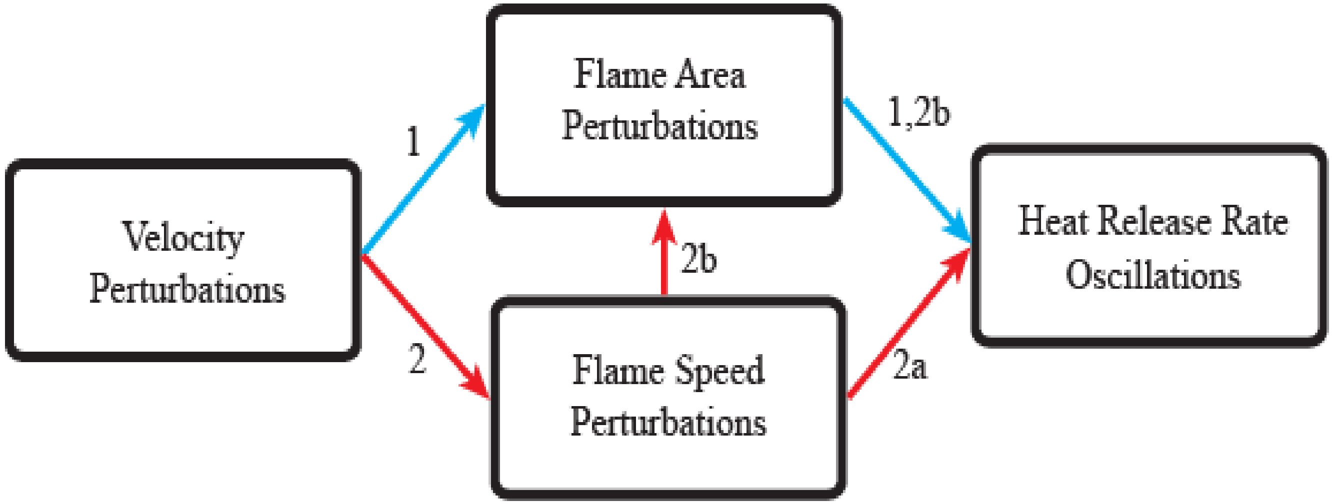


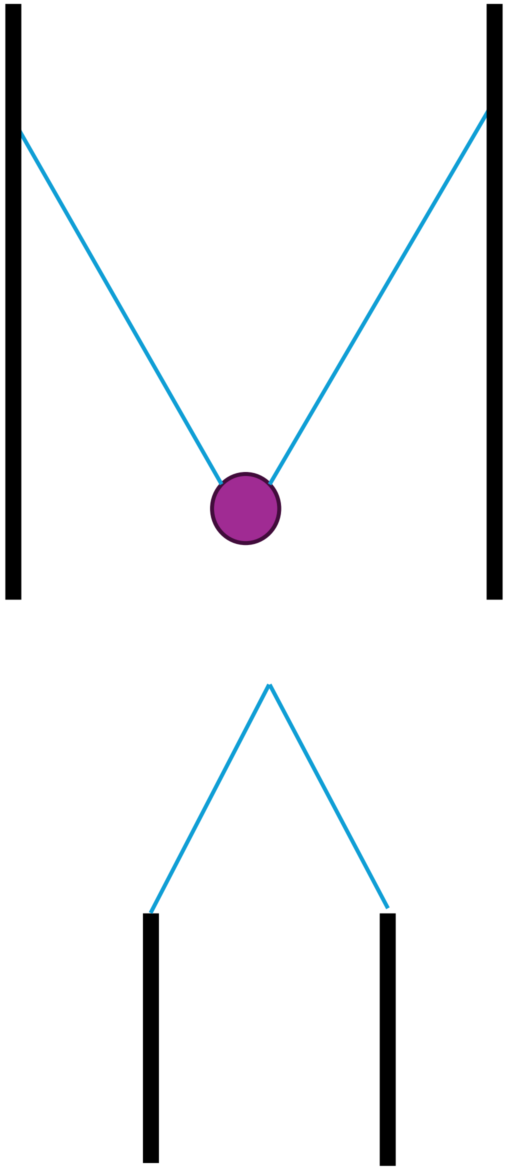




Shin, D. H., & Lieuwen, T. (2012). Flame wrinkle destruction processes in harmonically forced, laminar premixed flames. *Combustion and Flame*, *159*(11), 3312-3322.

Put a boundary condition on this problem and you can now explain the flame area pathway of velocity-coupled combustion instability

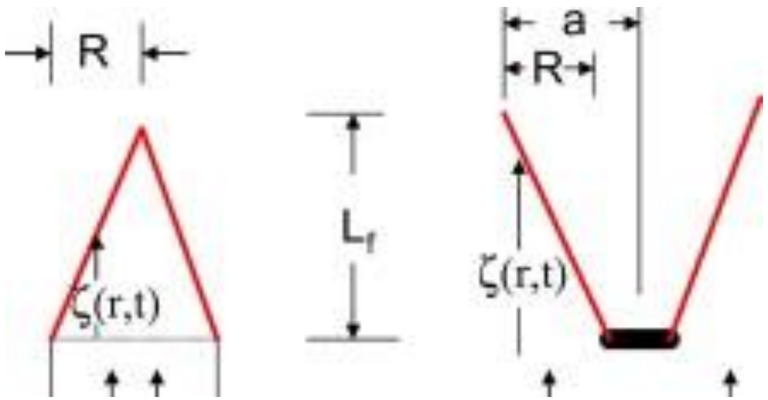




Source: Shreekrishna, *Response mechanisms of attached premixed flames to harmonic forcing*, Aerospace Engineering, Georgia Institute of Technology (2011)

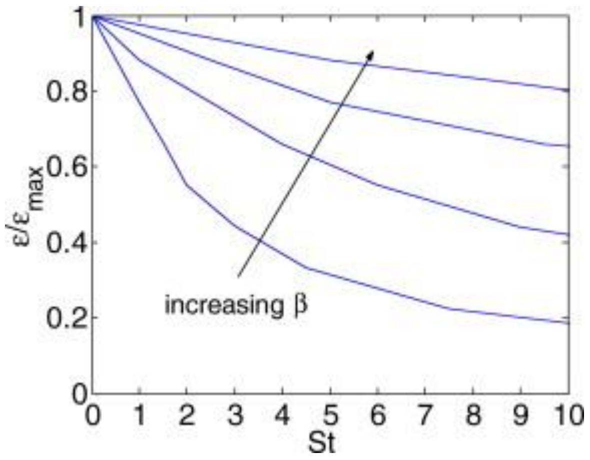
We can use a level-set formulation to better explain this – to the board!

The base shape of the flame in addition to the frequency of excitation determines the kinematic response of the flame to perturbations

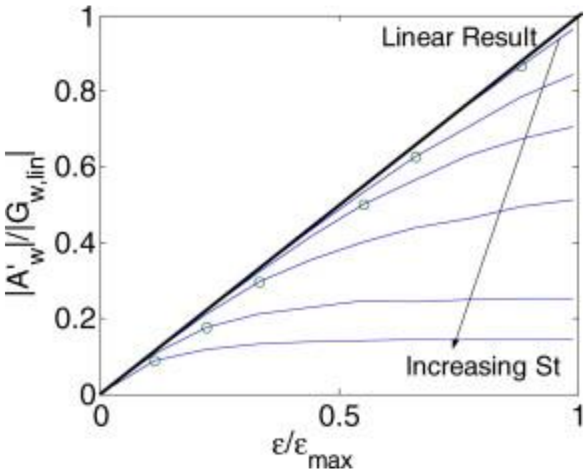


$$u_o / S_L = \sqrt{(L_f / R)^2 + 1}$$

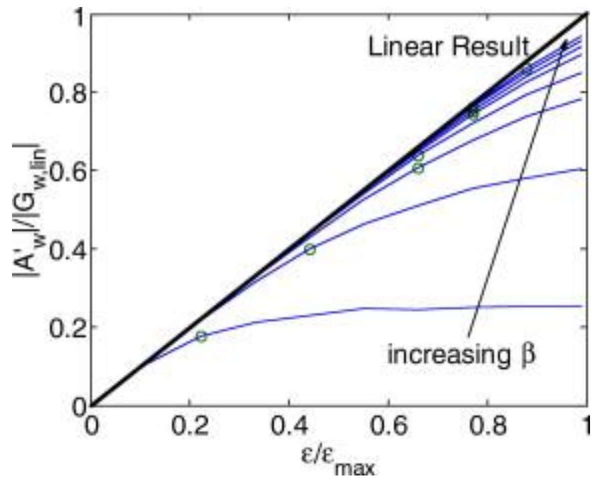
$$\beta \equiv L_f / R$$



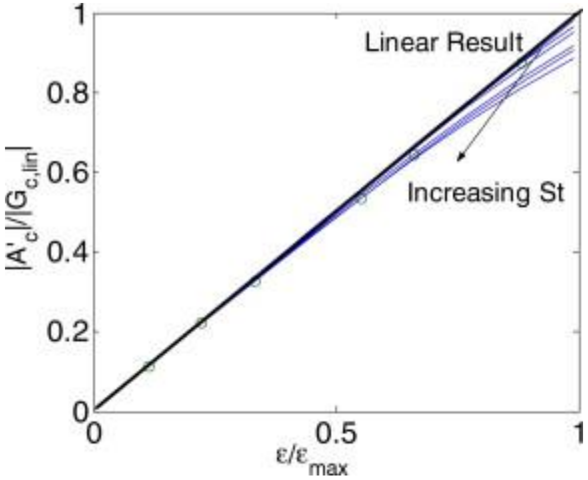
Dependence of lowest velocity amplitude at which flame front cusps are present on St (β = 1, 2, 4, and 8).



A' vs. u' for axisymmetric wedge flames at β = 1. Circle denotes lowest $\varepsilon/\varepsilon_{max}$ value where cusps appear.



A' vs. u' for axisymmetric wedge flames at St = 10 and β = 1–8. Circle denotes lowest $\varepsilon/\varepsilon_{max}$ value where cusps appear.



A' vs. u'for axisymmetric cone flames at β = 1. Circle denotes lowest $\varepsilon/\varepsilon_{max}$ value where cusps appear.

Lieuwen, T. (2005). Nonlinear kinematic response of premixed flames to harmonic velocity disturbances. *Proceedings of the Combustion Institute*, *30*(2), 1725-1732.

Key Point:

Flames act as low-pass filters and their shape (and robustness to stretch) determines what the cut-off frequency is.

What have we learned so far?

1. Identify the technologies in which combustion instability is an issue
2. Describe the basic thermoacoustic coupling process
3. Explain the kinematics of flame response to harmonic inputs
4. Assess a flame transfer function to identify the potential coupling processes
5. Use your understanding of flame response and thermoacoustic coupling to explain instability control techniques

Day 2: Tuesday, June 18

Welcome!

Review of key points

- Thermoacoustic instability is driven by a feedback process between pressure and heat release rate oscillations – the phase between these two determines whether the thermoacoustic coupling is driving or damping

- The final amplitude of the instability is a balance between driving and damping, located at a limit cycle amplitude that is determined by the level of nonlinearity in the flame response

- Several different coupling mechanisms exist to couple the acoustics and flame response – the mechanism of heat release rate oscillation is largely through area fluctuations where velocity fluctuations are present

- Flame kinematics determines these area fluctuations, where the flame responds as a low-pass filter to incoming oscillations

Course schedule

Monday:
— 1400-1500: Introduction and combustion instability in technologies
— 1500-1530: Thermoacoustic feedback
— 1530-1545: Break
— 1545-1730: Flame kinematics (with crafts!) and flame response

Tuesday:
— 1400-1530: Flame response and flame transfer functions
— 1530-1545: Break
— 1545-1630: Instability control
— 1630-1730: Advanced topics

Flame response is often described in terms of “flame transfer functions,” which describe the response of a flame to an input disturbance

Input velocity fluctuation → G(f) → Output heat release rate fluctuation

$$u(t) = \bar{u} + u'\cos(\omega t)$$
$$u(t) = \bar{u} + u'\cos(2\pi f t)$$

$$Q(t) = \bar{Q} + Q'\cos(\omega t)$$
$$Q(t) = \bar{Q} + Q'\cos(2\pi f t)$$

Fourier Transform

$$\hat{u}(f)$$

Fourier Transform

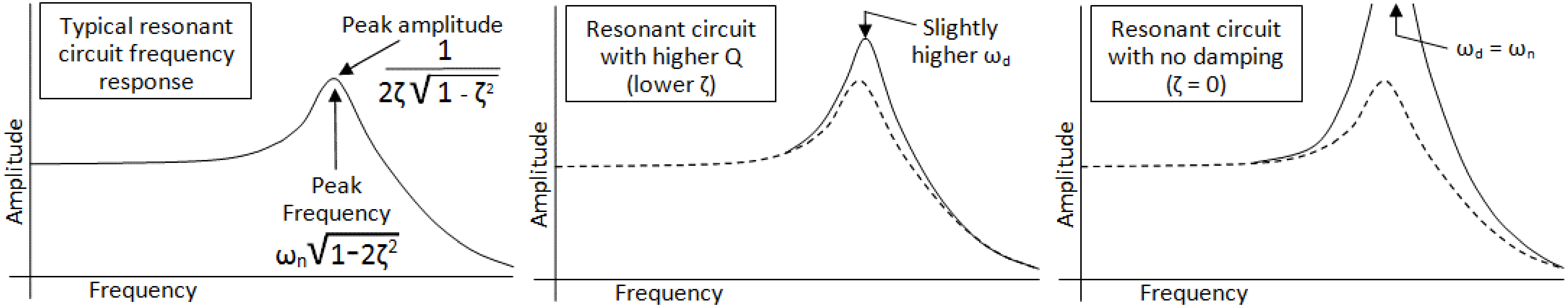
$$\hat{Q}(f)$$

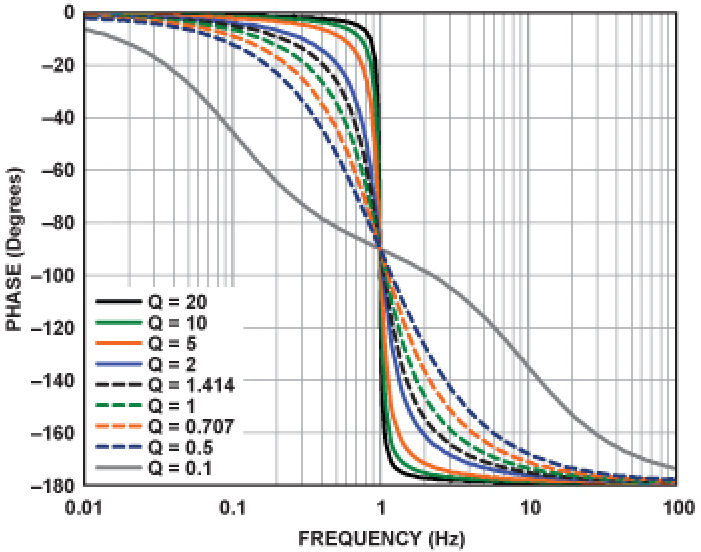
$$G(f) = \frac{\hat{Q}(f)/\bar{Q}}{\hat{u}(f)/\bar{u}}$$

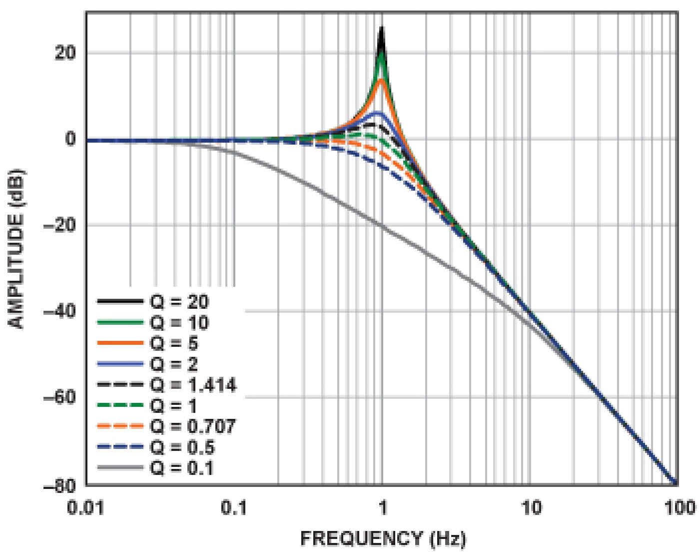
$$G(f) = Ae^{i\theta}$$

Gain Phase

We typically plot the transfer function using a Bode Diagram, which shows the frequency dependence of the gain and phase (thank you Stack Exchange)







https://electronics.stackexchange.com/questions/112521/resonant-frequency-from-bode-plot

Flame transfer functions are a useful construct for understanding thermoacoustic response as feedback occurs at specific resonant frequencies

Response of a flame at a given frequency:

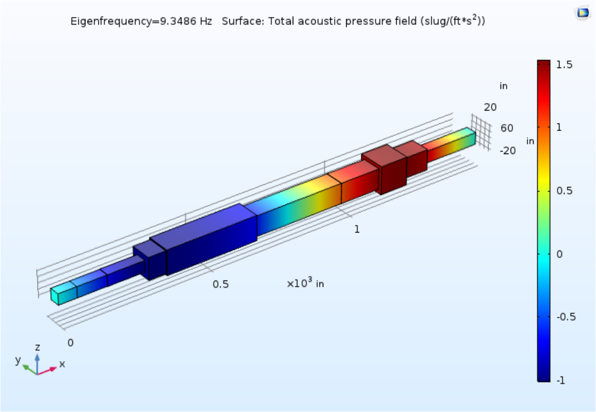
$$G(f) = \frac{\hat{Q}(f)/\bar{Q}}{\hat{u}(f)/\bar{u}}$$

If G(f) > 1 → amplification

If G(f) < 1 → not amplified

+

Knowledge of the acoustic modes of a system (pretty easy to calculate):



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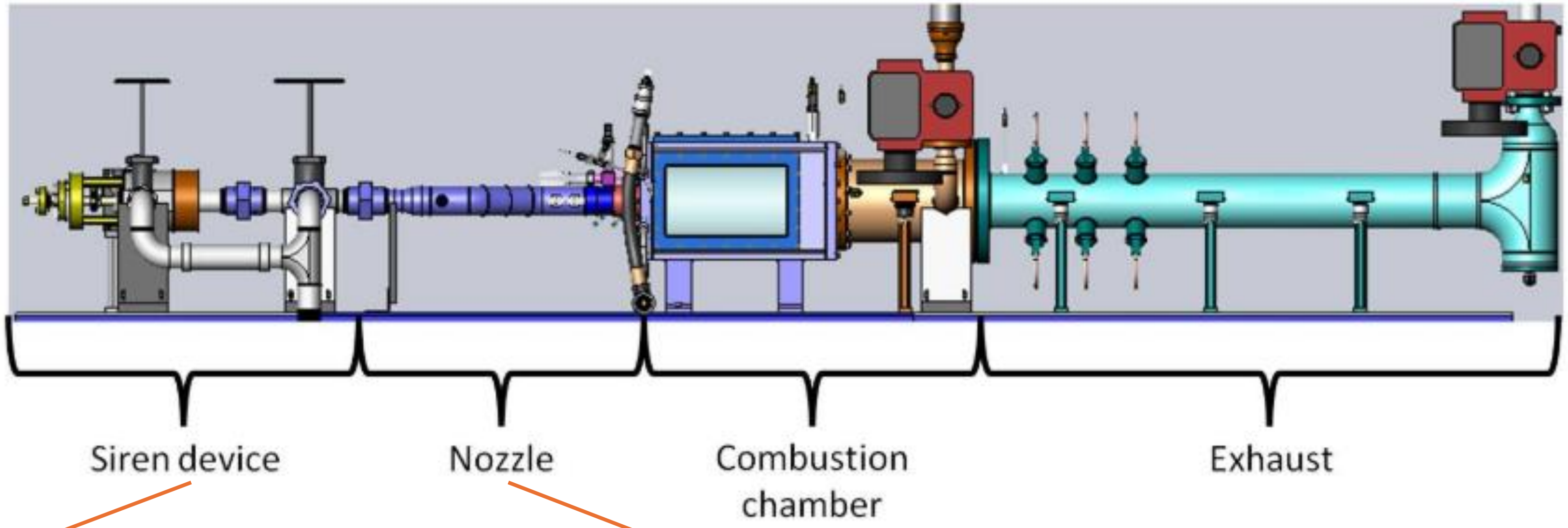
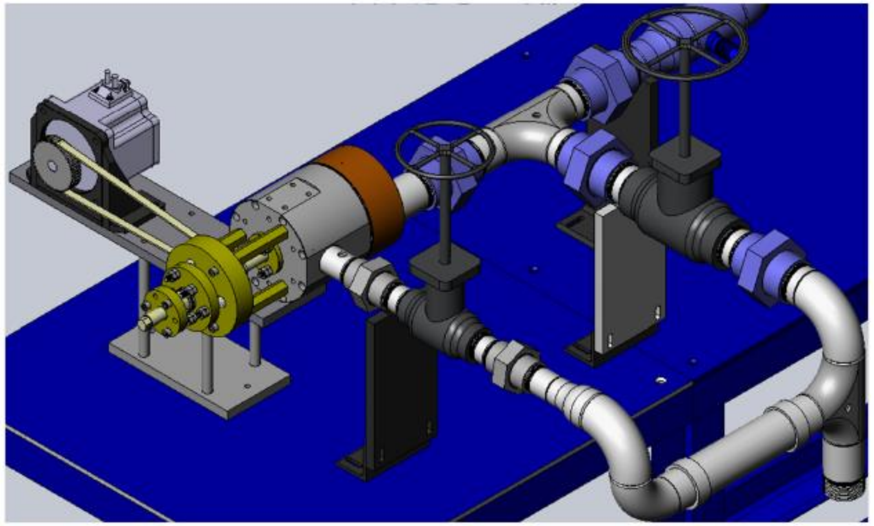
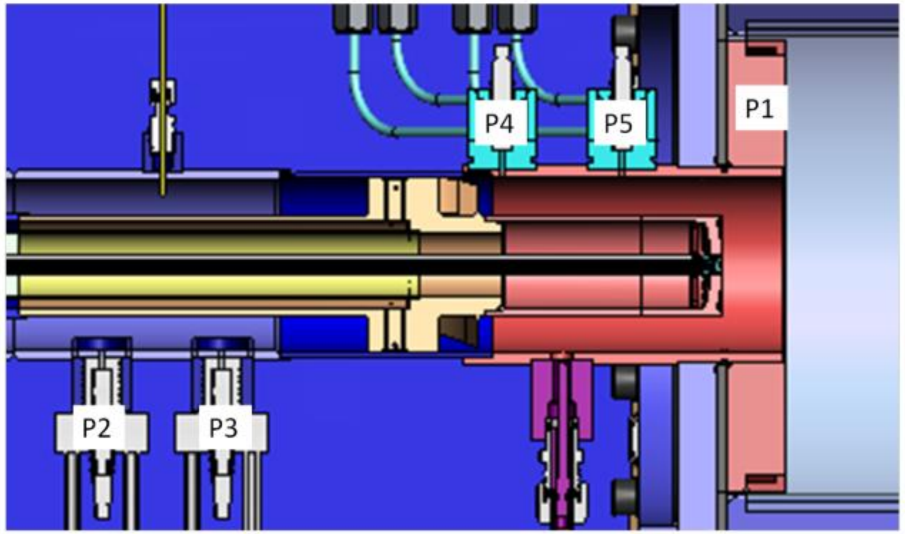
Information about what acoustic modes could couple with the flame

Flame transfer functions are typically measured using chemiluminescence as a marker of heat release rate

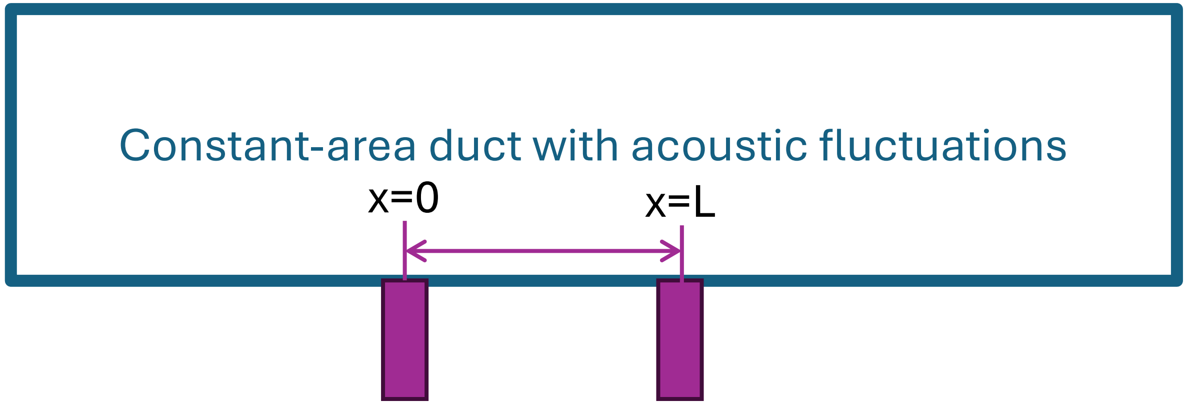






Bunce, N. "Flame transfer function measurements and mechanisms in a single-nozzle combustor," PhD thesis, Pennsylvania State University

Acoustic velocity fluctuation is calculated using a multi-microphone method, which uses conservation of momentum from pressure transducer measurements



$$P(x,t) = Re\{[P_+ e^{ikx} + P_- e^{-ikx}]e^{-i2\pi f t}\}$$

$$\hat{P}(x,f) = \hat{P}_+ e^{ikx} + \hat{P}_- e^{-ikx}$$

$P_1(t)$ $\quad$ $P_2(t)$

$\hat{P}_1(f)$ $\quad$ $\hat{P}_2(f)$

$$\hat{P}_1 = \hat{P}_+ + \hat{P}_-$$

$$\hat{P}_2 = \hat{P}_+ e^{ikL} + \hat{P}_- e^{-ikL}$$

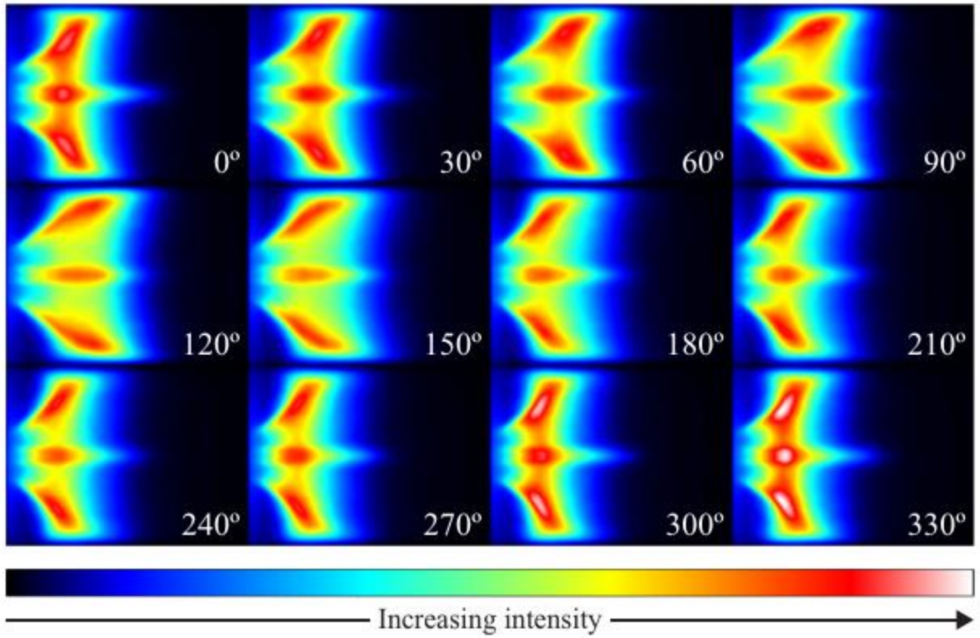
$$\begin{bmatrix} \hat{P}_1 \\ \hat{P}_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ e^{ikL} & e^{-ikL} \end{bmatrix} \boxed{\begin{bmatrix} \hat{P}_+ \\ \hat{P}_- \end{bmatrix}}$$

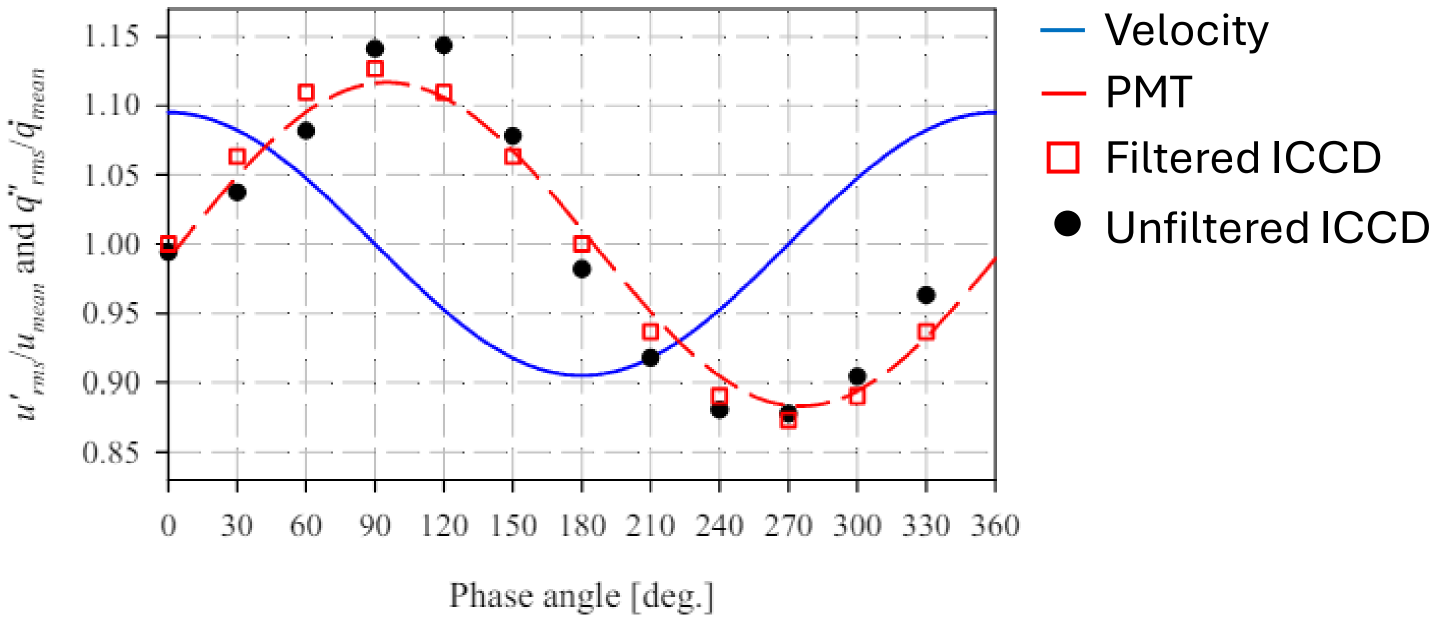
Math this

Momentum conservation says:

$$\hat{u} = \frac{1}{i\omega\rho}\frac{d\hat{P}}{dx} = \frac{1}{\rho c}[\hat{P}_+ e^{ikx} - \hat{P}_- e^{-ikx}]$$

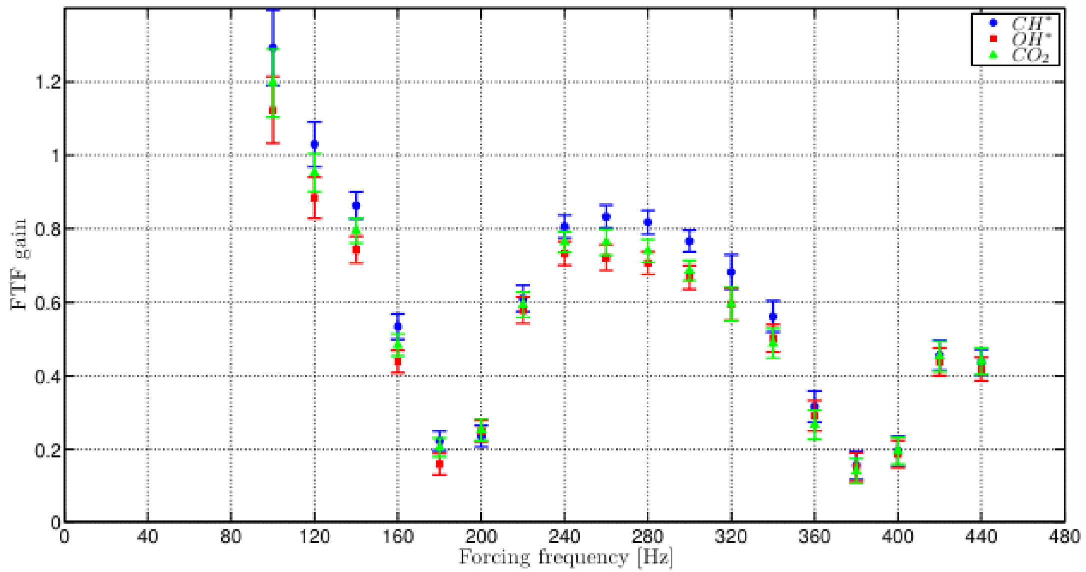
The measurement is done at a stable condition and a linear perturbation (~5%) is provided at a range of frequencies, measuring pressure and chemiluminescence

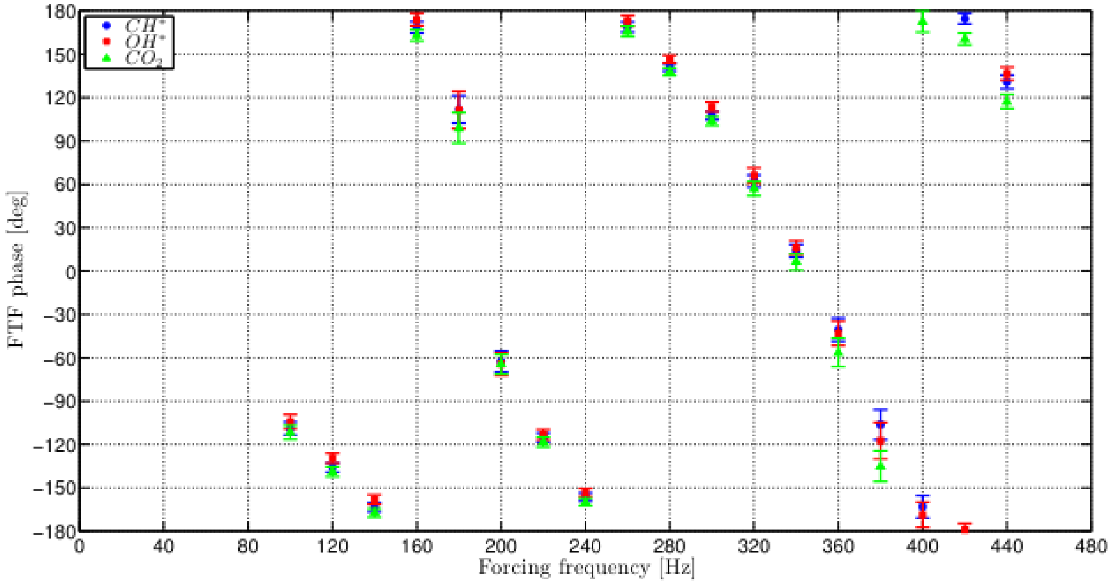


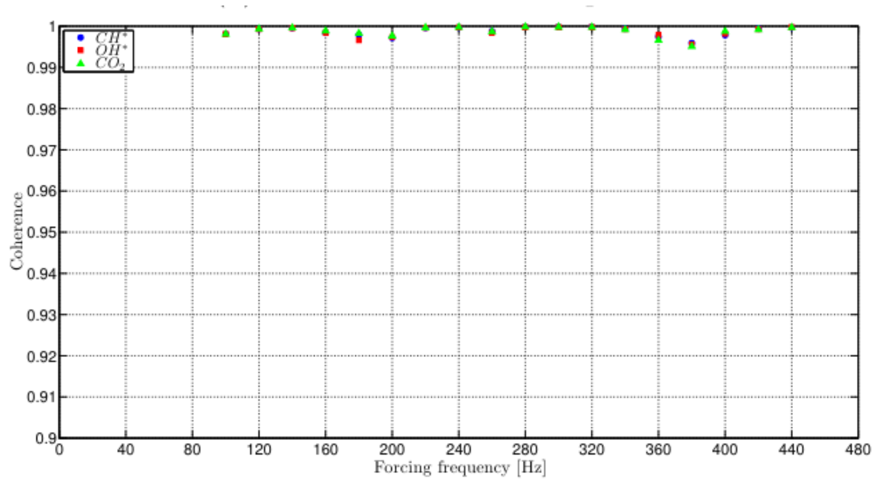


Peluso, S. "An Experimental Study of Flame Response Mechanisms in a Lean-premixed Gas Turbine Combustor," PhD Thesis, Pennsylvania State University

Resulting flame transfer function shows response of the flame to coherent oscillations at a range of frequencies





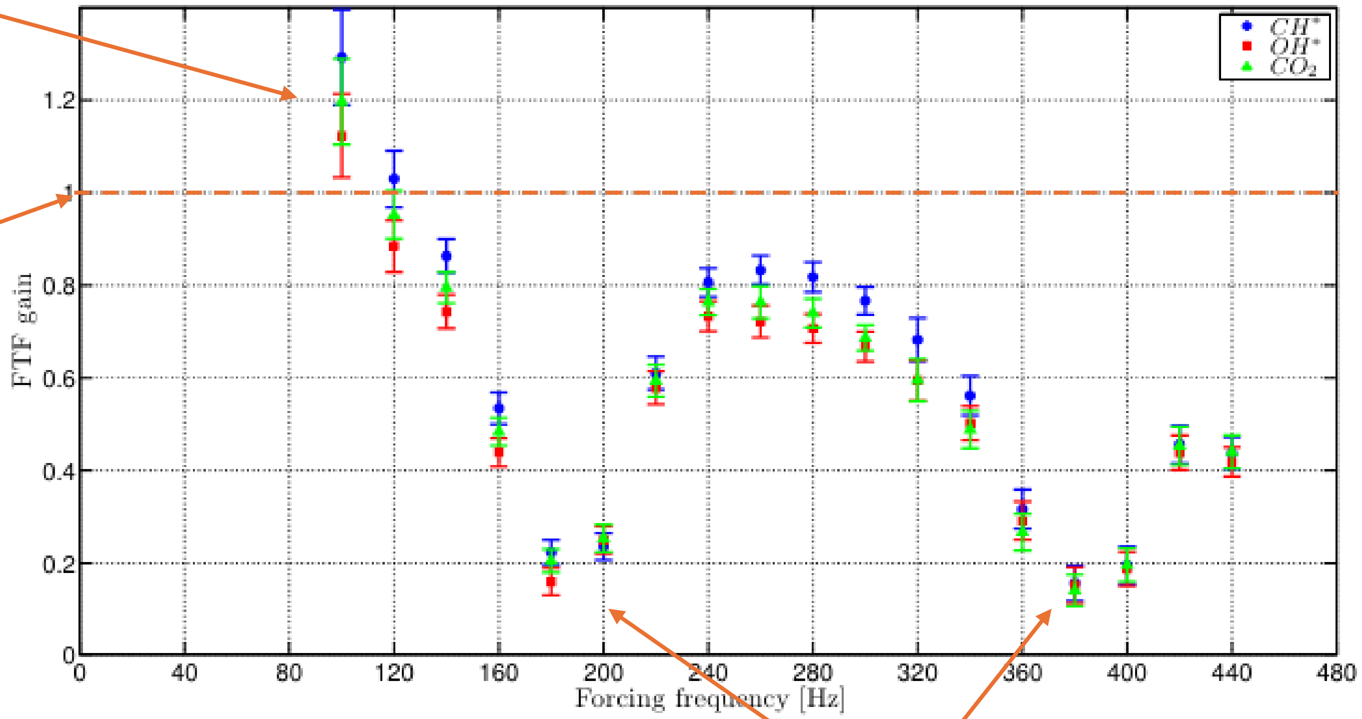


Bunce, N. "Flame transfer function measurements and mechanisms in a single-nozzle combustor," PhD thesis, Pennsylvania State University

A few interesting features:

Flame acts as amplifier

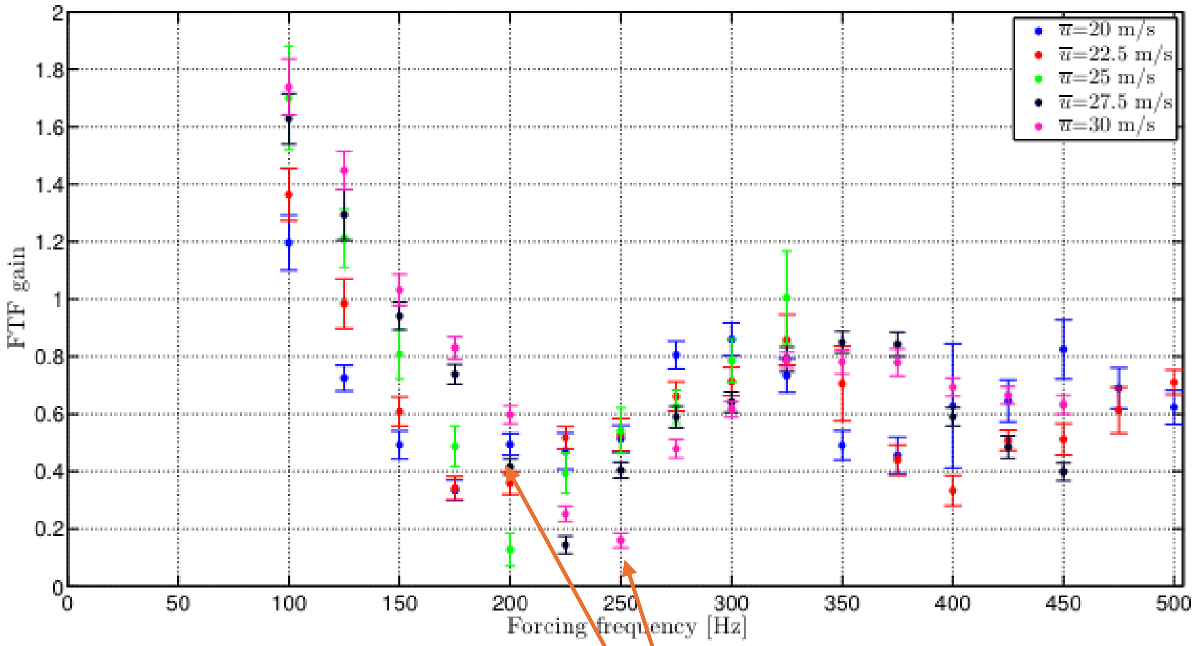
At zero frequency,
FTF gain will equal 0

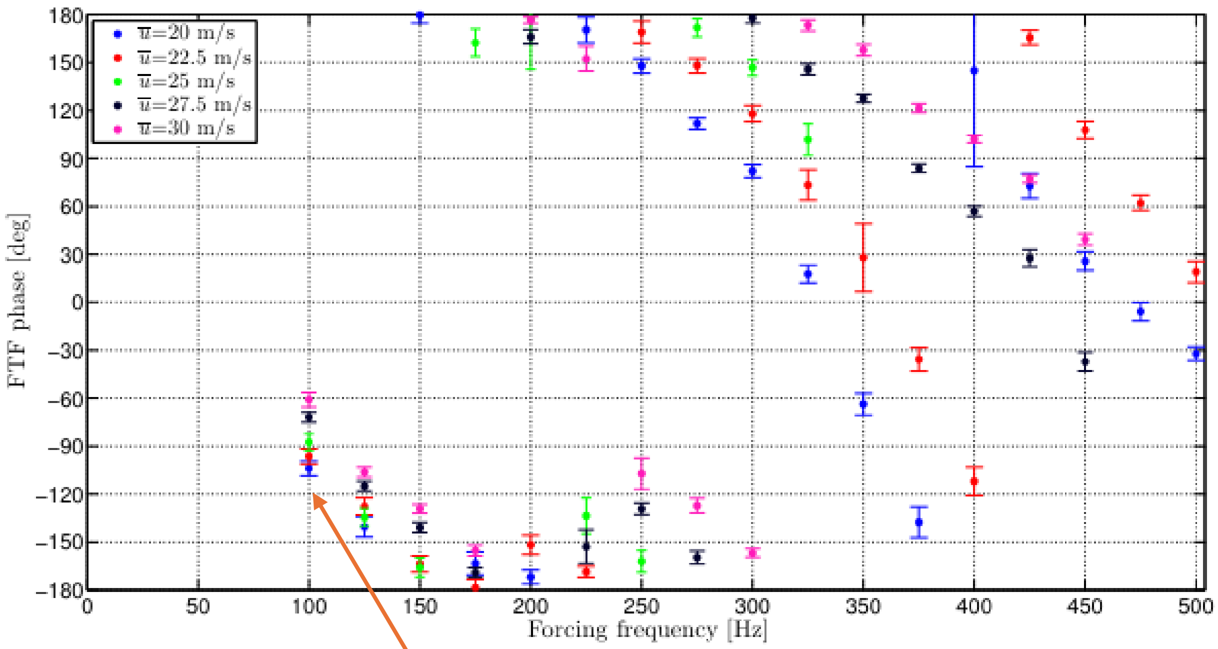


Presence of nodes suggests
cancellation process

Bunce, N. “Flame transfer function measurements and mechanisms in a single-nozzle combustor,” PhD thesis, Pennsylvania State University

Impact of variation in velocity:



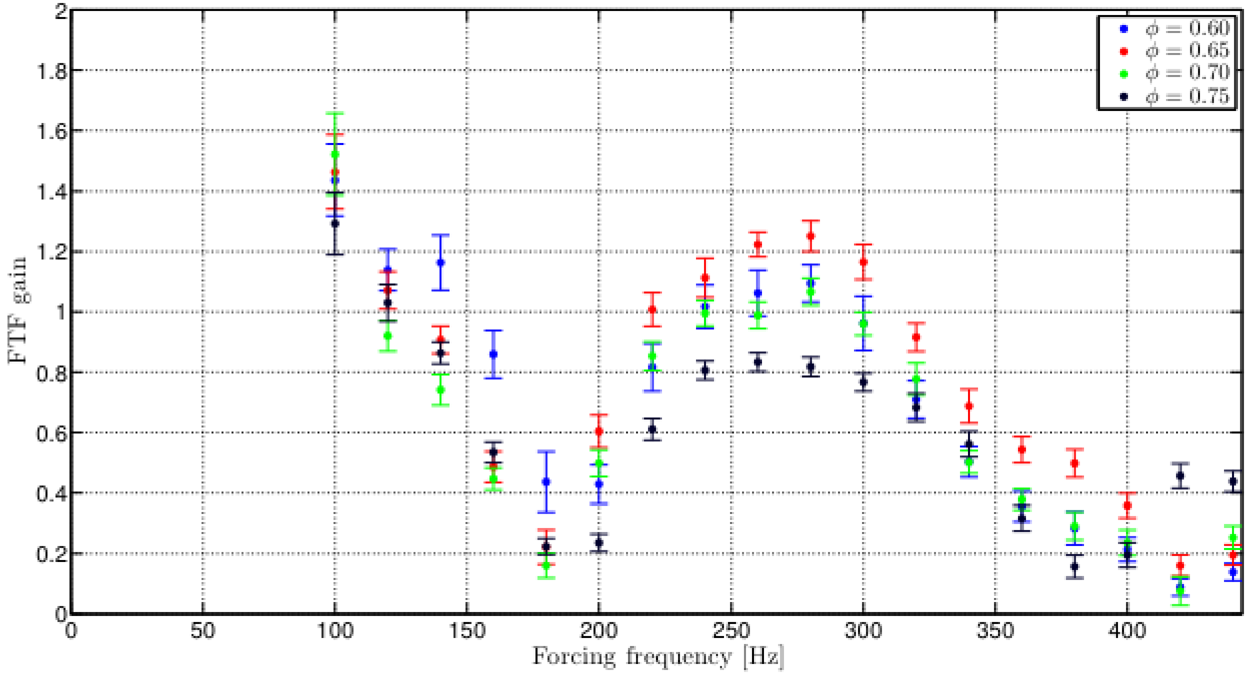


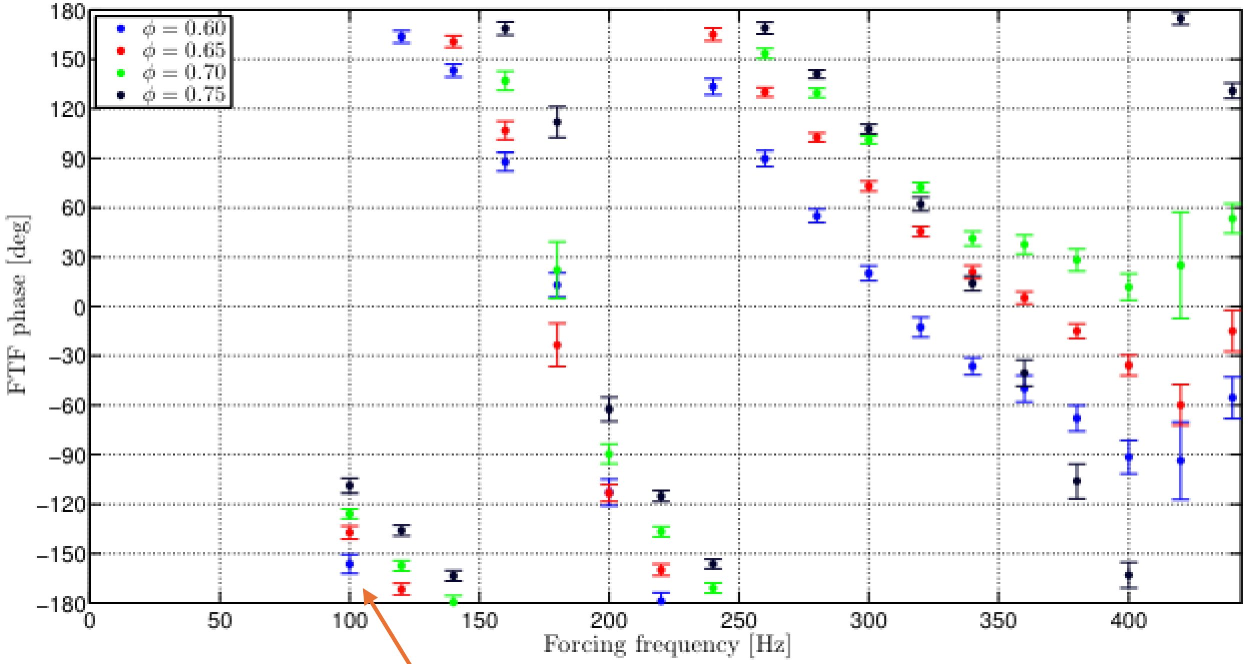
Nodes move to higher frequency as the velocity increases – does this make sense?

Phase difference decreases with increasing velocity – does this make sense?

Bunce, N. "Flame transfer function measurements and mechanisms in a single-nozzle combustor," PhD thesis, Pennsylvania State University

Impact of equivalence ratio:

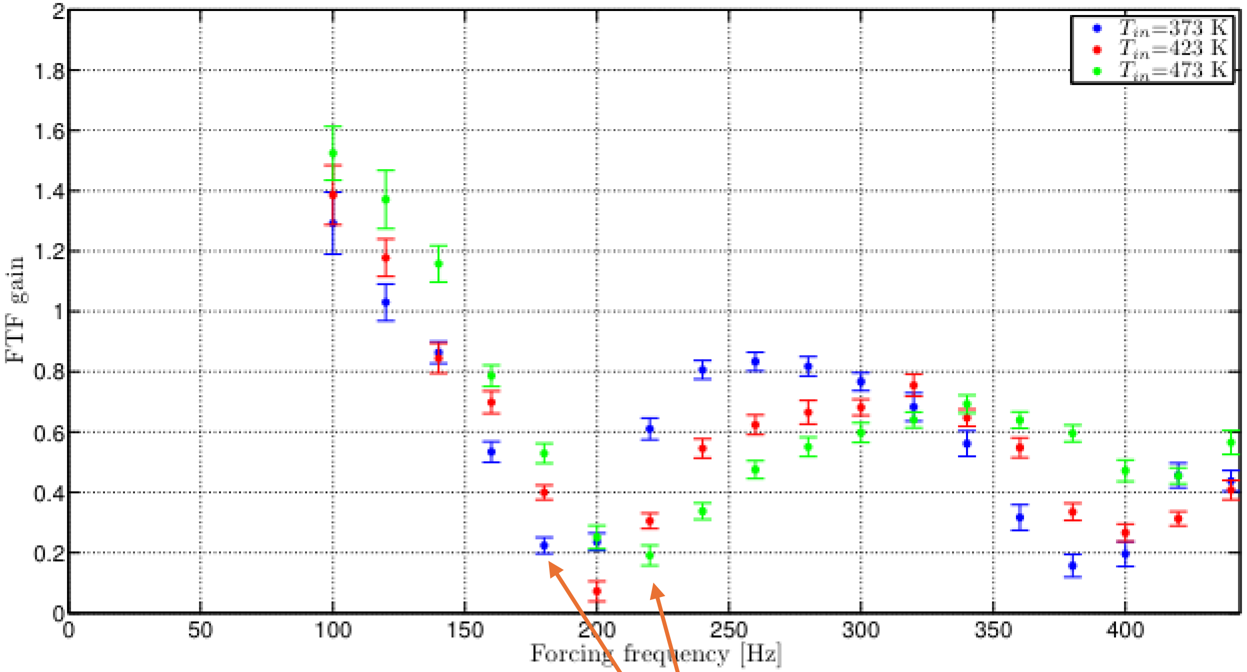


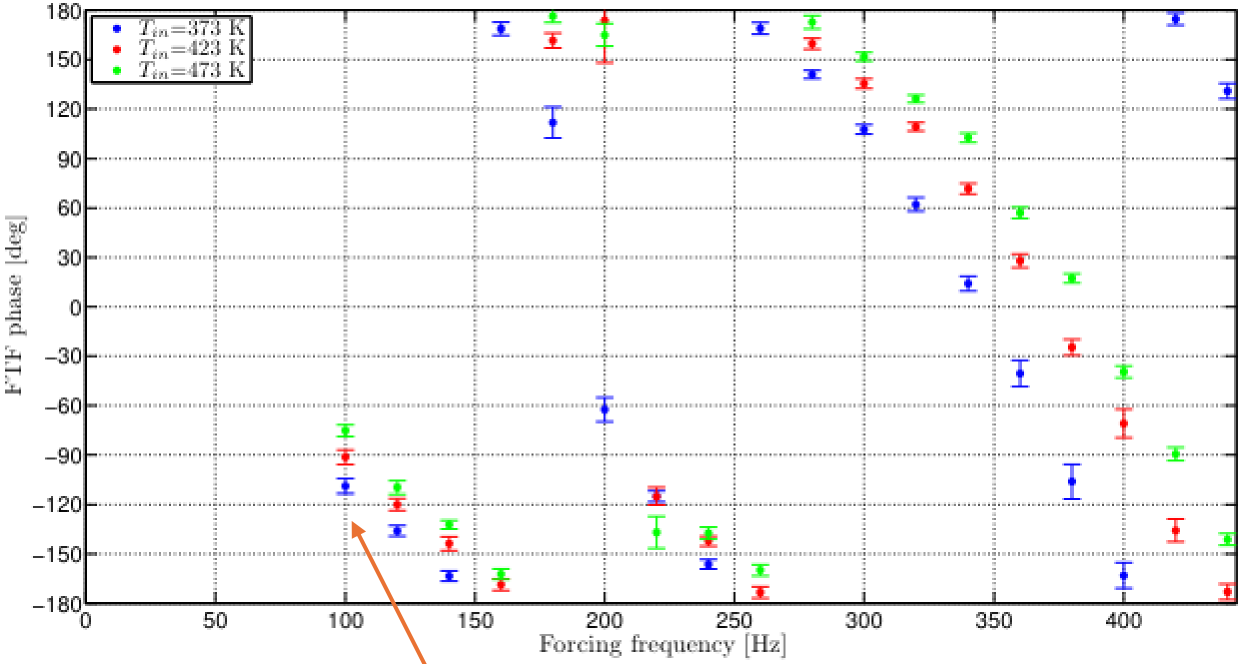


Phase difference decreases with increasing equivalence ratio – does this make sense?

Bunce, N. "Flame transfer function measurements and mechanisms in a single-nozzle combustor," PhD thesis, Pennsylvania State University

Impact of preheat temperature:



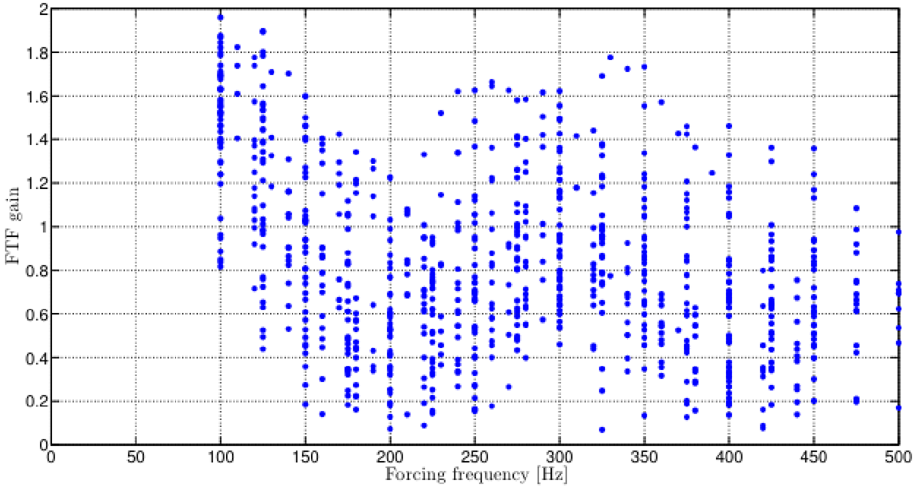


Nodes move to higher frequency as the temperature increases – does this make sense?

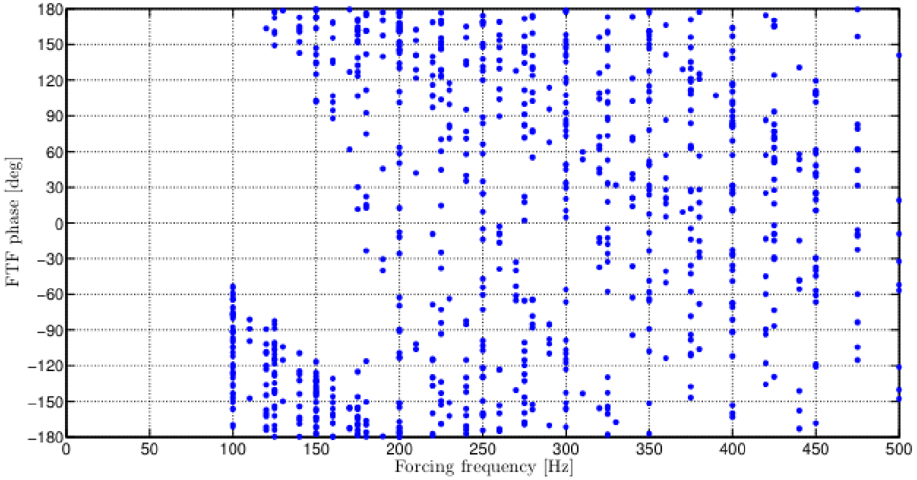
Phase difference decreases with increasing temperature – does this make sense?

Bunce, N. "Flame transfer function measurements and mechanisms in a single-nozzle combustor," PhD thesis, Pennsylvania State University

Non-dimensionalizing the frequency into Strouhal number tells you whether the mechanisms responsible at these different conditions are the same

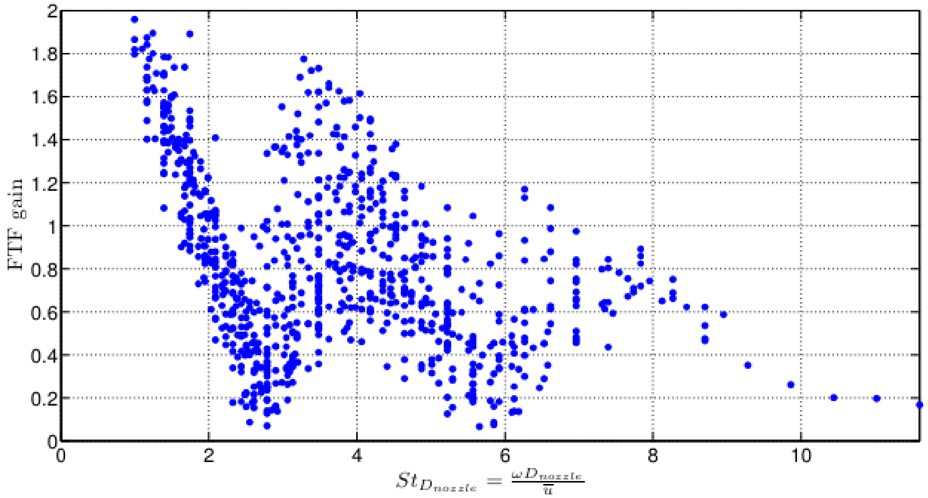


(a) Flame transfer function gain.

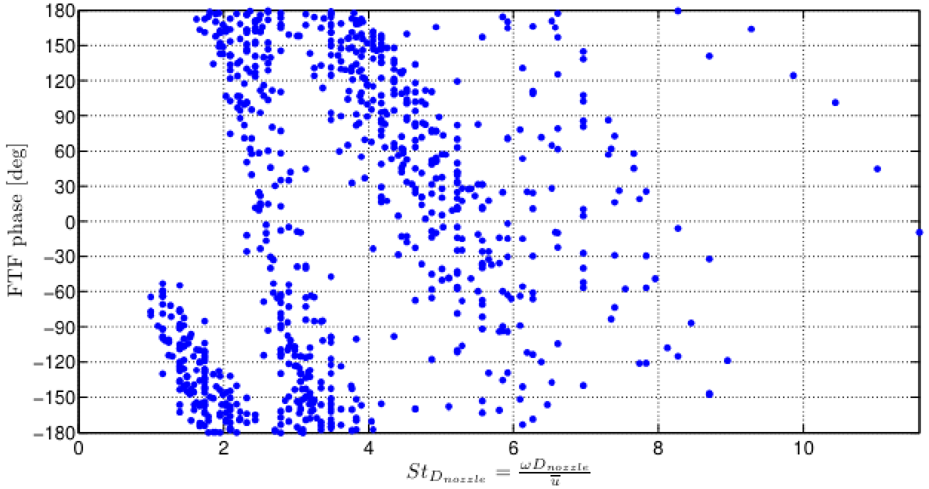


(b) Flame transfer function phase.

$$St = \frac{f L_f}{\bar{u}}$$



(a) Flame transfer function gain.



(b) Flame transfer function phase.

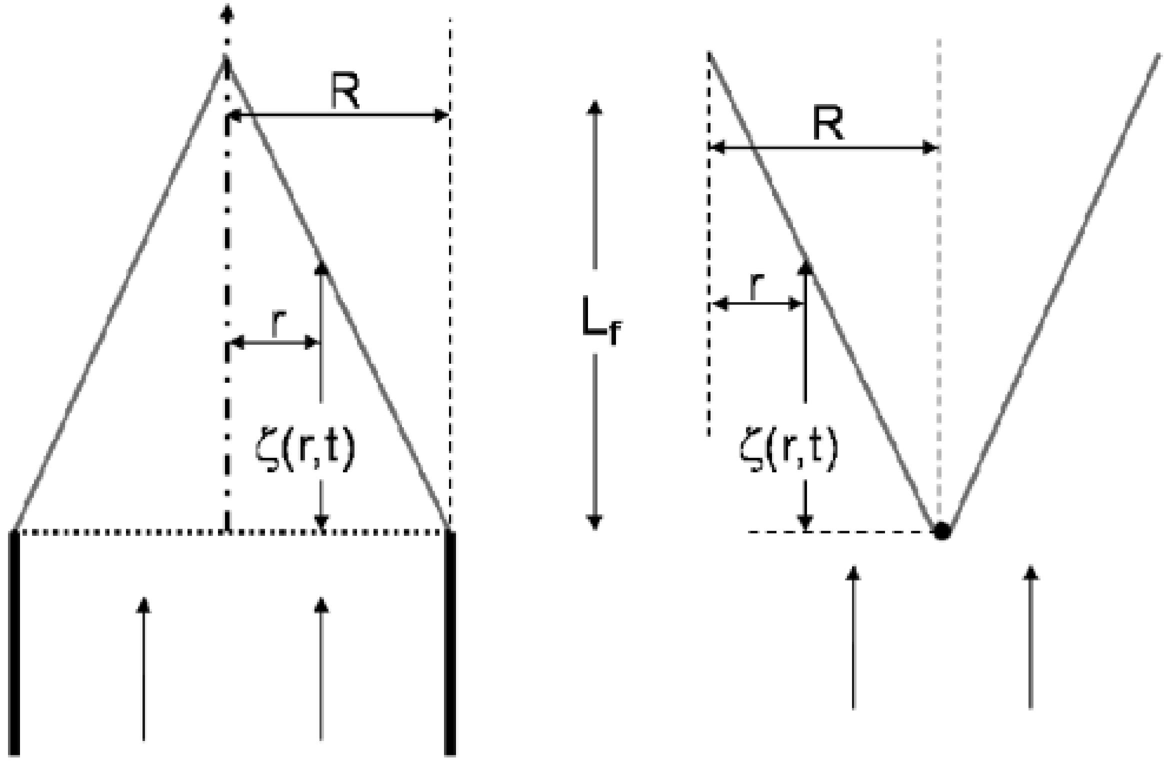
Bunce, N. “Flame transfer function measurements and mechanisms in a single-nozzle combustor,” PhD thesis, Pennsylvania State University

So what's up with this cancellation phenomenon?

Back to theory!

The level-set formulation can provide direct insight into the cancellation process – to the board!

One of the best starter references on this type of work:
Preetham, Santosh, H., & Lieuwen, T. (2008). Dynamics of laminar premixed flames forced by harmonic velocity disturbances. *Journal of Propulsion and Power*, *24*(6), 1390-1402.



Two different disturbance sources have varying amplitudes and phases as a function of frequency

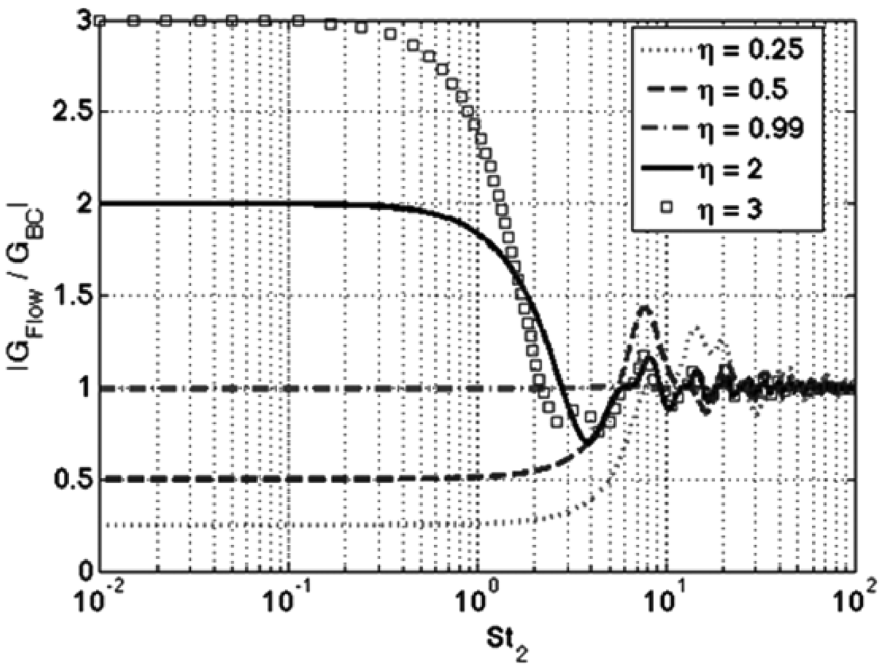


Fig. 6 Strouhal number dependence of the magnitude of the ratio of the transfer functions due to the flow-nonuniformity and boundary-condition terms for different values of η.

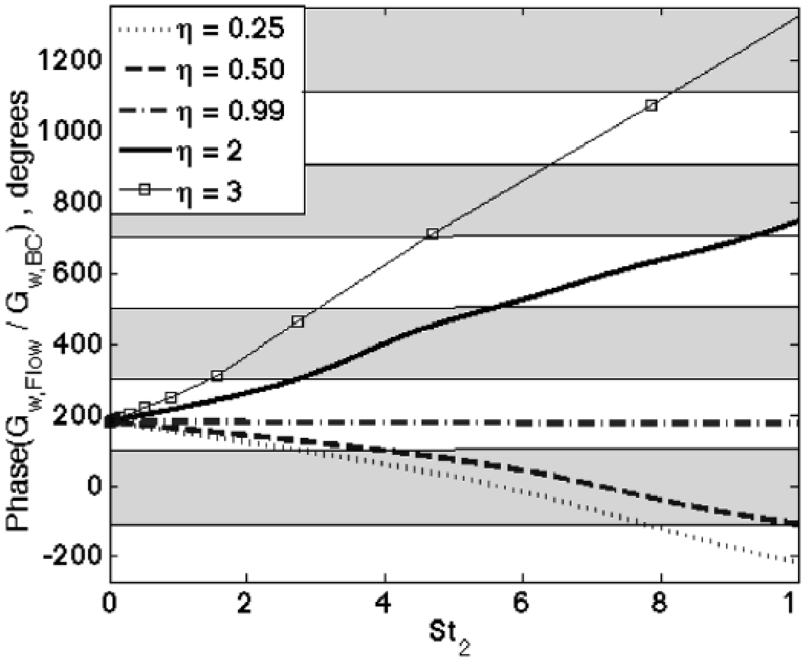


Fig. 7 Strouhal number dependence of the phase of the ratio of the transfer functions due to the flow-forcing and boundary-condition terms for wedge flames. Shaded regions indicate points at which boundary-condition and flow-nonuniformity terms are in phase.

Preetham, Santosh, H., & Lieuwen, T. (2008). Dynamics of laminar premixed flames forced by harmonic velocity disturbances. *Journal of Propulsion and Power*, *24*(6), 1390-1402.

Superimposed, these disturbances result in nodes and anti-nodes in the total flame area fluctuation transfer function

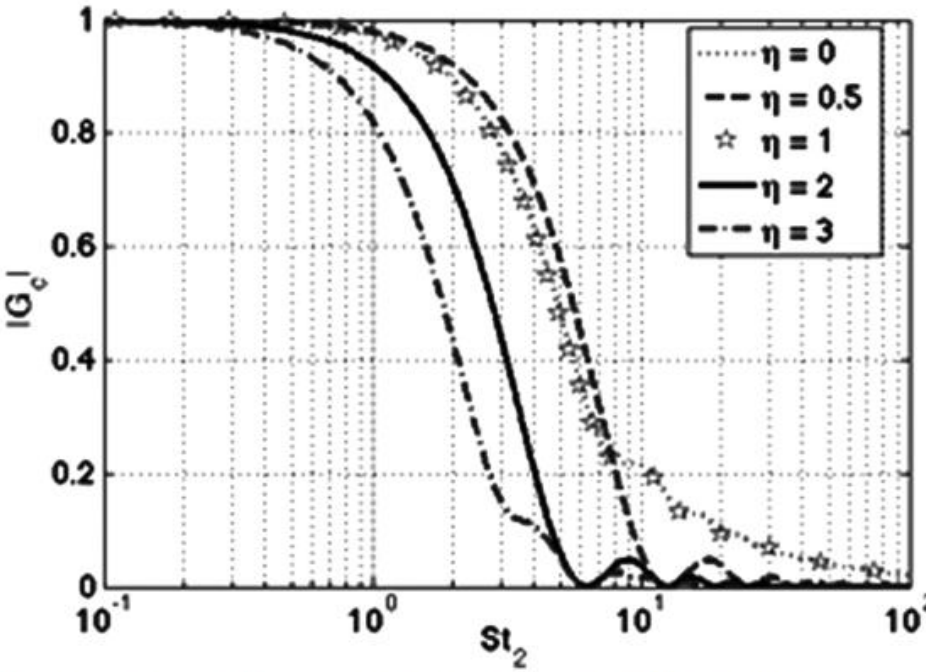


Fig. 8 Axisymmetric conical linear transfer-function $G_c(St_2, \eta)$ magnitude dependence upon the reduced Strouhal number St_2 for different values of η.

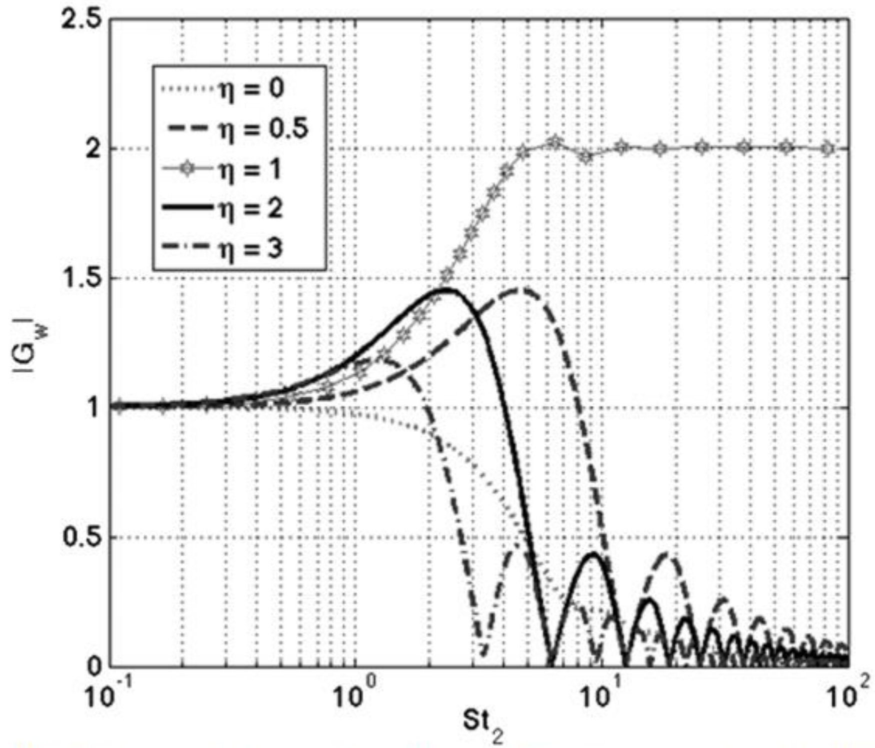


Fig. 10 Axisymmetric wedge linear transfer-function $G_w(St_2, \eta)$ amplitude dependence upon the reduced Strouhal number St_2 for different values of η.

Preetham, Santosh, H., & Lieuwen, T. (2008). Dynamics of laminar premixed flames forced by harmonic velocity disturbances. *Journal of Propulsion and Power*, *24*(6), 1390-1402.

Theory also shows the impact that the nonlinear kinematic restoration effects can have as a function of disturbance amplitude on FTF gain

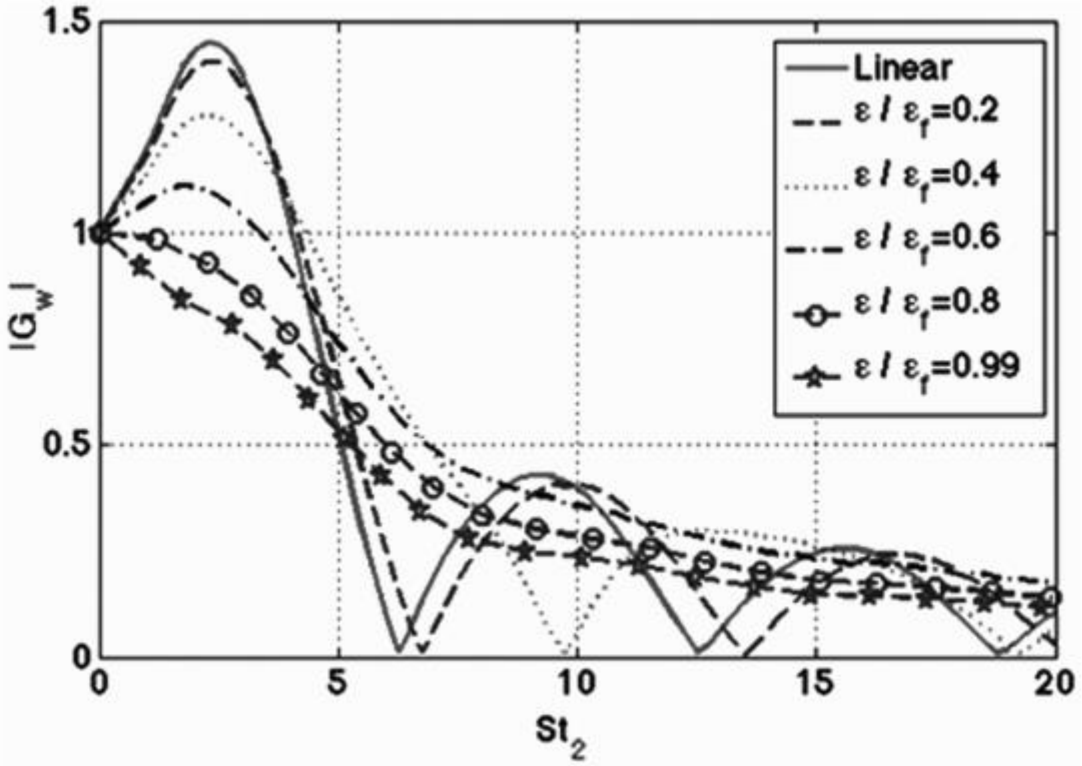


Fig. 19 Strouhal number dependence of the magnitude of the flame-area velocity transfer function for the axisymmetric wedge flame; $\beta = 2$ and $\eta = 2$.

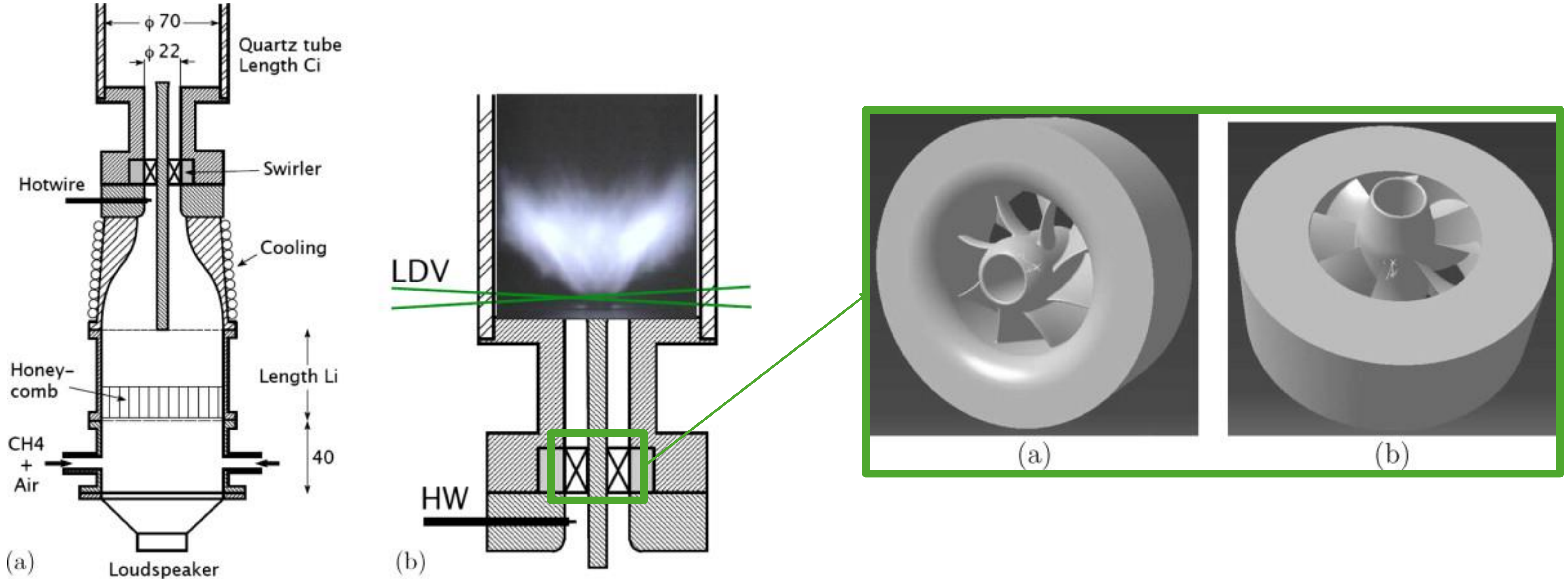
Preetham, Santosh, H., & Lieuwen, T. (2008). Dynamics of laminar premixed flames forced by harmonic velocity disturbances. *Journal of Propulsion and Power*, *24*(6), 1390-1402.

Key Point:

The nodes and anti-nodes of your flame transfer functions tell you something about interference of multiple disturbances.

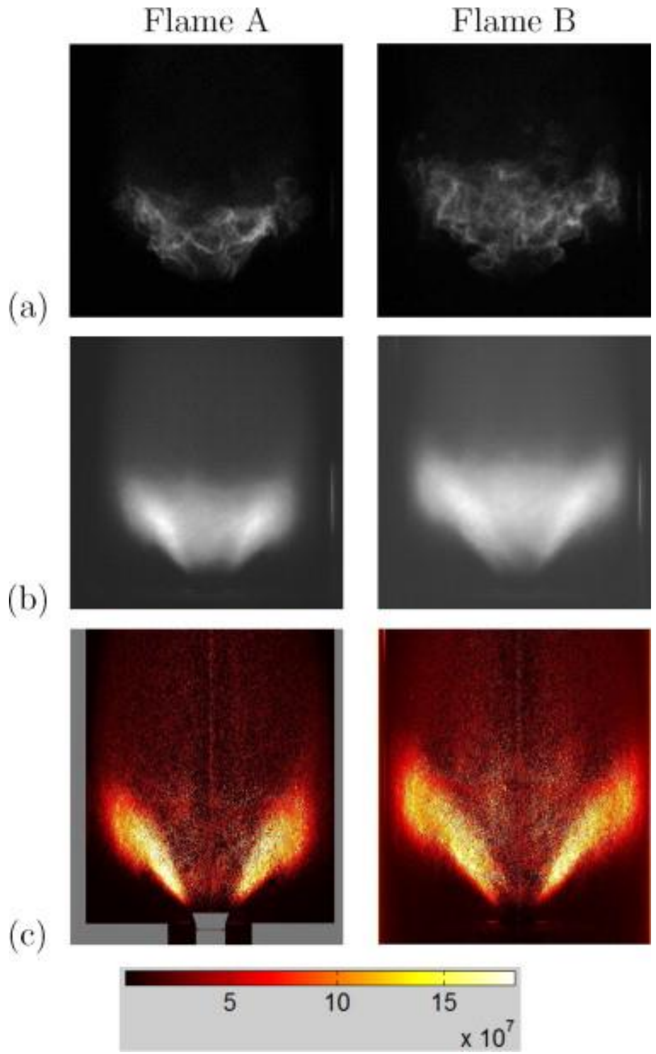
What does this look like in experiments?

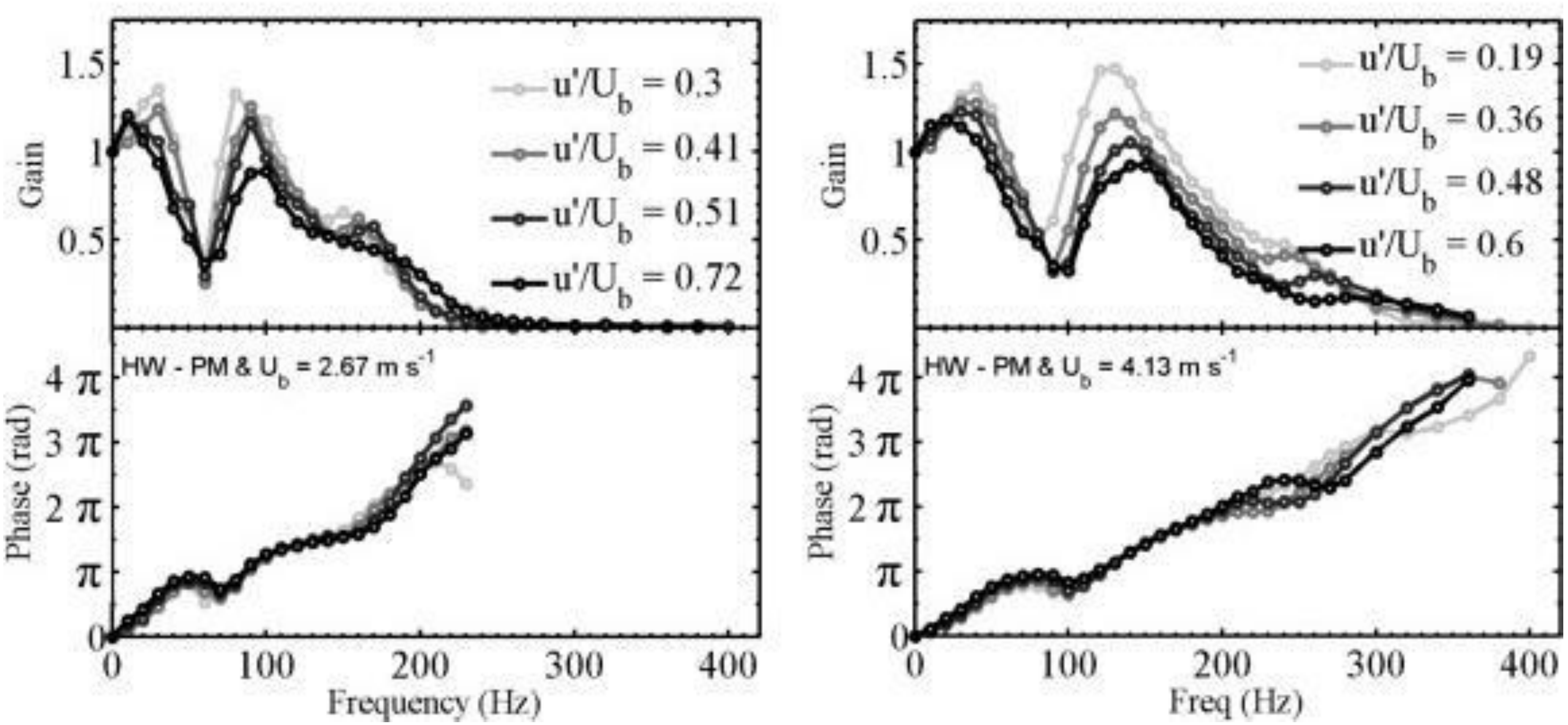
Example 1: Swirl-stabilized flame from Palies et al.



Palies, P., Durox, D., Schuller, T., & Candel, S. (2010). The combined dynamics of swirler and turbulent premixed swirling flames. Combustion and flame, 157(9), 1698-1717.

They measured flame describing functions for two different flames and saw significant constructive/destructive interference in the FDF





Palies, P., Durox, D., Schuller, T., & Candel, S. (2010). The combined dynamics of swirler and turbulent premixed swirling flames. Combustion and flame, 157(9), 1698-1717.

SIDEBAR

Flame describing functions

A flame describing function is a function of both the frequency and amplitude of the input disturbance and so captures nonlinear behavior

Input velocity fluctuation → $G(f,\varepsilon)$ → Output heat release rate fluctuation

$$u(t) = \bar{u} + u'(\varepsilon)\cos(\omega t)$$
$$u(t) = \bar{u} + u'(\varepsilon)\cos(2\pi f t)$$

$$Q(t) = \bar{Q} + Q'(\varepsilon)\cos(\omega t)$$
$$Q(t) = \bar{Q} + Q'(\varepsilon)\cos(2\pi f t)$$

Fourier Transform

$$\hat{u}(f,\varepsilon)$$

Fourier Transform

$$\hat{Q}(f,\varepsilon)$$

$$G(f,\varepsilon) = \frac{\hat{Q}(f,\varepsilon)/\bar{Q}}{\hat{u}(f,\varepsilon)/\bar{u}}$$

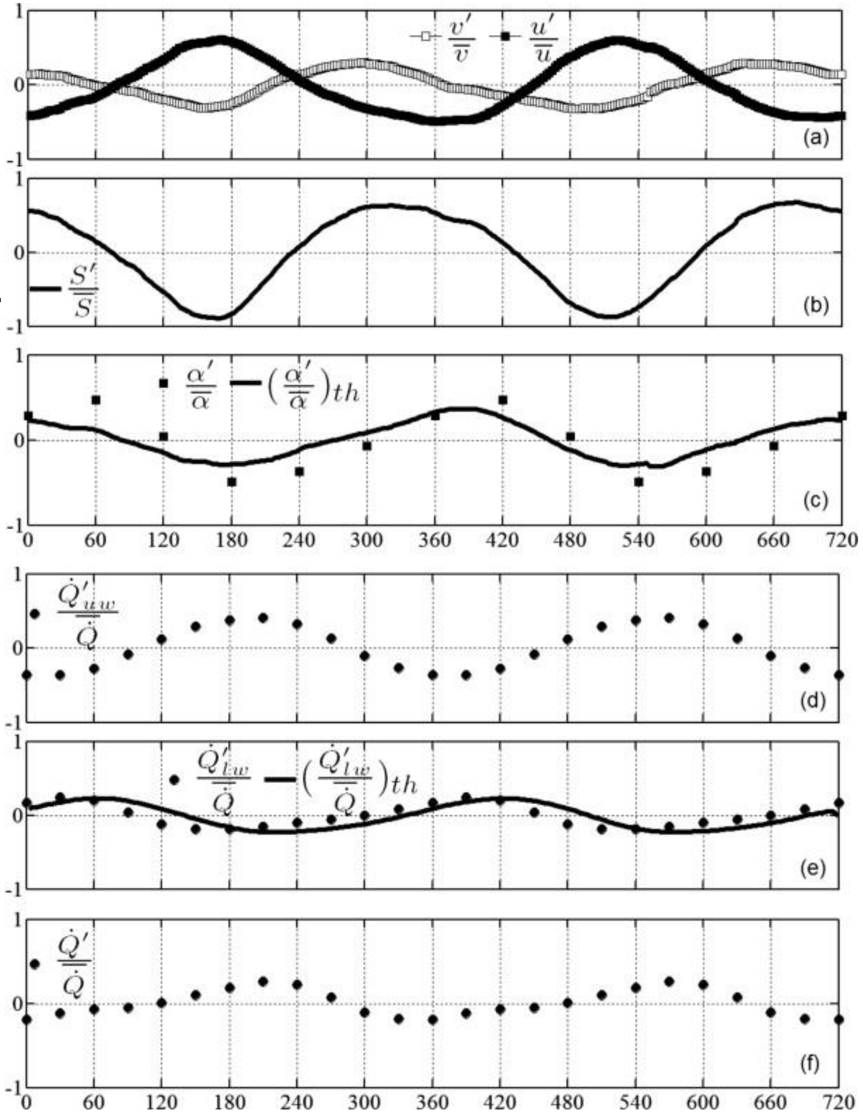
SIDEBAR OVER

Back to the example

Careful measurement of the velocity field and theory from Cumpsty showed that there are fluctuations in axial velocity and swirling velocity

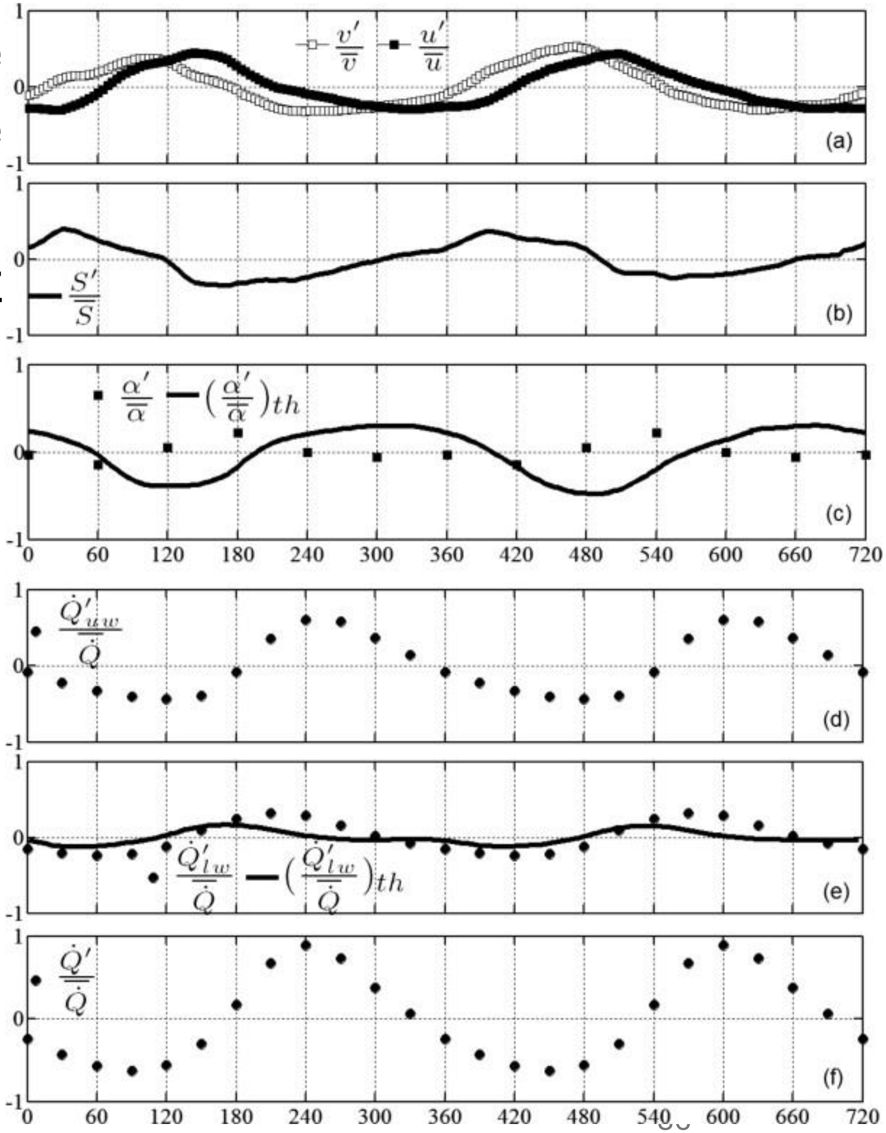
Destructive interference

60 Hz



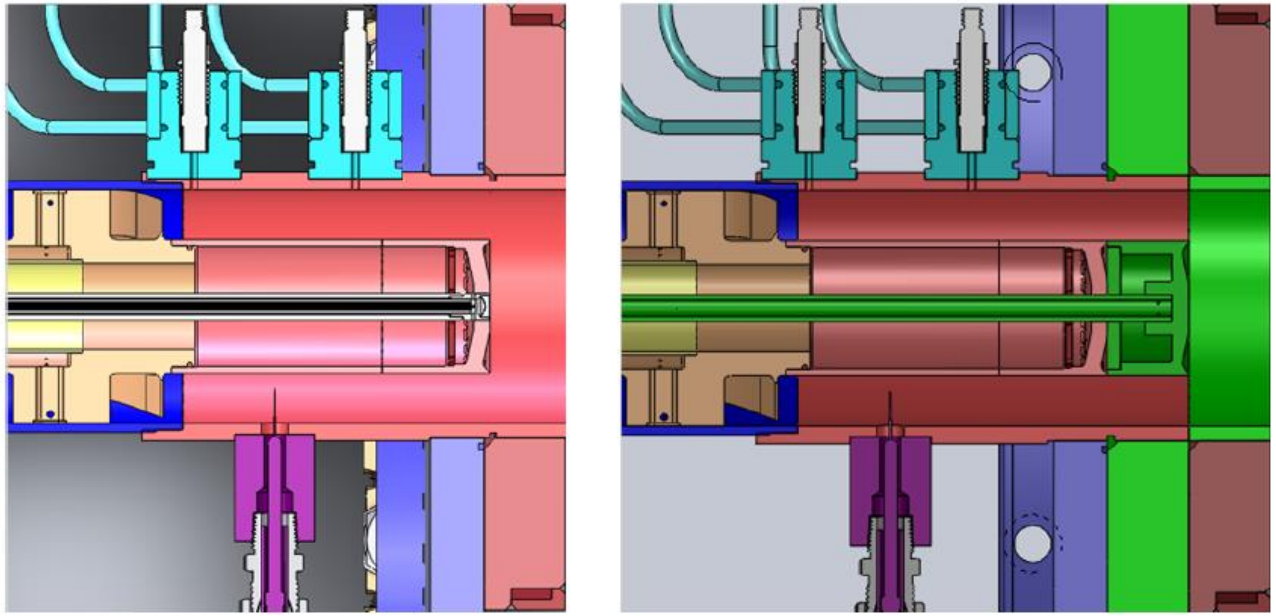
Constructive interference

90 Hz

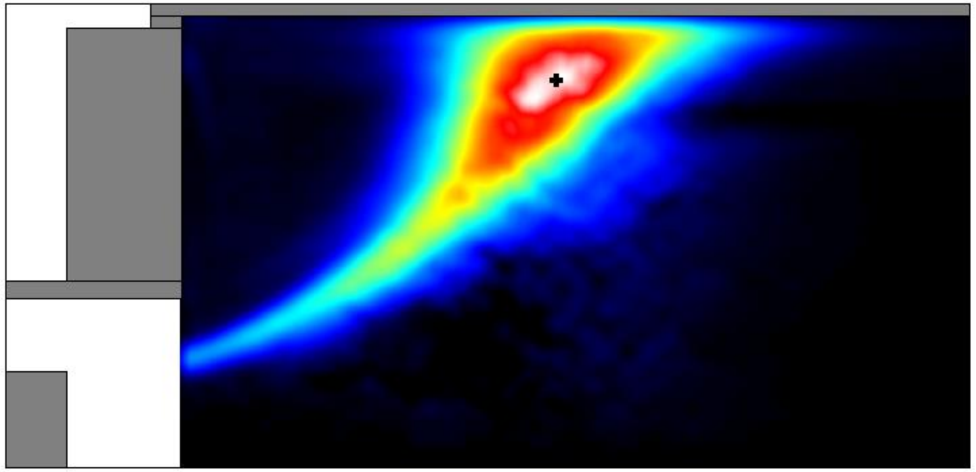


Palies, P., Durox, D., Schuller, T., & Candel, S. (2010). The combined dynamics of swirler and turbulent premixed swirling flames. Combustion and flame, 157(9), 1698-1717.

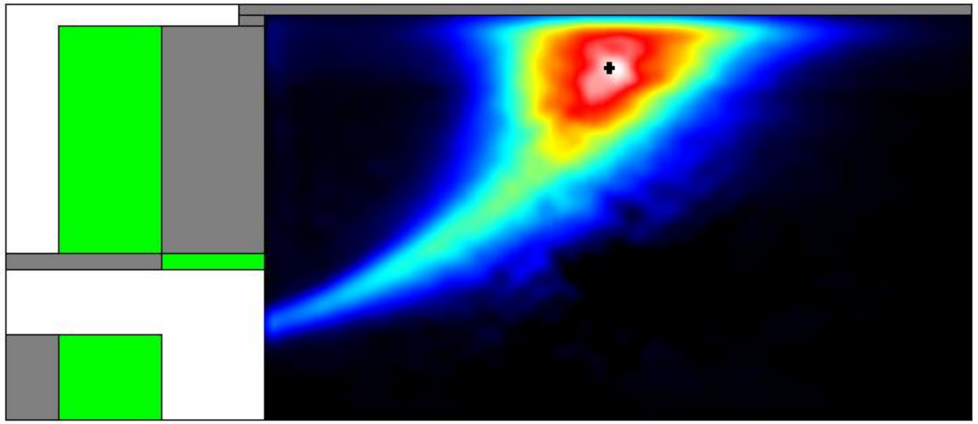
Example 2: Impact of time delay on disturbance interference in a swirl-stabilized flame (Bunce and Santavicca)



(a) Original geometry. (b) Modified geometry.



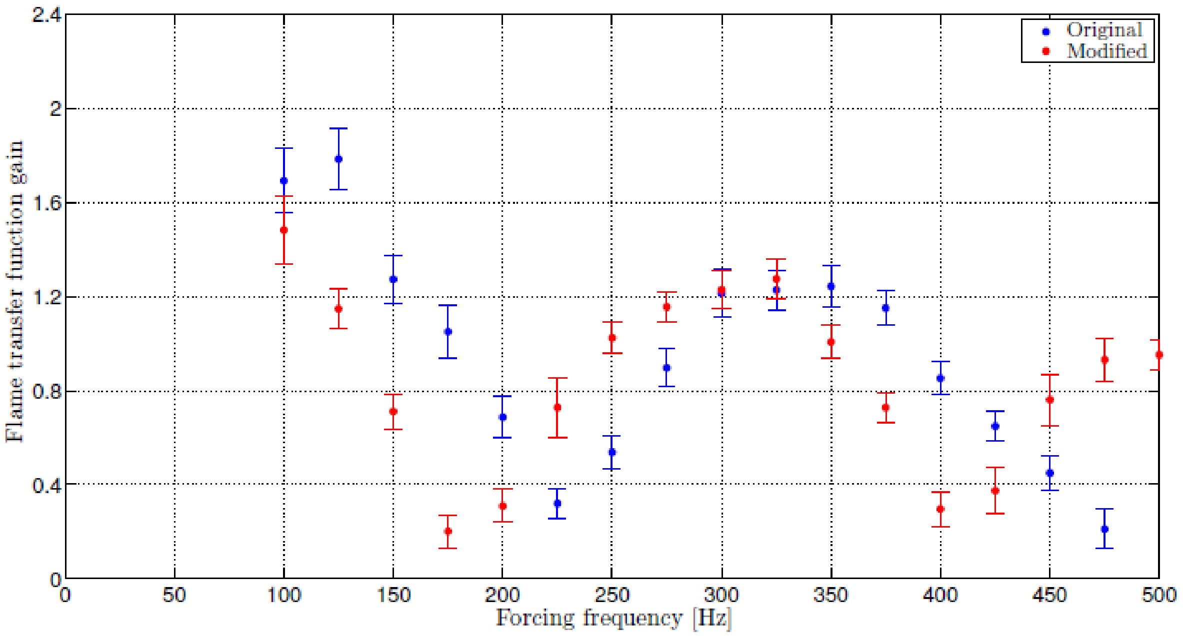
(a) Original geometry.

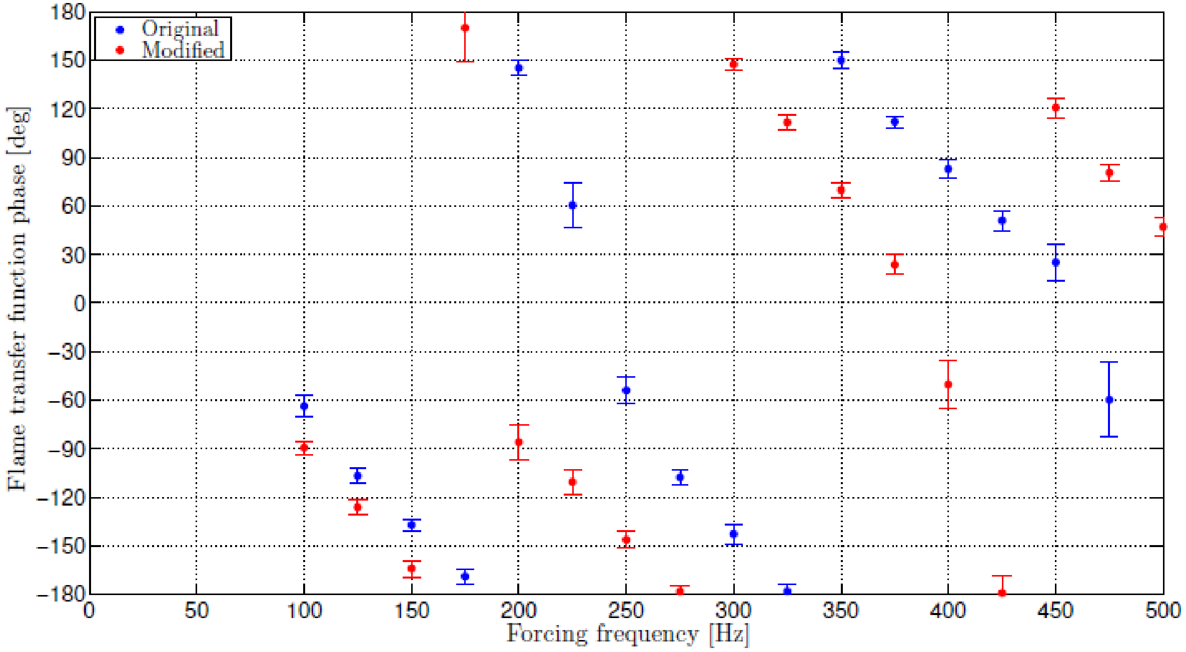


(b) Modified geometry.

Bunce, N. “Flame transfer function measurements and mechanisms in a single-nozzle combustor,” PhD thesis, Pennsylvania State University

Flame location modification led to changes in the location of the nodes and anti-nodes in the flame transfer function

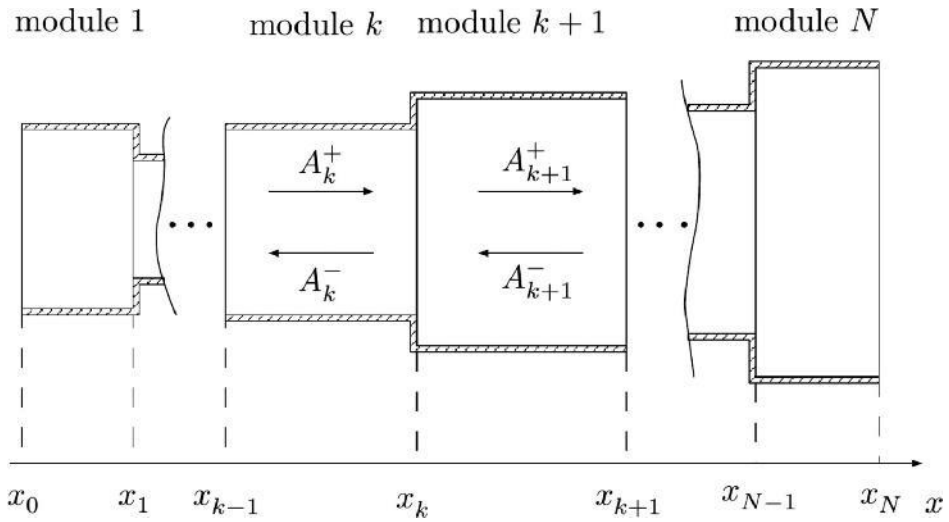




Bunce, N. "Flame transfer function measurements and mechanisms in a single-nozzle combustor," PhD thesis, Pennsylvania State University

So what do I do with this FTF?

Network models are a common way that flame transfer functions can be used to predict instability in complex combustion systems



Very cool, open-source tool:
https://www.oscilos.com/

$$p_k(x,t) = \bar{p}_k + p'_k(x,t) = \bar{p}_k + A_k^+(t-\tau_k^+) + A_k^-(t-\tau_k^-)$$

$$u_k(x,t) = \bar{u}_k + u'_k(x,t) = \bar{u}_k + \frac{1}{\bar{\rho}_k\bar{c}_k}\left[A_k^+(t-\tau_k^+) - A_k^-(t-\tau_k^-)\right]$$

$$\rho_k(x,t) = \bar{\rho}_k + \rho'_k(x,t) = \bar{\rho}_k + \frac{1}{\bar{c}_k^2}\left[A_k^+(t-\tau_k^+) + A_k^-(t-\tau_k^-)\right] - \frac{1}{\bar{c}_k^2}E_k(t-\tau_k^s)$$

Flame treated as an interface:

$$\rho_{k+1}(x_k,t)u_{k+1}(x_k,t) = \rho_{k+\frac{1}{2}}(x_k,t)u_{k+\frac{1}{2}}(x_k,t)$$

$$p_{k+1}(x_k,t) + \rho_{k+1}(x_k,t)u_{k+1}^2(x_k,t) = p_{k+\frac{1}{2}}(x_k,t) + \rho_{k+\frac{1}{2}}(x_k,t)u_{k+\frac{1}{2}}^2(x_k,t)$$

$$\rho_{k+1}(x_k,t)u_{k+1}(x_k,t)H_{k+1}(x_k,t) = \rho_{k+\frac{1}{2}}(x_k,t)u_{k+\frac{1}{2}}(x_k,t)H_{k+\frac{1}{2}}(x_k,t) + \dot{q}(t)$$

$$p_k(x_k,t) = \rho_k(x_k,t)R_{g,1}T_k(x_k,t), \quad p_{k+\frac{1}{2}}(x_k,t) = \rho_{k+\frac{1}{2}}(x_k,t)R_{g,2}T_{k+\frac{1}{2}}(x_k,t)$$

$$\bar{H} = \int_{\bar{T}_0}^{\bar{T}} C_p \mathrm{d}T + \frac{1}{2}\bar{u}_k^2$$

Course schedule

Monday:
— 1400-1500: Introduction and combustion instability in technologies
— 1500-1530: Thermoacoustic feedback
— 1530-1545: Break
— 1545-1730: Flame kinematics (with crafts!) and flame response

Tuesday:
— 1400-1530: Flame response and flame transfer functions
— 1530-1545: Break
— 1545-1630: Instability control
— 1630-1730: Advanced topics

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The Rayleigh Index can tell us a lot about the potential methods of instability control

$$RI = \int_V \int_T p'(\vec{x},t)\,\dot{q}'(\vec{x},t)\,dt\,dV$$

Increase the damping so that it suppresses the instability

De-phase the p’ and q’ over an acoustic cycle

Move the pressure anti-node relative to the flame

Change the heat release rate spatial distribution

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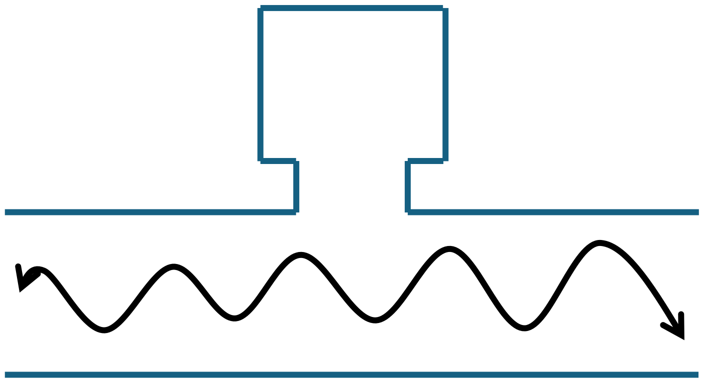
Increase damping: damping is achieved through both direct damping devices as well as combustor design

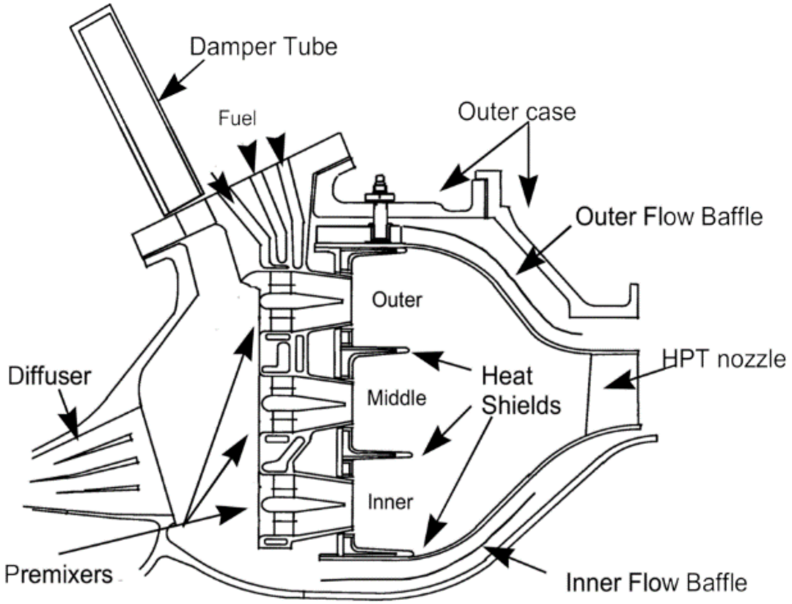
Damping devices:

– Helmholtz resonators



He

– Quarter wave tubes



Resonant frequency: $f = \frac{c}{4L}$

Mongia, H. C., Held, T. J., Hsiao, G. C., & Pandalai, R. P. (2005). Incorporation of combustion instability issues into design process: GE aeroderivative and aero engines experience. *Progress in astronautics and aeronautics*, *210*, 43.

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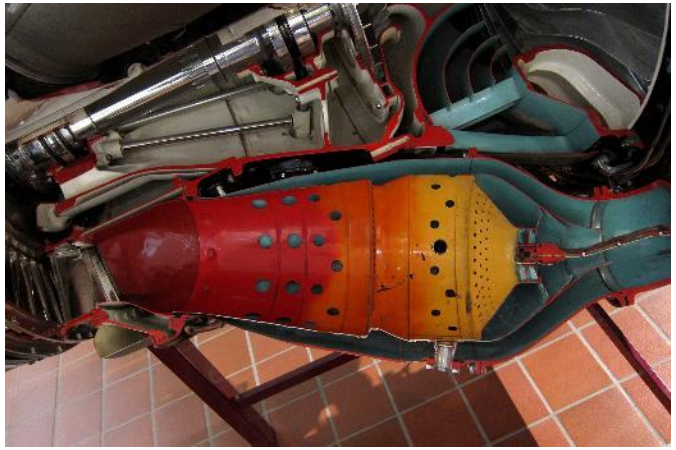
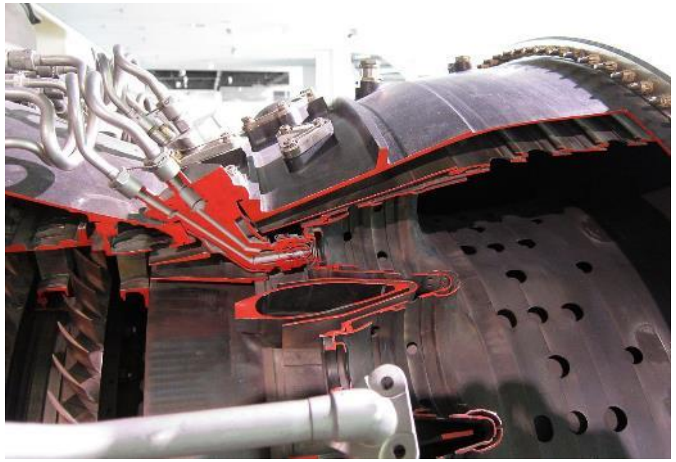
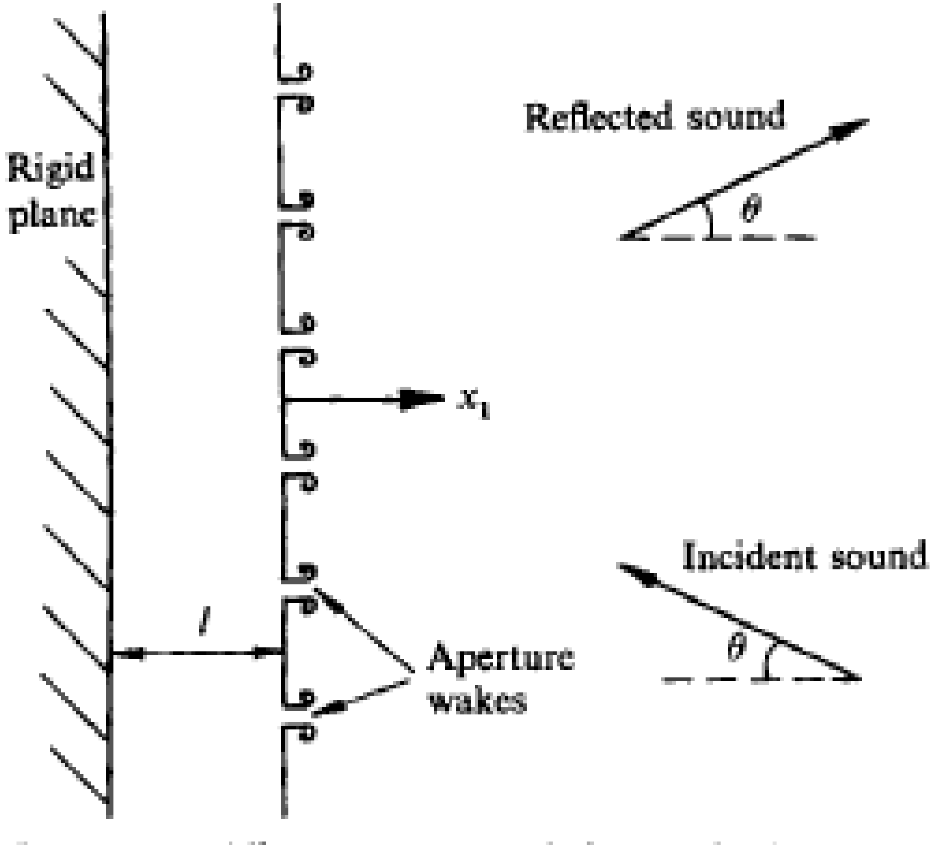


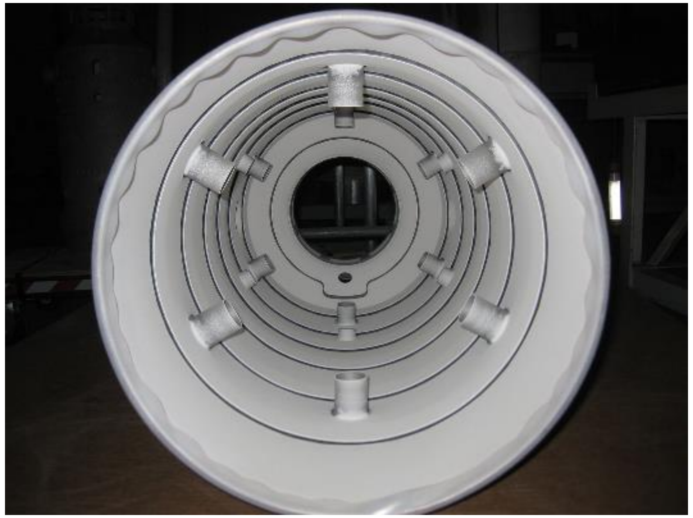
Wikipedia: Rolls-Royce Nene turbojet



Wikipedia: P&W JT9D



Source: Hughes, I. J., and Dowling, A. P., (1990) "The absorption of sound by perforated linings," *Journal of Fluid Mechanics*, vol. 218, p. 299-335



https://atshouston.com/project/siemens-westinghouse-gas-turbine-combustor-basket/

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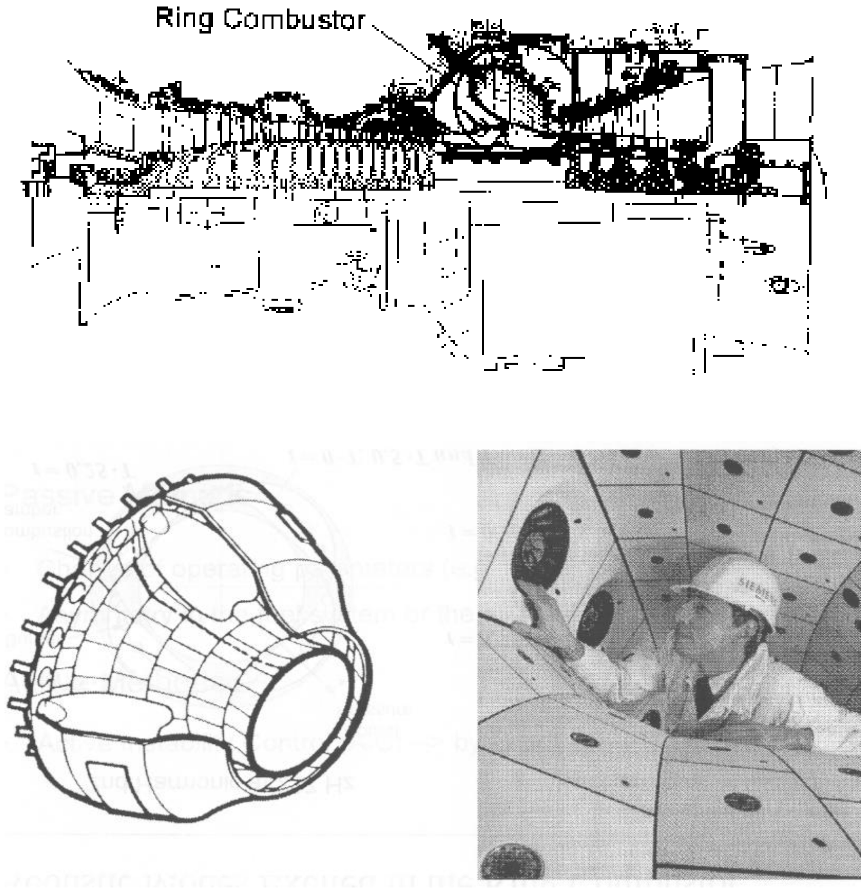
Change the heat release rate spatial distribution

Example 1: Circumferential modes were broken by a technique called “symmetry breaking,” where the symmetry of Q’ is changed relative to acoustic field

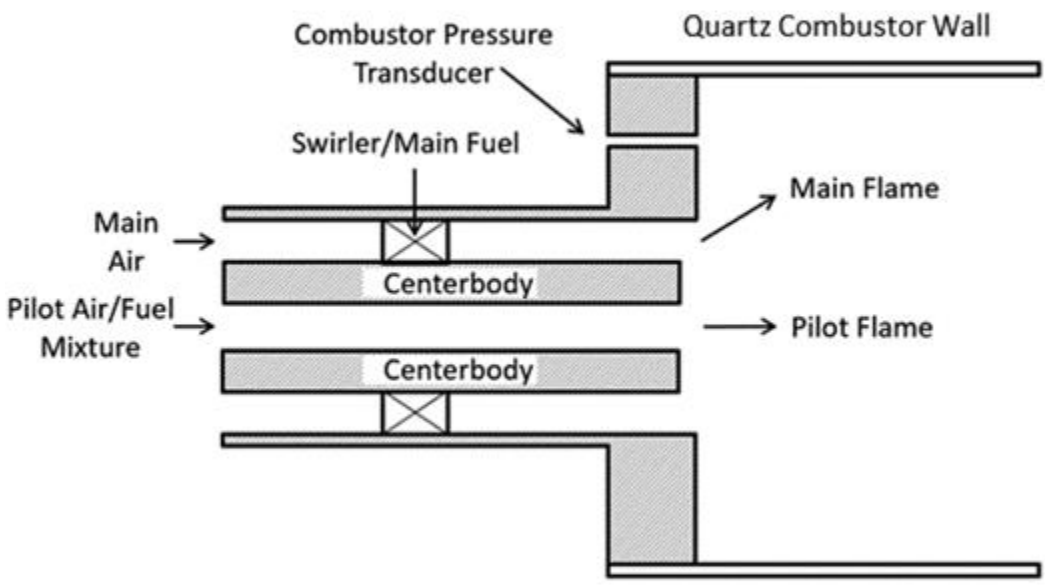
Siemens VX4.3A

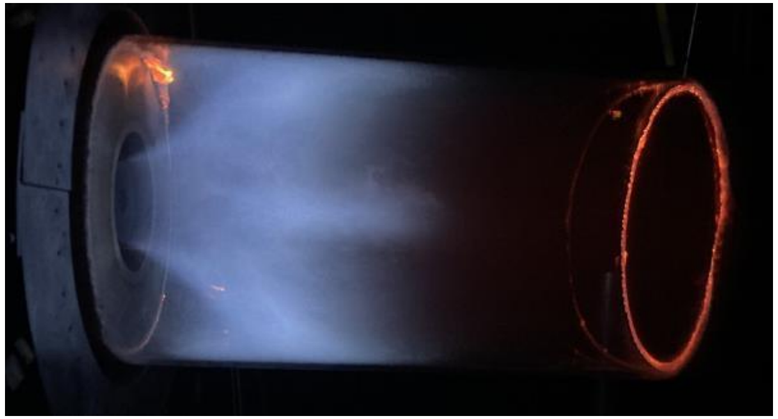


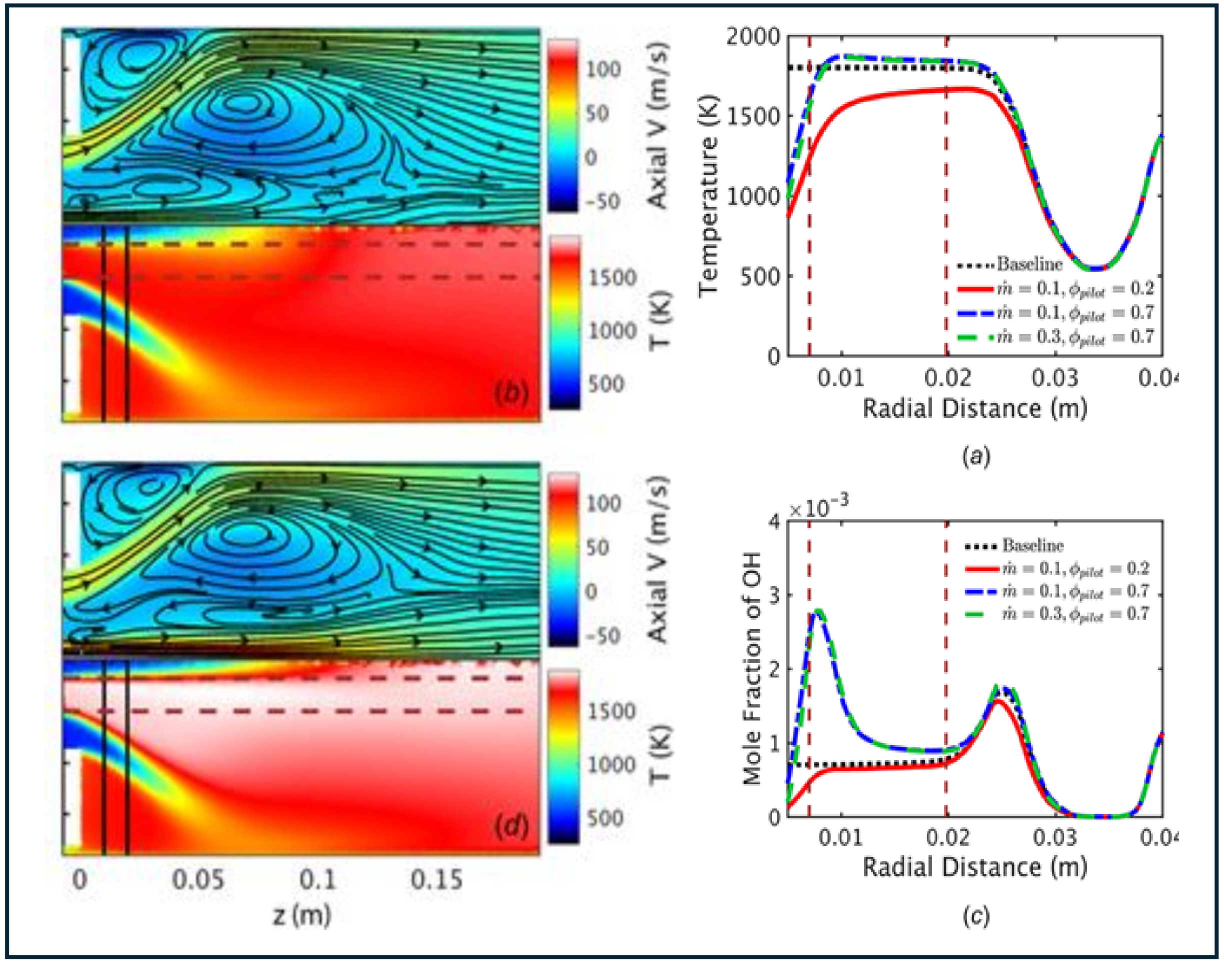
From “Application of Active Combustion Instability Control to a Heavy Duty Gas Turbine”, J. Hermann et al., presentation at 1997 Workshop on Dynamics and Control of Combustion Instabilities in Propulsion and Power Systems

From “Suppression of Dynamic Combustion Instabilities by Passive and Active Means”, by P. Berenbrink and S. Hoffman, ASME Paper 2000-GT-0079

Example 2: Piloting can change both the shape and strength of the flame, making it less susceptible to combustion instability

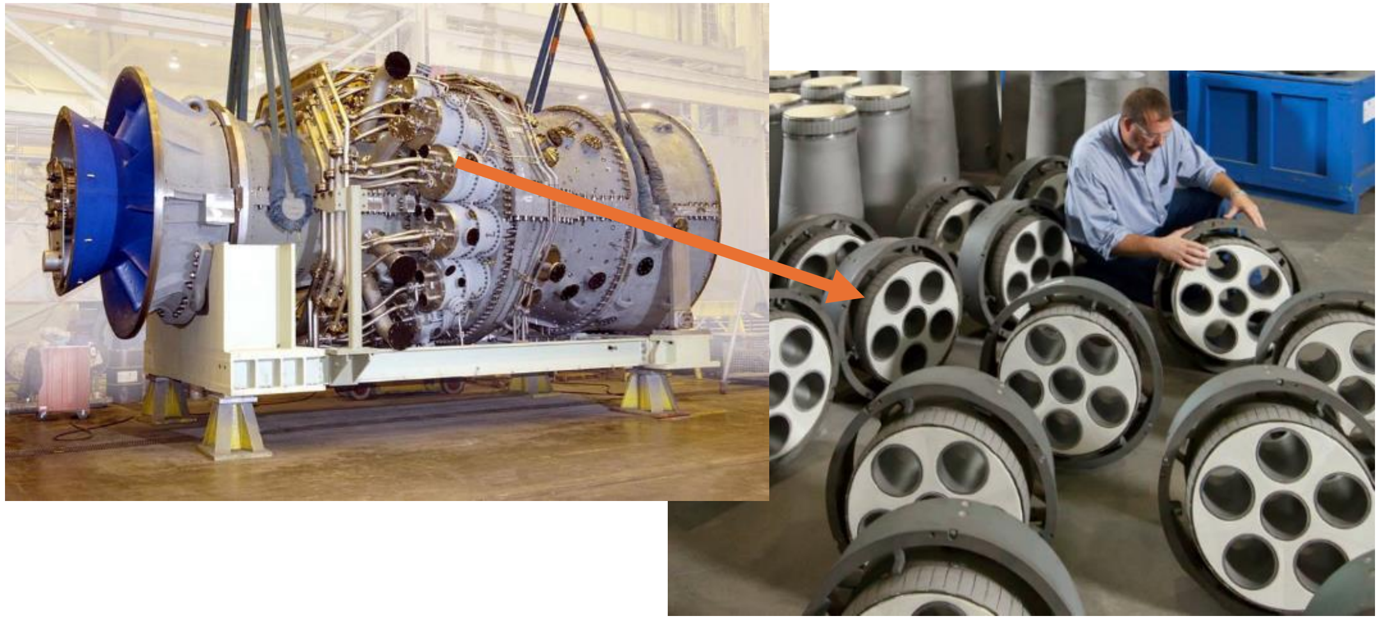




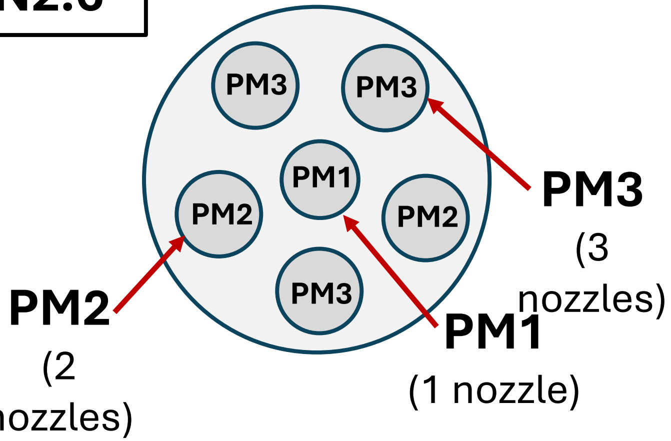


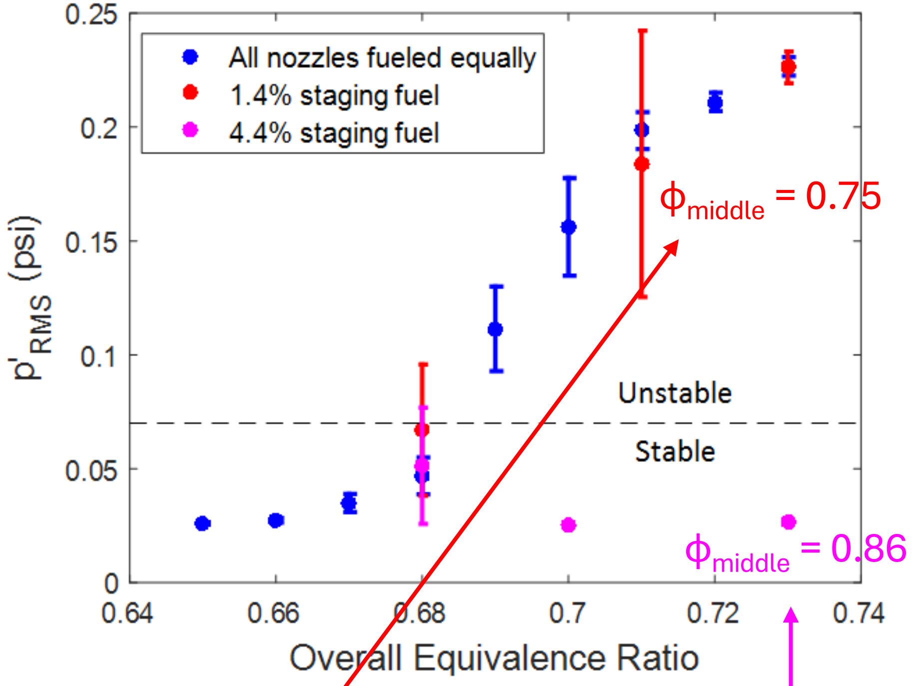
Li, J., Kwon, H., Seksinsky, D., Doleiden, D., O'Connor, J., Xuan, Y., ... & Blust, J. (2022). Describing the mechanism of instability suppression using a central pilot flame with coupled experiments and simulations. *Journal of Engineering for Gas Turbines and Power*, *144*(1), 011015.

Example 3: Fuel staging us commonly used in systems with multiple flames to redistribute heat release rate and de-phase heat release rate oscillations



GE DLN2.6





Small amounts of staging fuel: Unsatisfactory suppression

Higher amounts of staging fuel: Instability completely suppressed

Source: http://www.ccj-online.com/bg/companies/allied/
http://news.thomasnet.com/companystory/GE-s-F-Technology-Fleet-of-Gas-Turbines-Tops-20-Million-Hours-of-Operation-Worldwide-524223

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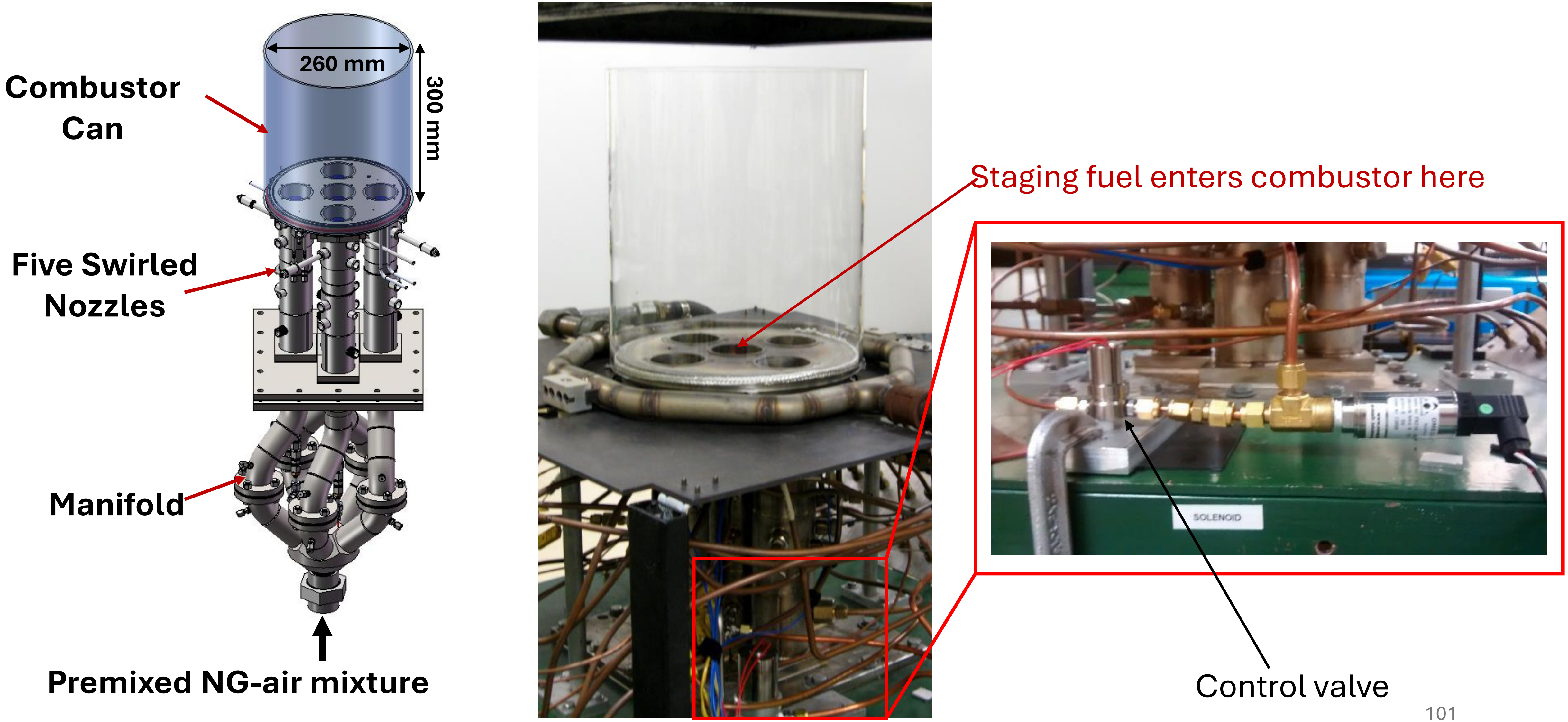
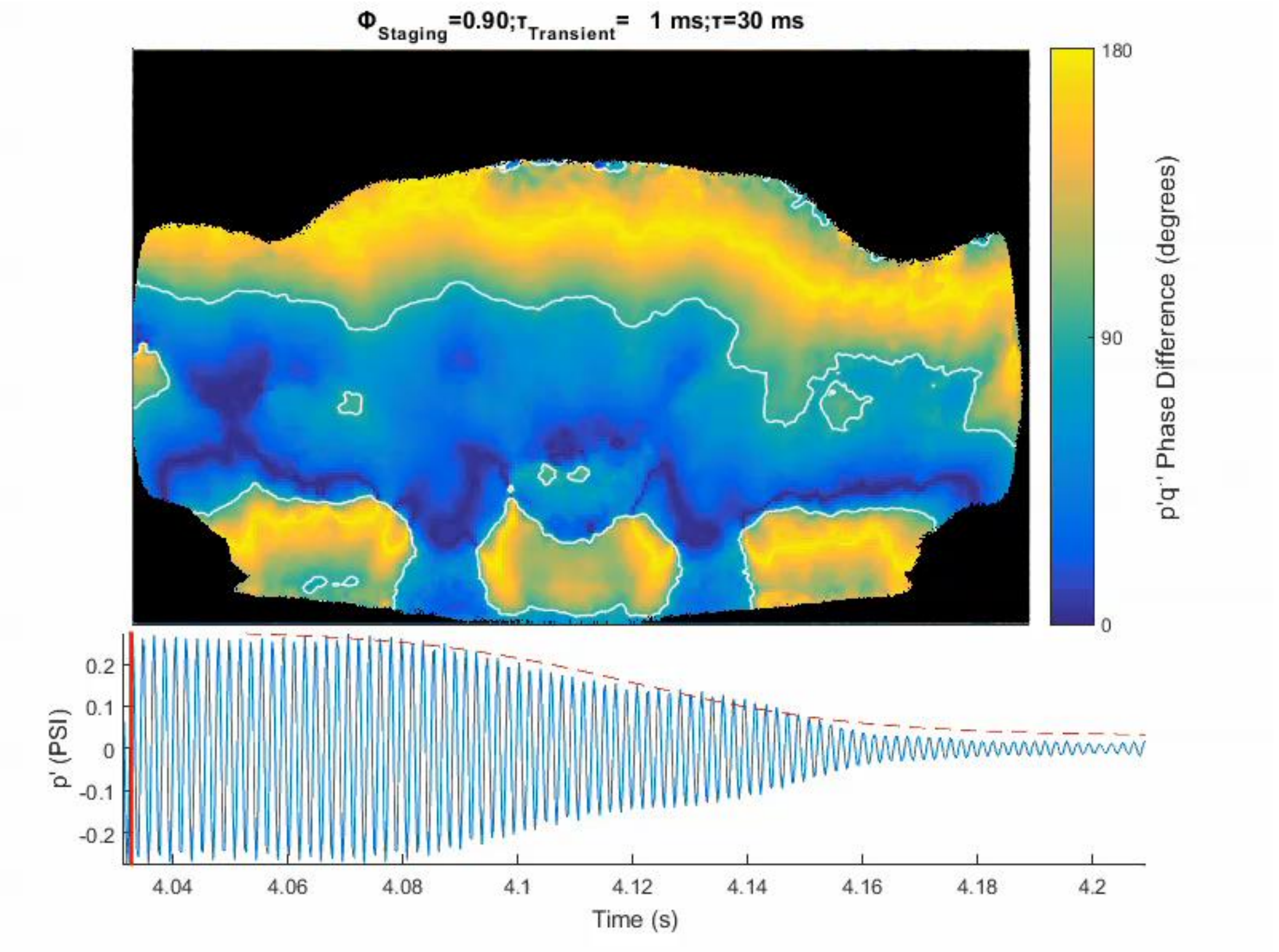


As we increased the equivalence ratio of the inner nozzle, the equivalence ratio of the outer nozzles decrease and the instability is suppressed

Video credit: Wyatt Culler

Culler, W., Chen, X., Samarasinghe, J., Peluso, S., Santavicca, D., & O'Connor, J. (2018). The effect of variable fuel staging transients on self-excited instabilities in a multiple-nozzle combustor. Combustion and Flame, 194, 472-484.

High-speed imaging allows us to calculate the local Rayleigh Index, showing the regions of driving in the flame



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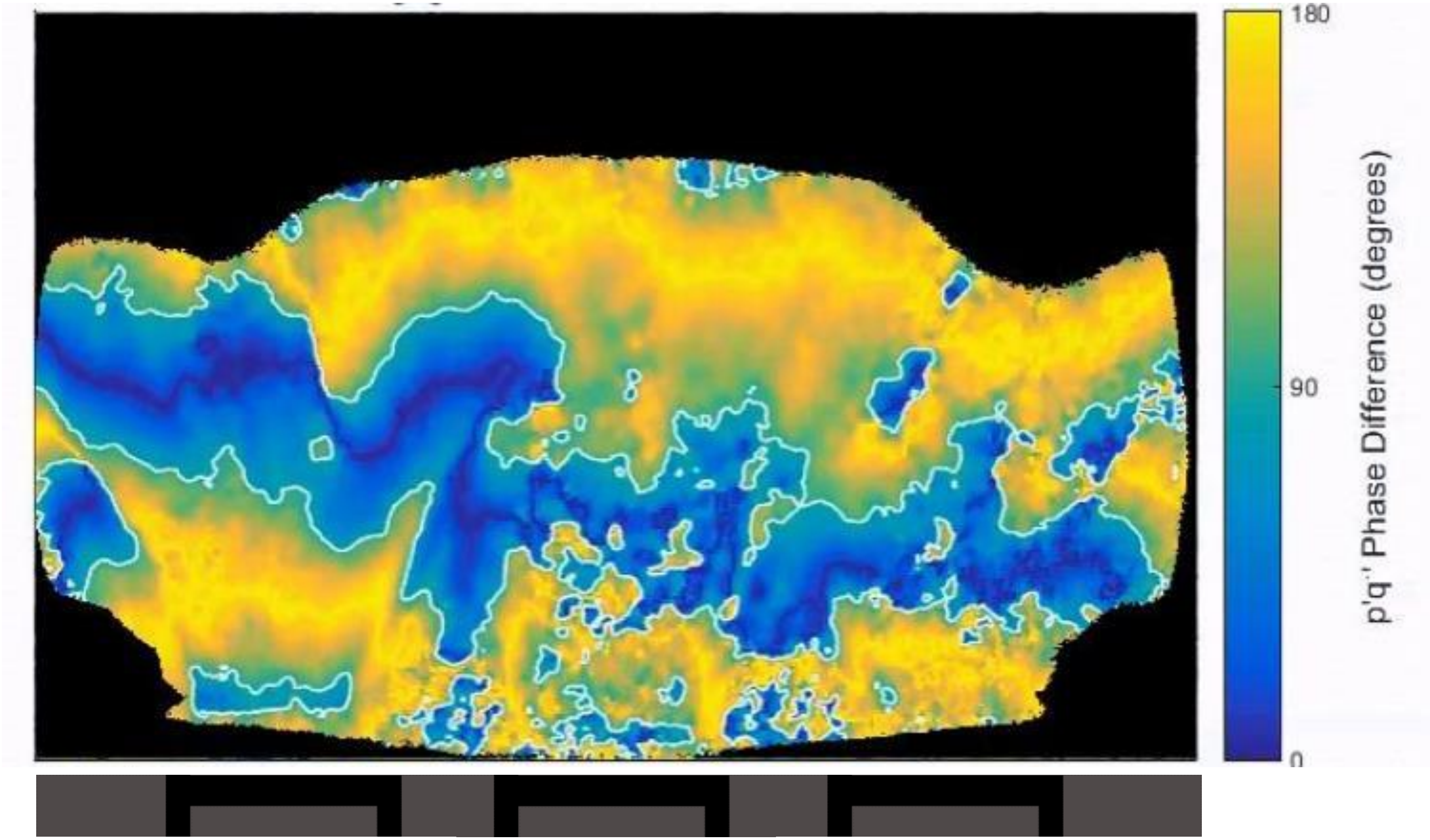
We noticed that through time, there seemed to be an imbalance in the phasing of the flame oscillations during the staging transient



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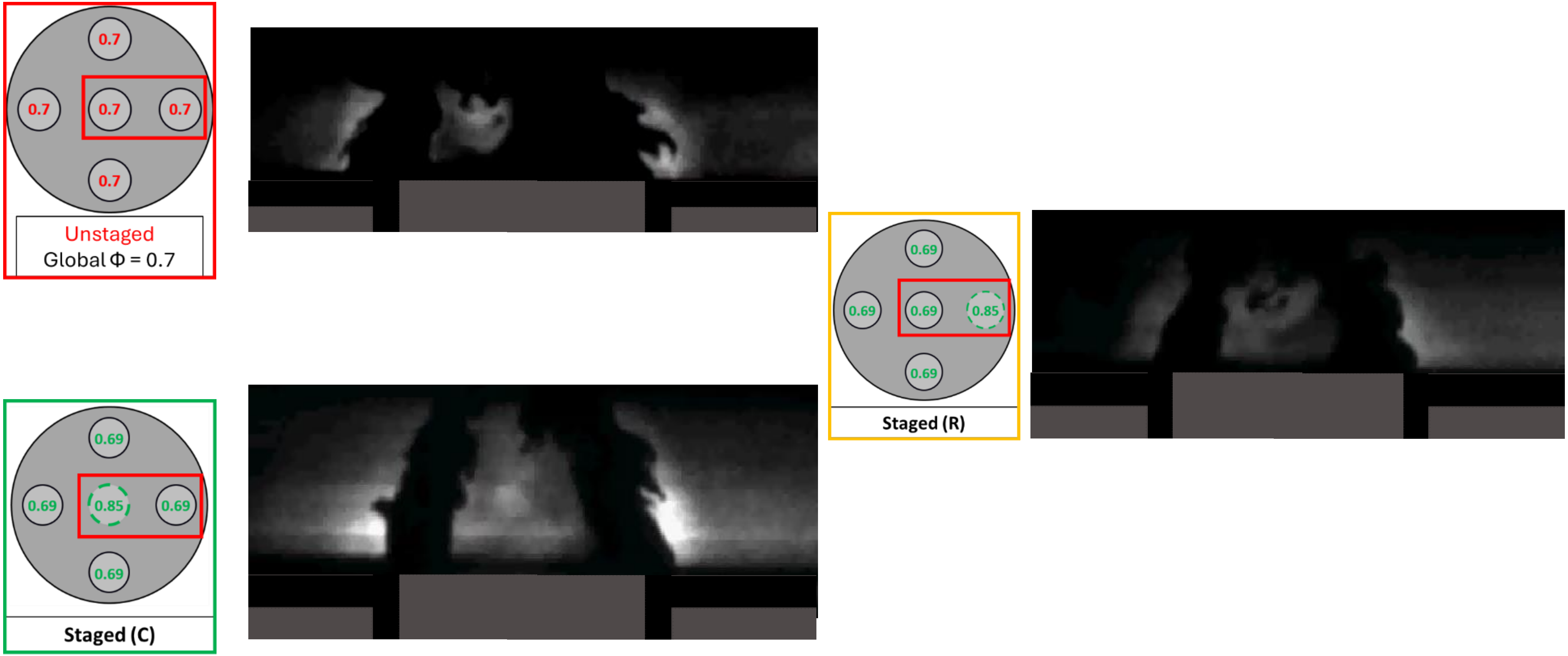
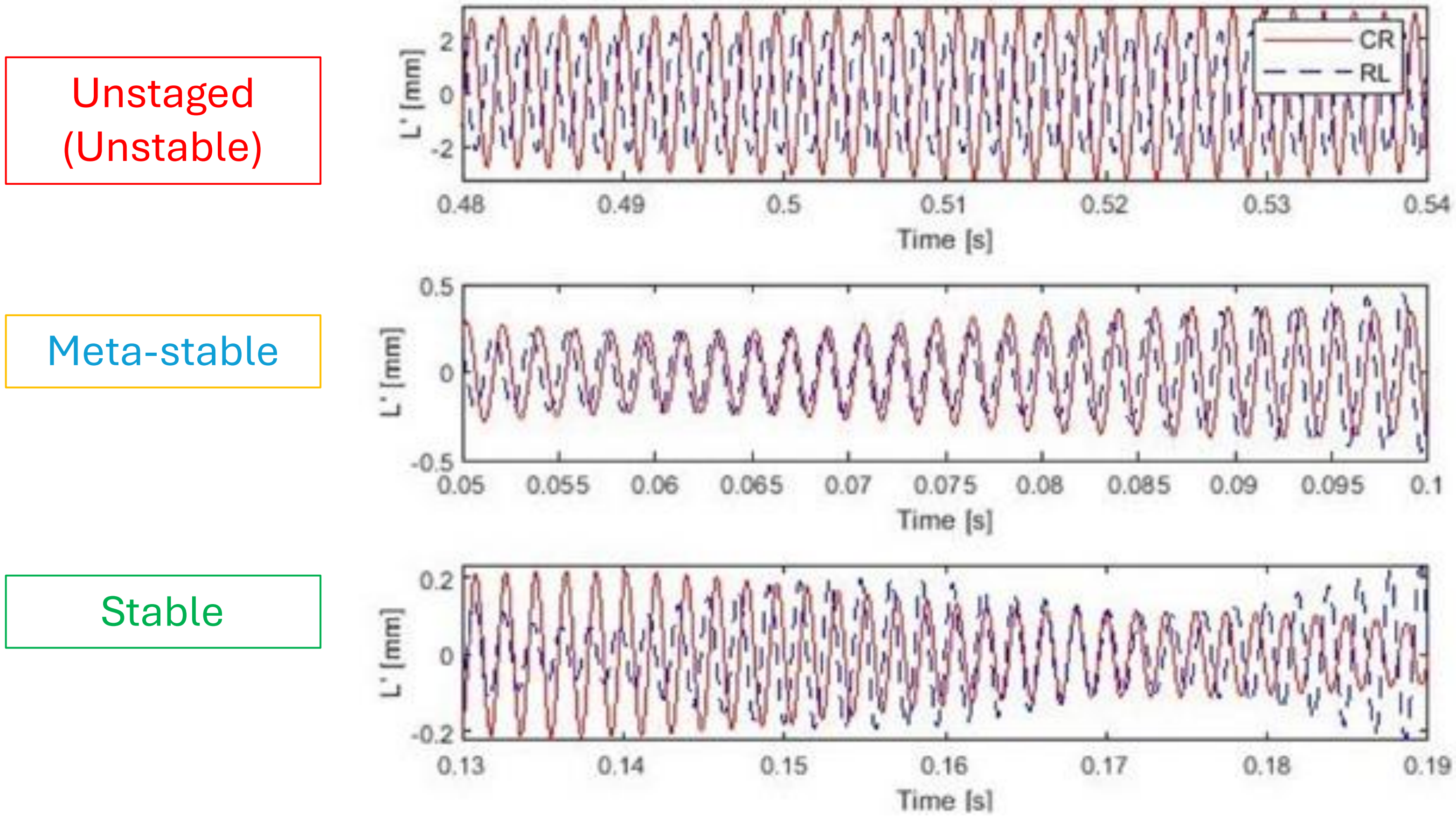
High-speed PLIF imaging allowed us to look at the flame edge oscillation in three cases: unstable, meta-stable, and stable



Doleiden, D., Culler, W., Tyagi, A., Peluso, S., & O'Connor, J. (2019). Flame edge dynamics and interaction in a multinozzle can combustor with fuel staging. *Journal of Engineering for Gas Turbines and Power*, *141*(10), 101009.

The phase oscillation between the flame edges suggest that staging changes the phase of oscillation between adjacent flames, leading to Q' cancellation



Flame Edge Oscillation Time Series

Doleiden, D., Culler, W., Tyagi, A., Peluso, S., & O'Connor, J. (2019). Flame edge dynamics and interaction in a multinozzle can combustor with fuel staging. *Journal of Engineering for Gas Turbines and Power*, *141*(10), 101009.

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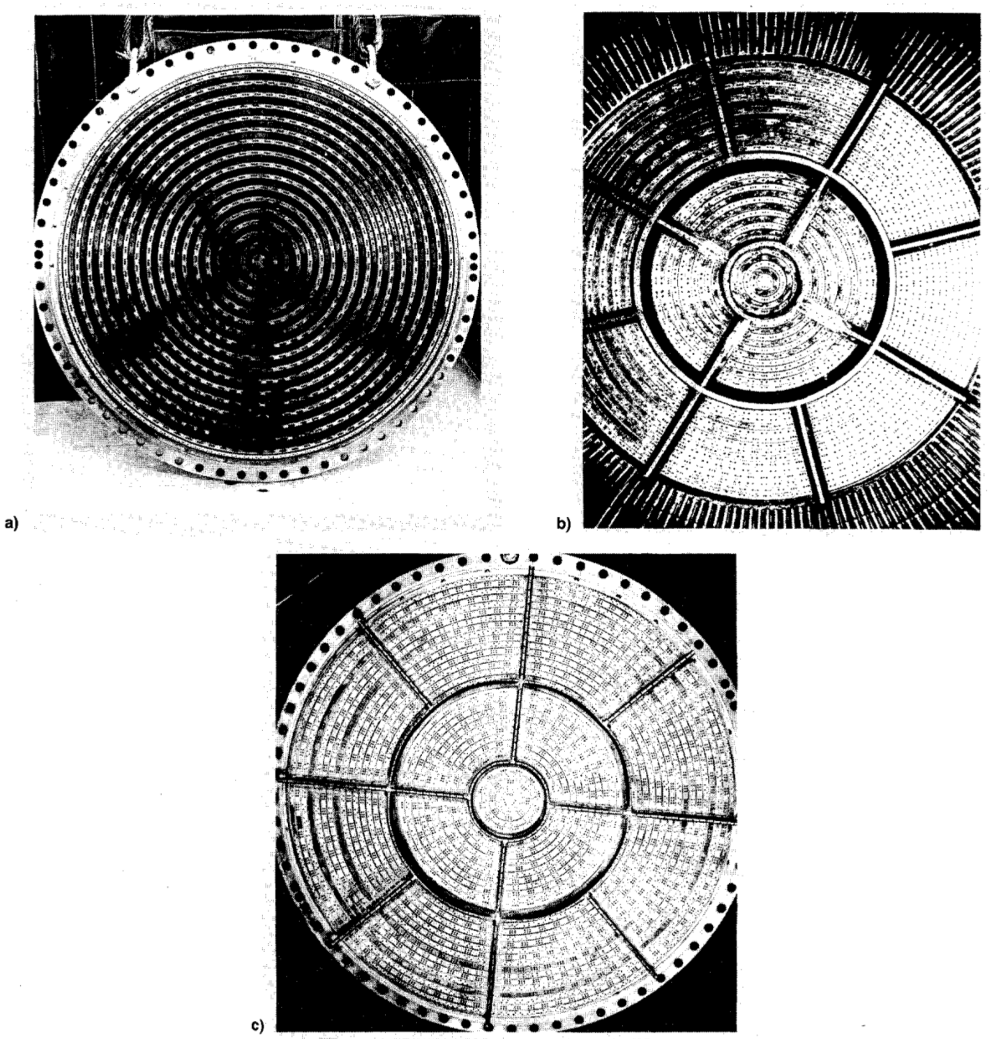
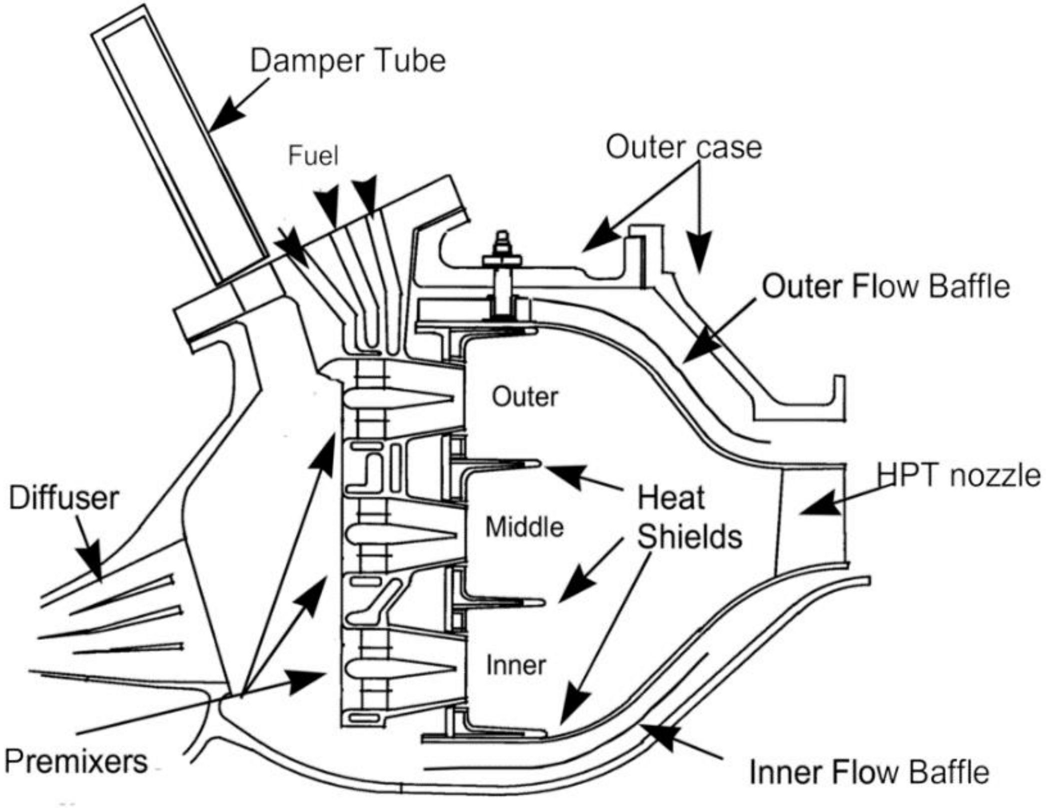
Change the heat release rate spatial distribution

Baffles are commonly used to change the shape of the acoustic mode compared to the flame by impeding its oscillation in certain directions



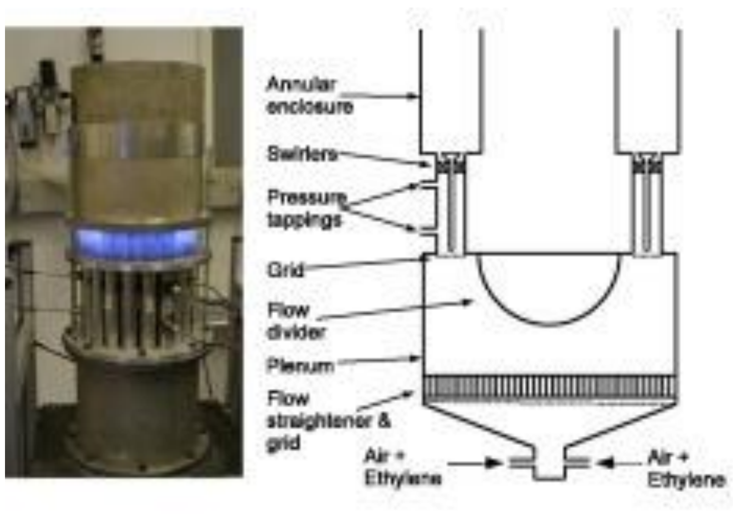
Oefelein, J. C., & Yang, V. (1993). Comprehensive review of liquid-propellant combustion instabilities in F-1 engines. *Journal of Propulsion and Power*, 9(5), 657-677.



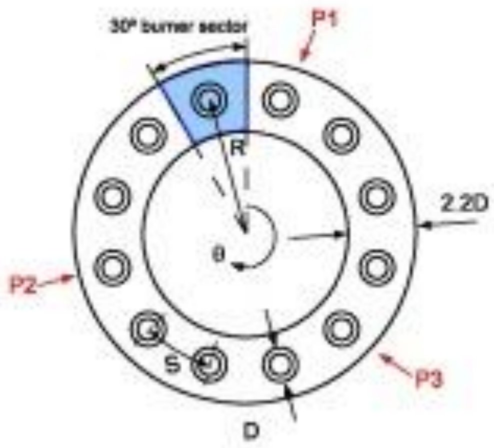
Mongia, H. C., Held, T. J., Hsiao, G. C., & Pandalai, R. P. (2005). Incorporation of combustion instability issues into design process: GE aeroderivative and aero engines experience. *Progress in astronautics and aeronautics*, *210*, 43.

Example: Annular combustor with transverse mode and baffles

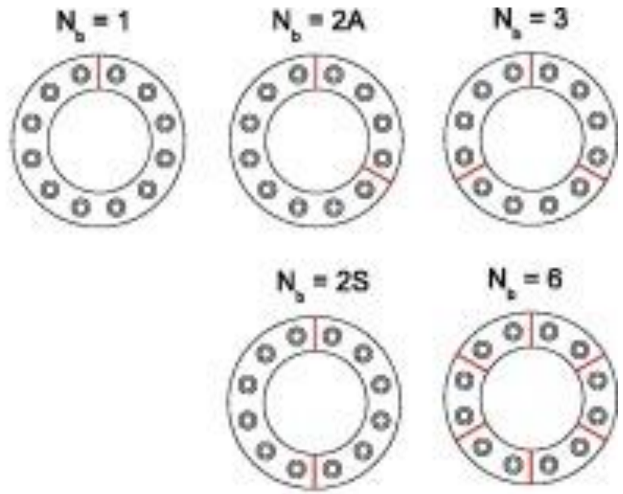
Dawson, J. R., & Worth, N. A. (2015). The effect of baffles on self-excited azimuthal modes in an annular combustor. *Proceedings of the Combustion Institute*, *35*(3), 3283-3290.



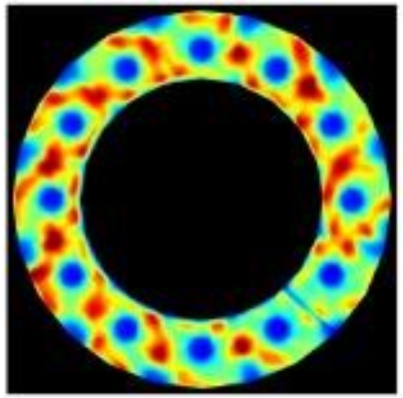
(a)



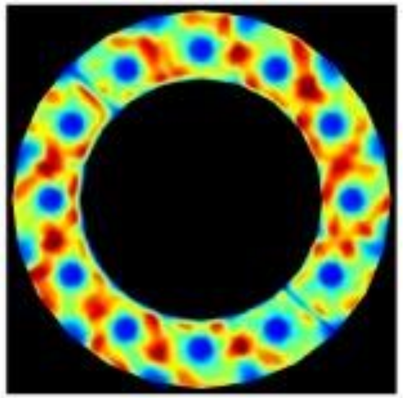
(b)



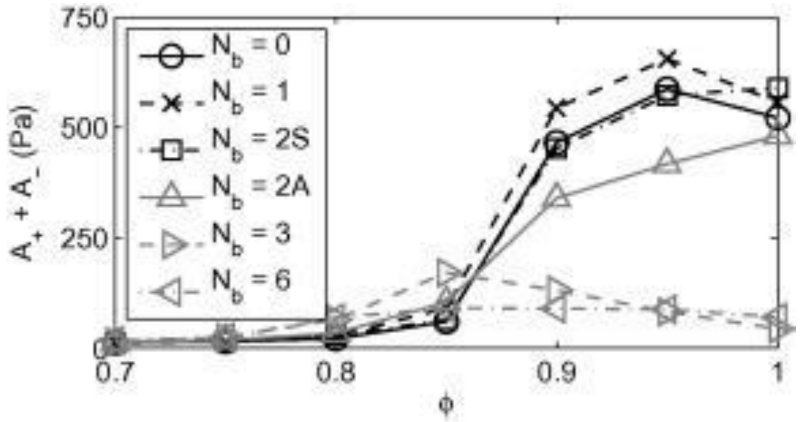
(c)



(a) $N_b = 1$



(b) $N_b = 2(s)$



(a) Amplitude response

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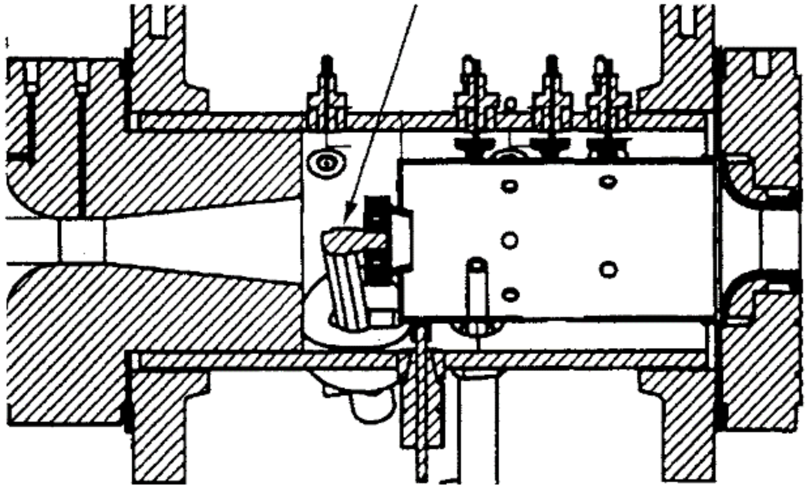
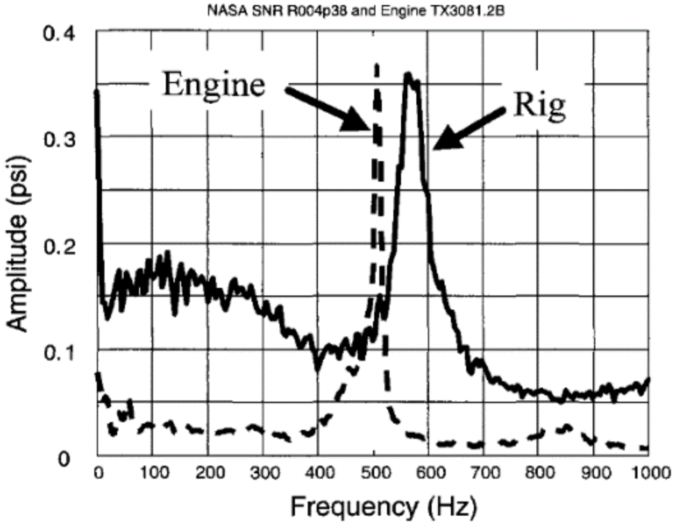
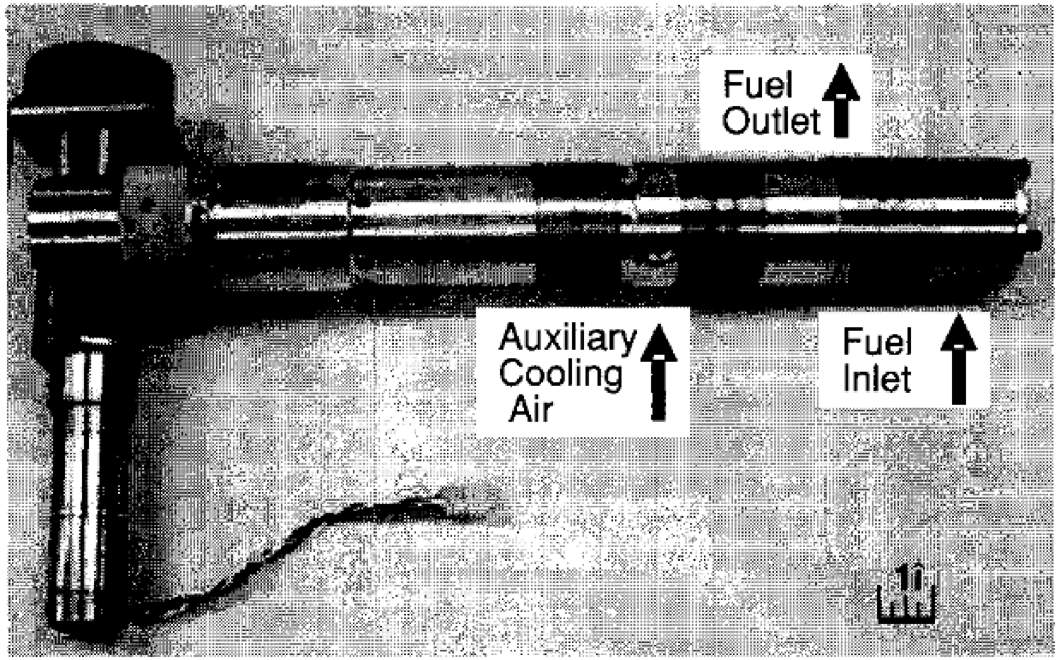
Move the pressure anti-node relative to the flame

Change the heat release rate spatial distribution

Active control is the way to active de-phase the oscillations between p' and Q' in a combustor, but is very difficult to achieve in practice







Cohen, J. M., Proscia, W., & DeLaat, J. (2005). Characterization and control of aeroengine combustion instability: Pratt & Whitney and NASA experience. *Progress in Astronautics and Aeronautics*, *210*, 113.

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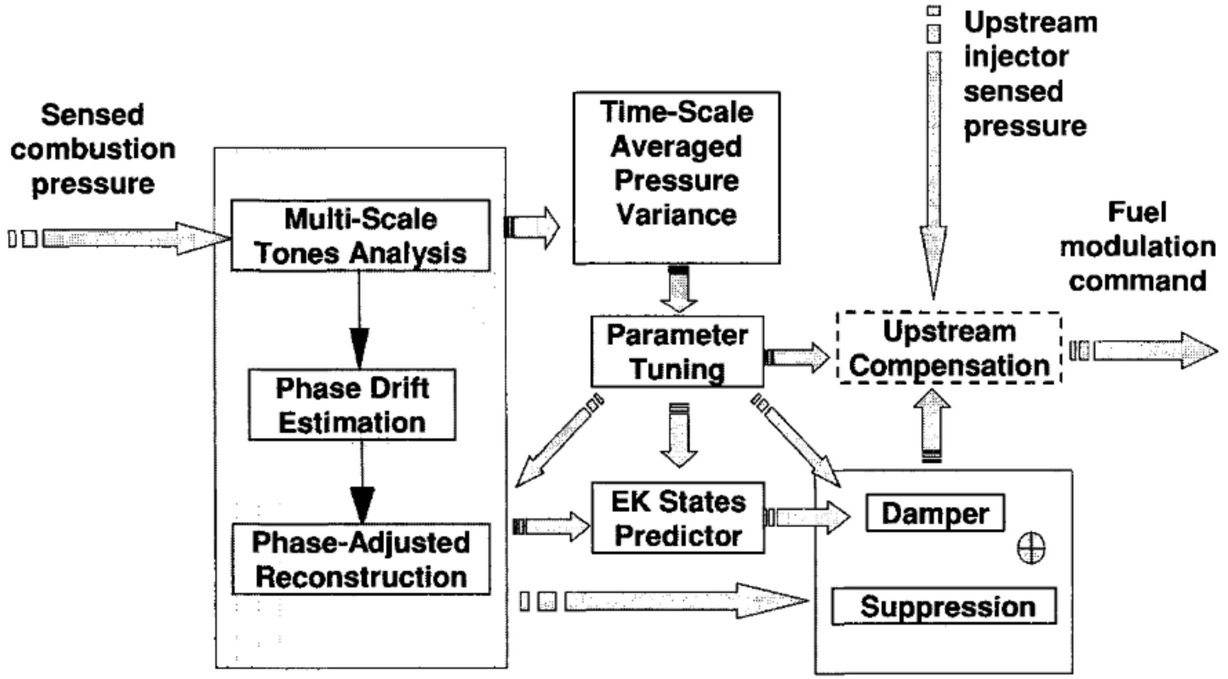
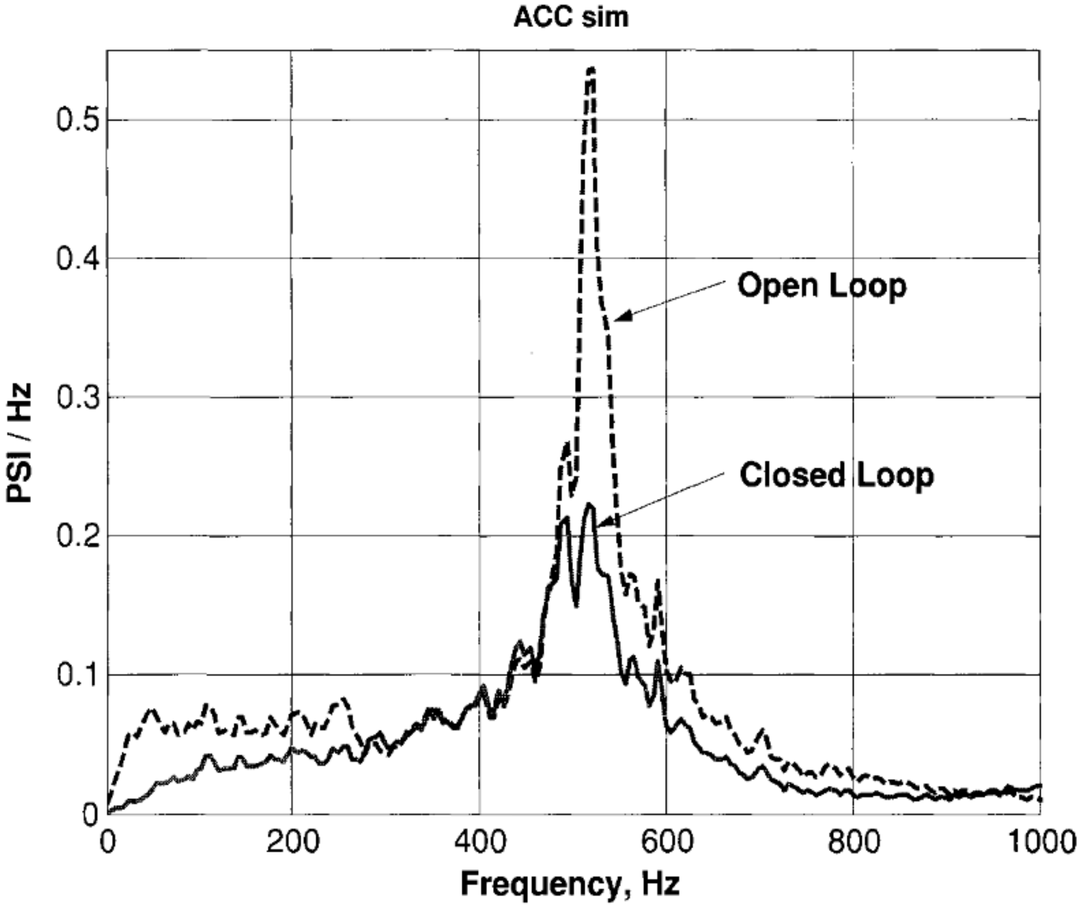
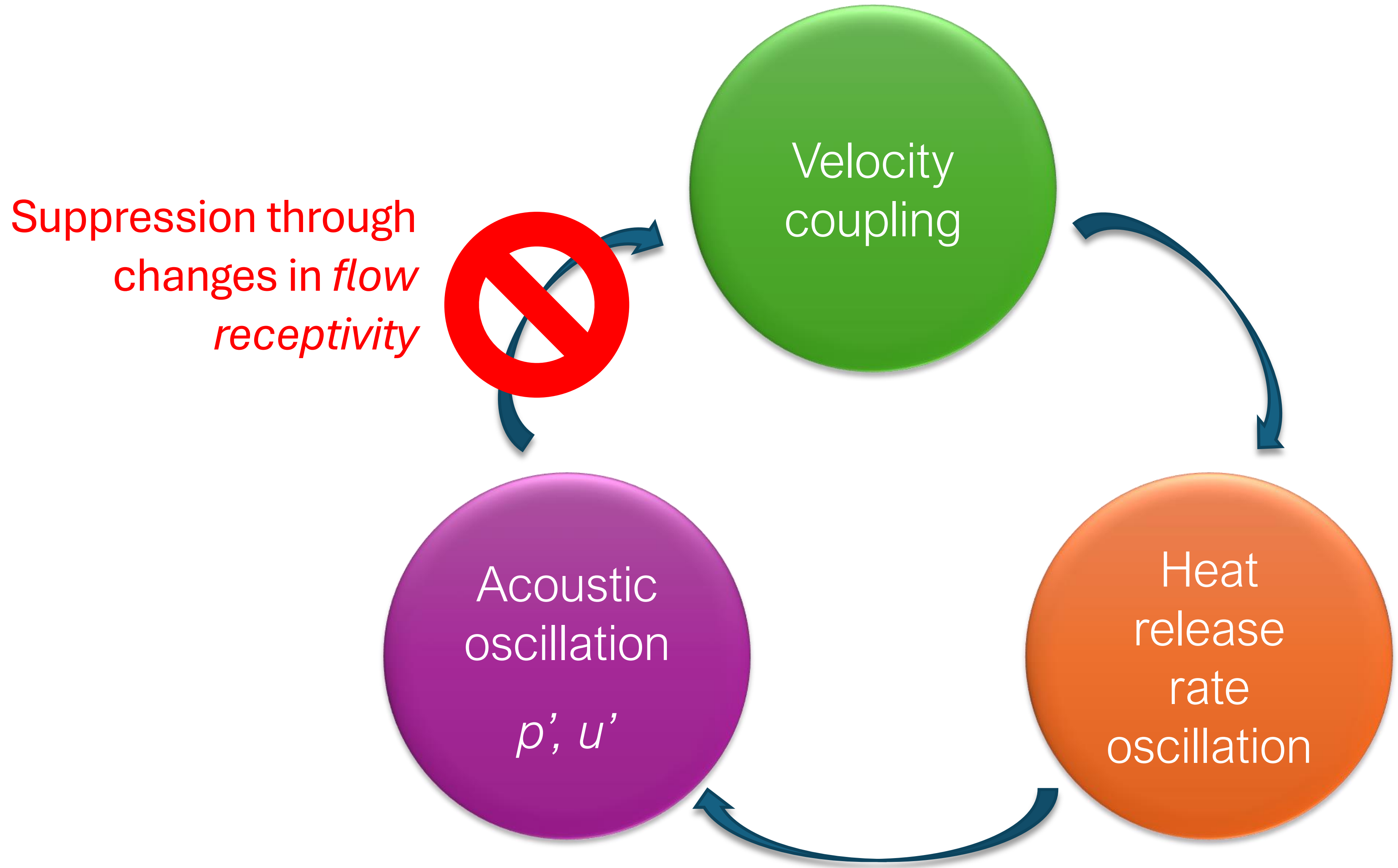




Cohen, J. M., Proscia, W., & DeLaat, J. (2005). Characterization and control of aeroengine combustion instability: Pratt & Whitney and NASA experience. *Progress in Astronautics and Aeronautics*, *210*, 113.

Bonus suppression method!

What if we could stop the feedback loop not by changing the flame or acoustics, but changing the flow?



SIDEBAR

Flow receptivity

In a stable flow, disturbances are damped and do not grow in space or time

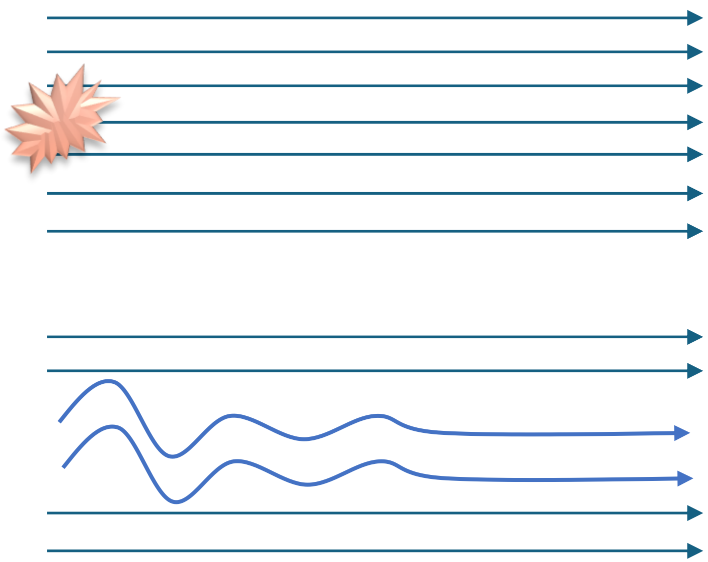
Stable Flow

In a stable flow, disturbances are damped and do not grow in space or time

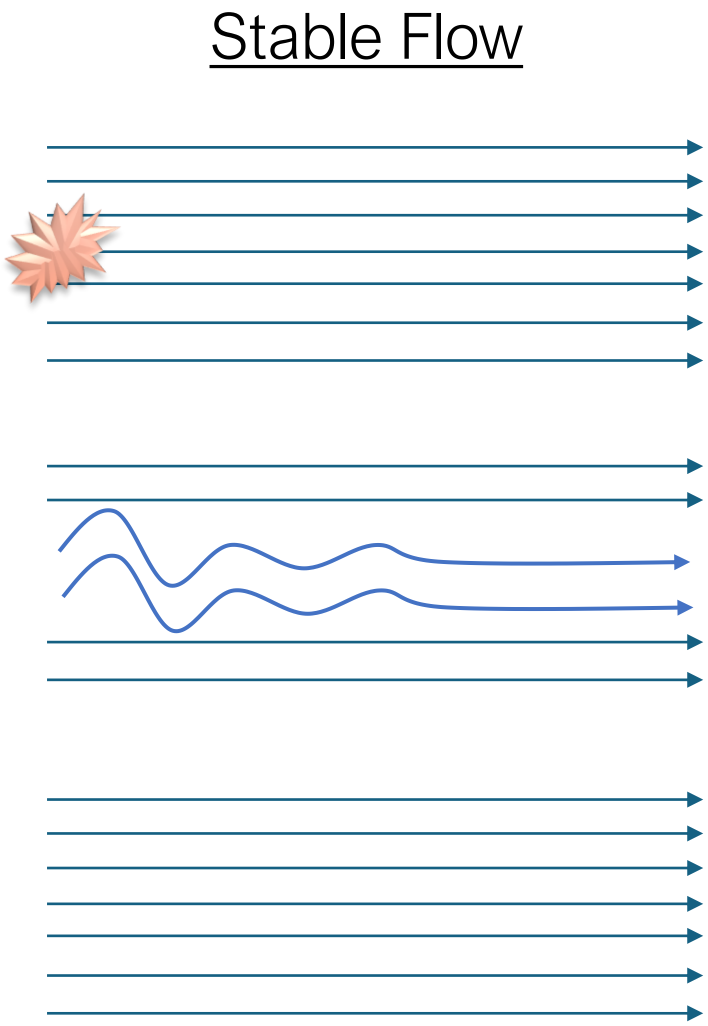
Stable Flow

In a stable flow, disturbances are damped and do not grow in space or time

Stable Flow

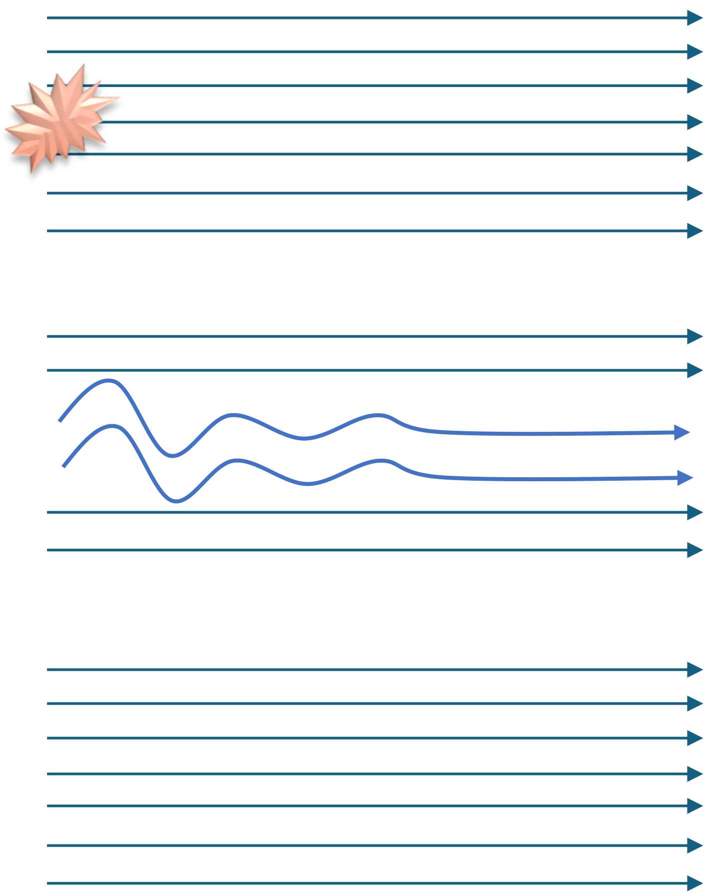


In a stable flow, disturbances are damped and do not grow in space or time



Some flows are disturbance amplifiers, where input disturbances grow in space as they’re convected

Stable Flow



Disturbance Amplifier

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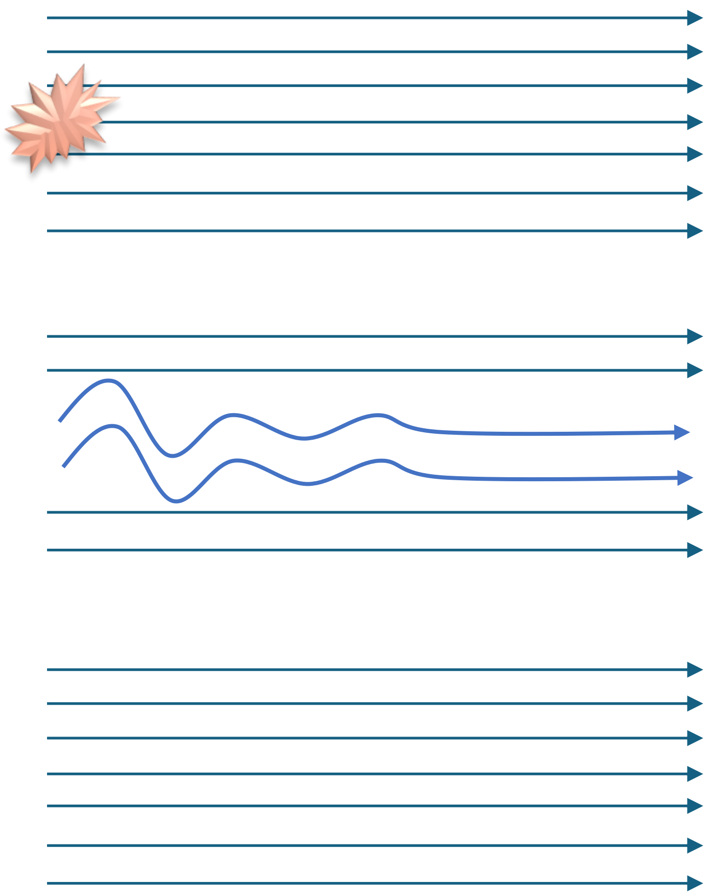
Stable Flow



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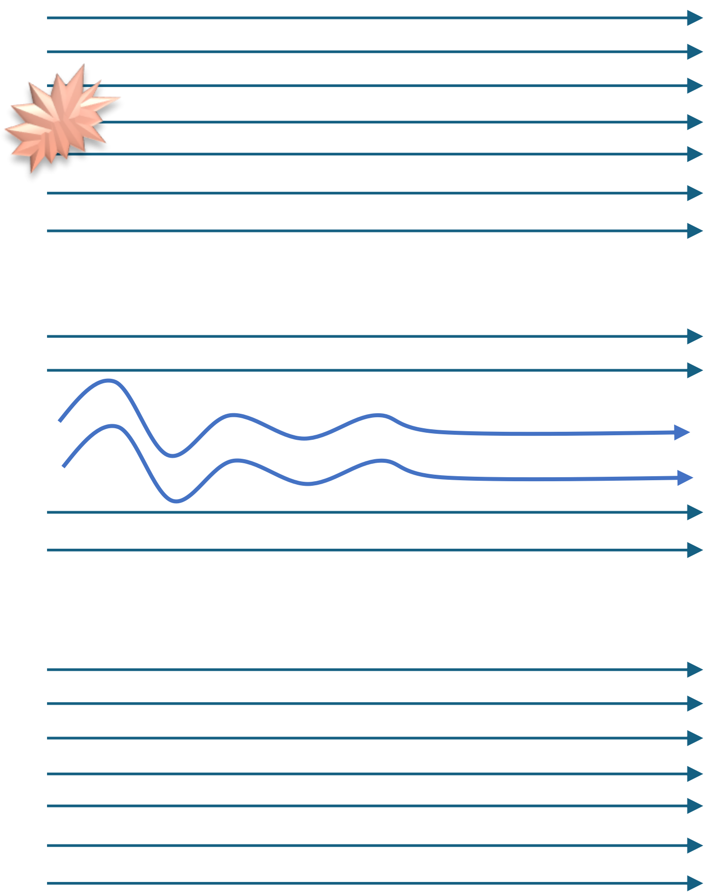
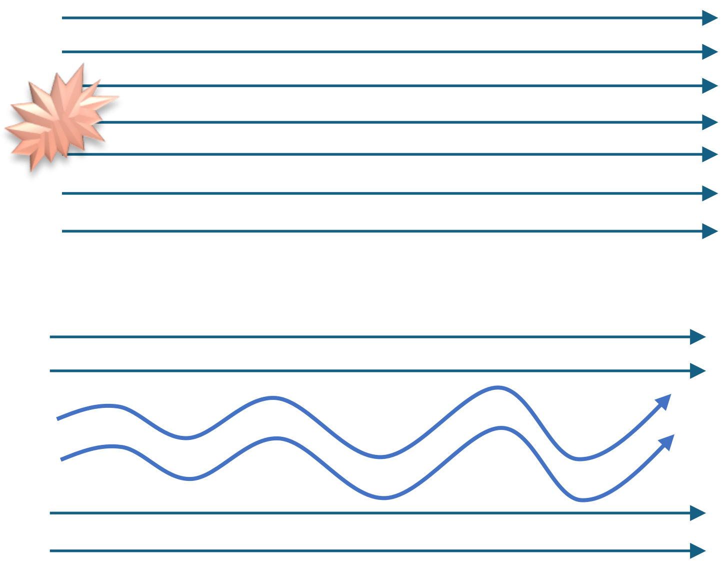
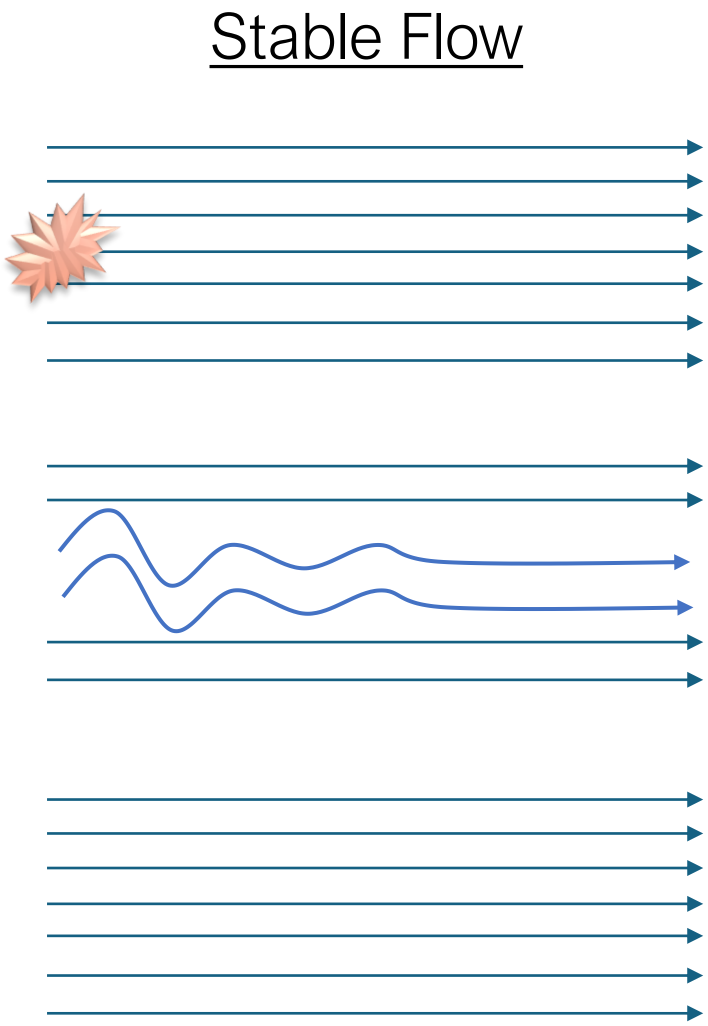
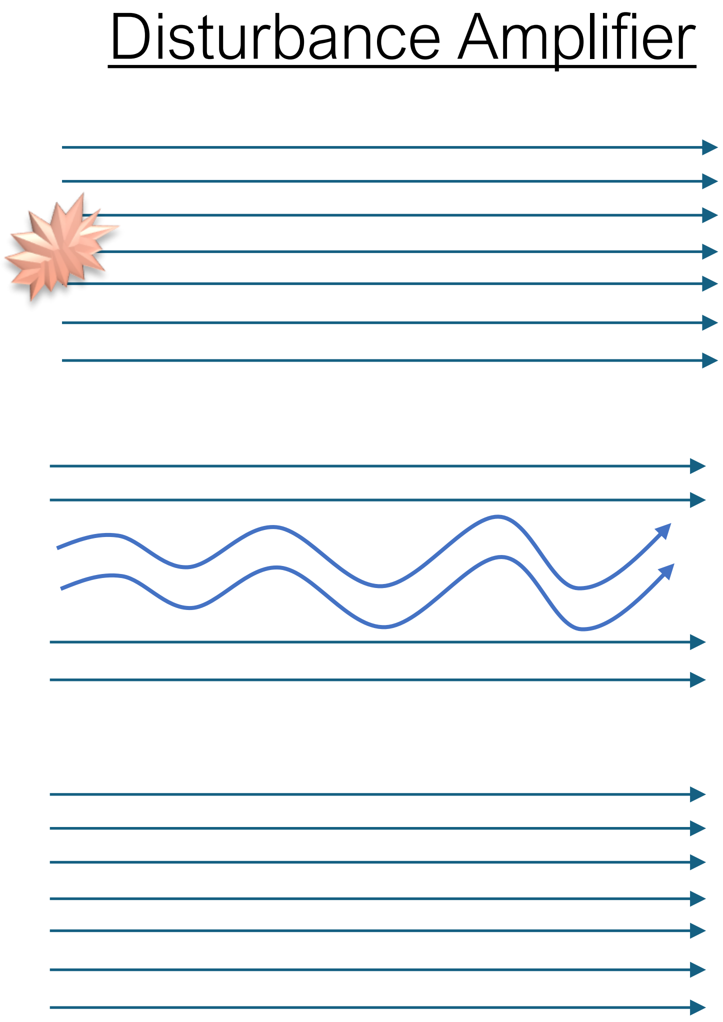


Disturbance Amplifier



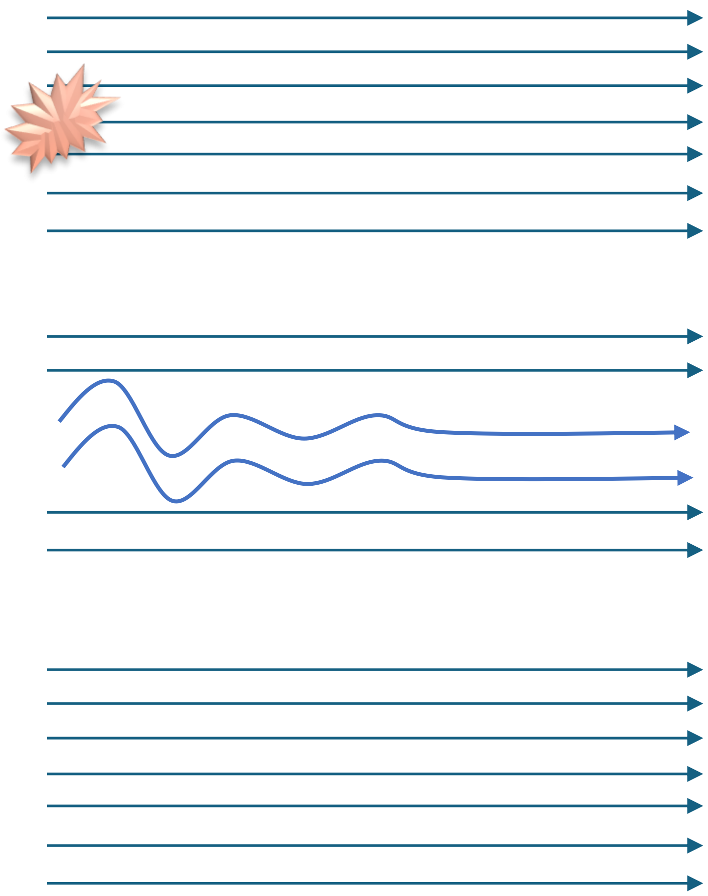
These disturbance amplifiers require continual excitation for oscillations to occur



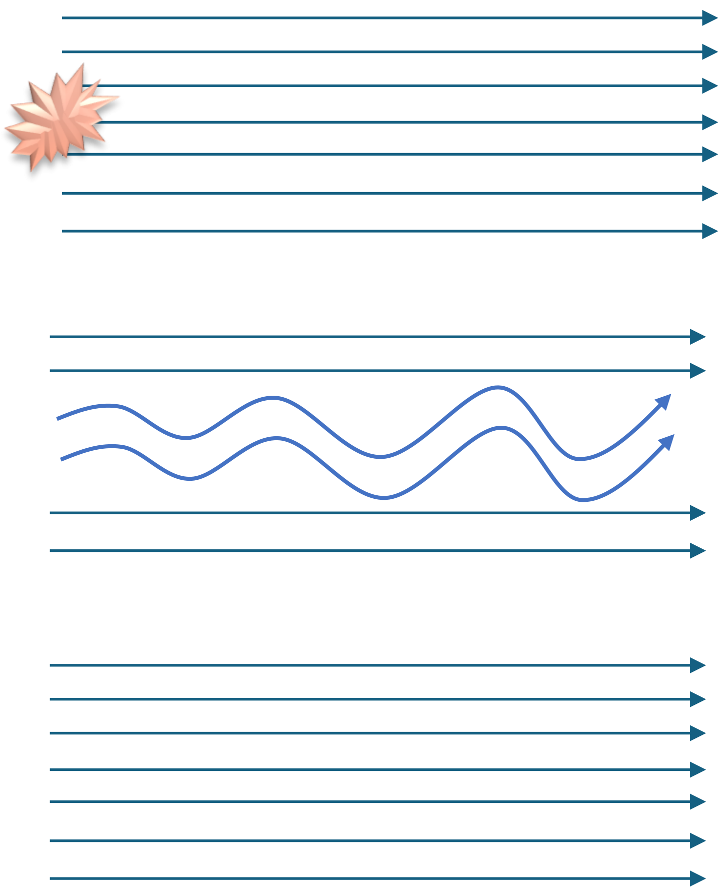


Finally, some flows are self-excited and display oscillations without the need for an input disturbance

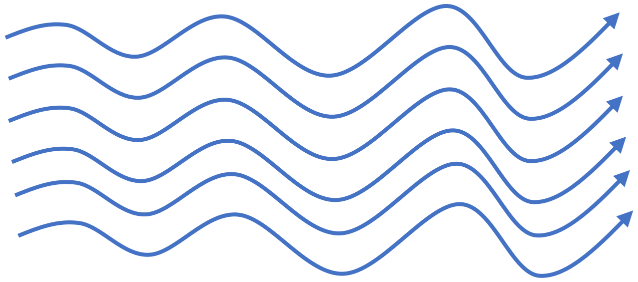
Stable Flow



Disturbance Amplifier

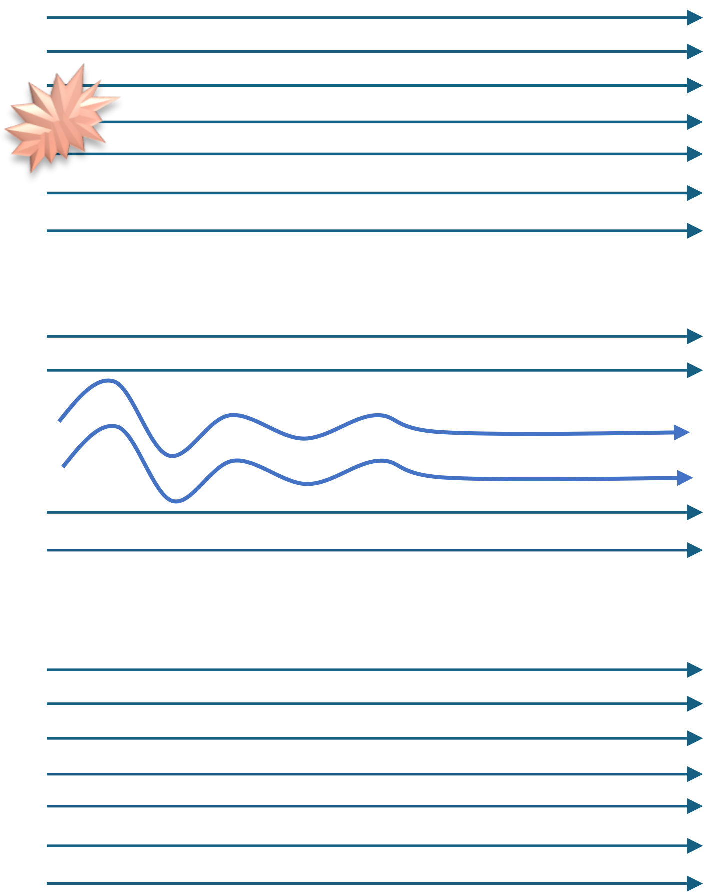


Self-Excited Flow

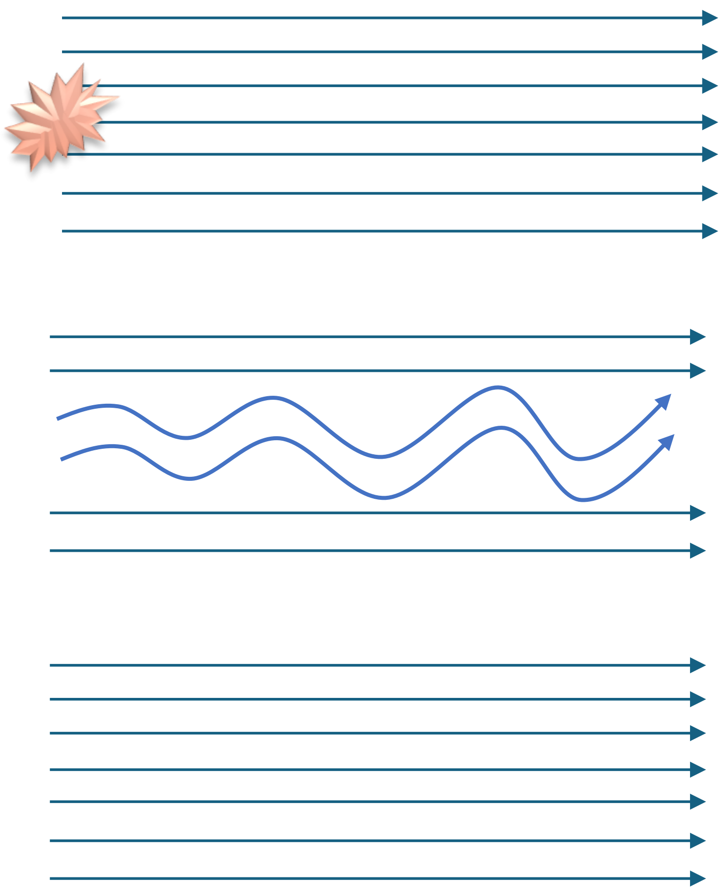


If you add additional disturbances to these self-excited flows, it's not always clear what's going to happen…

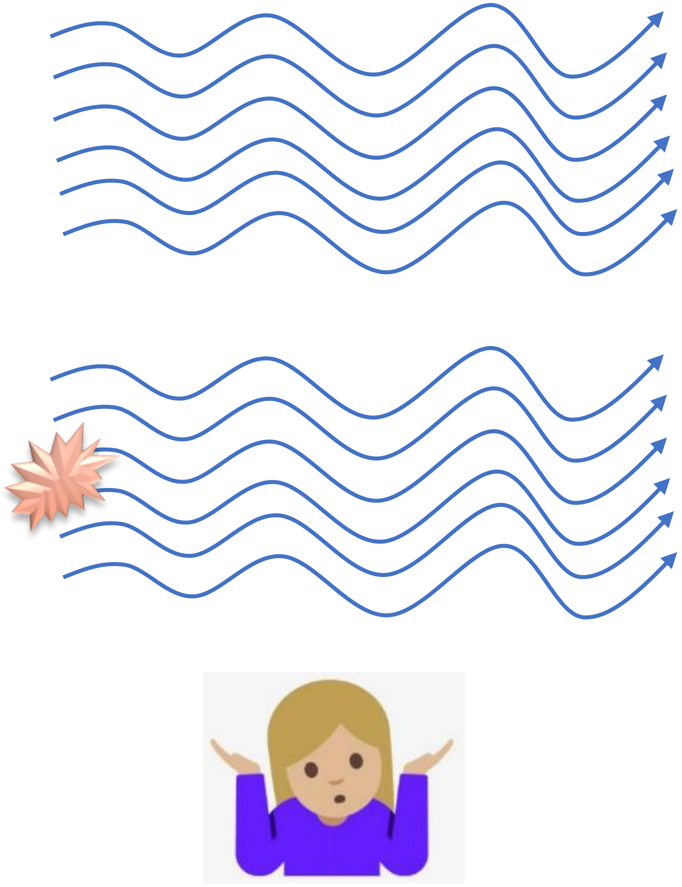
Stable Flow



Disturbance Amplifier



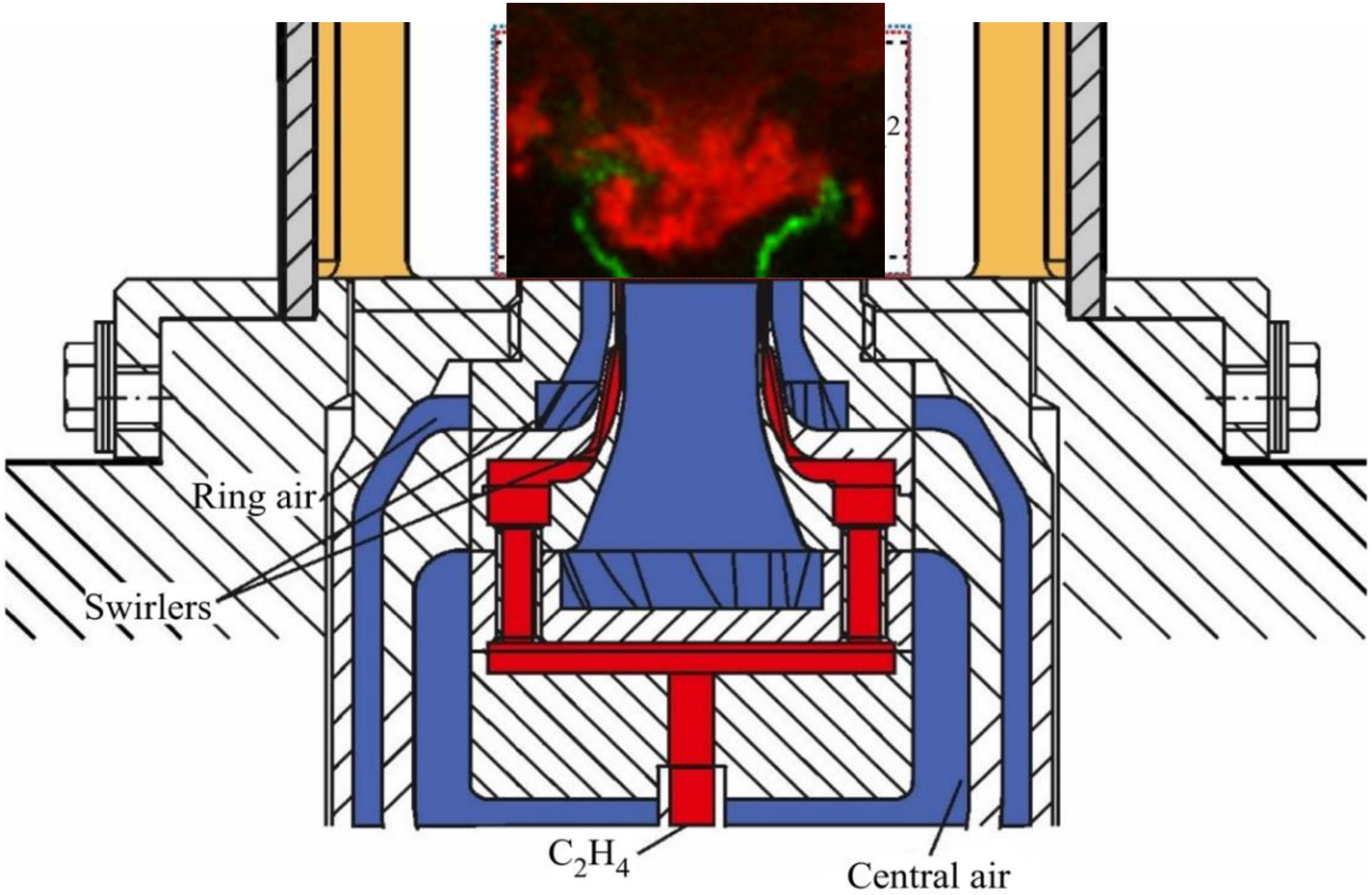
Self-Excited Flow



SIDEBAR OVER

Back to bonus suppression mechanism

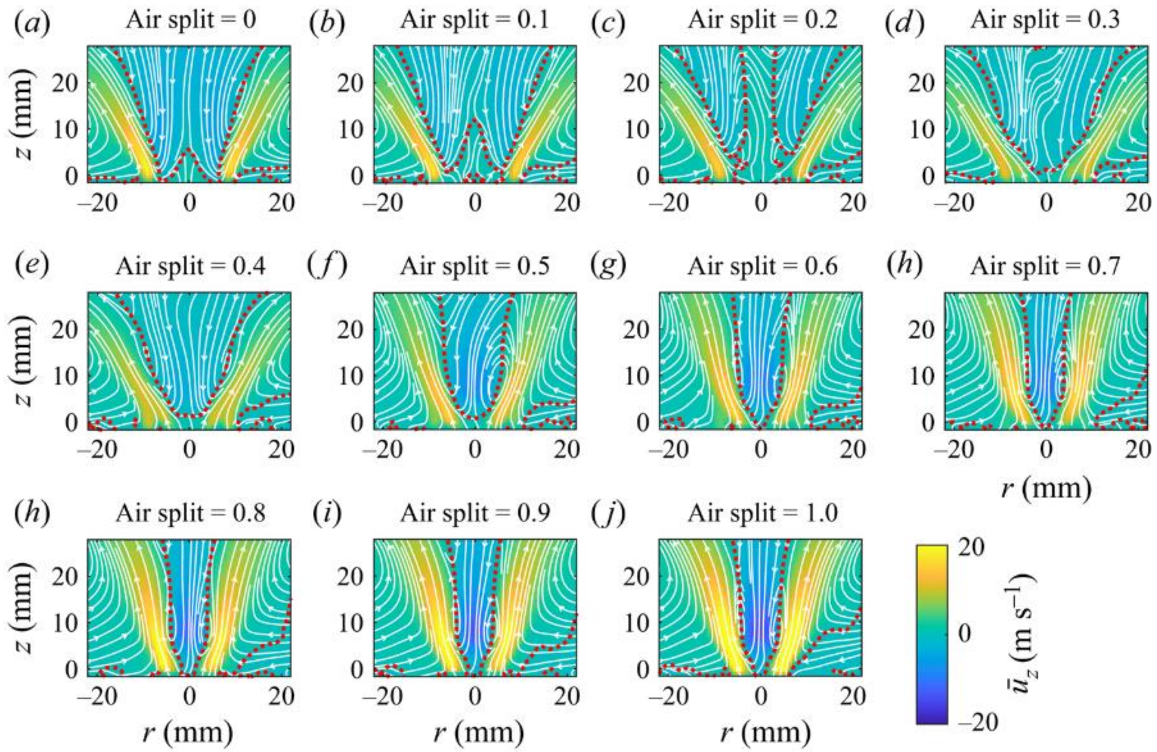
Experiments done at DLR-Stuttgart showed both hydrodynamic and thermoacoustic instability as a function of air split through the nozzle



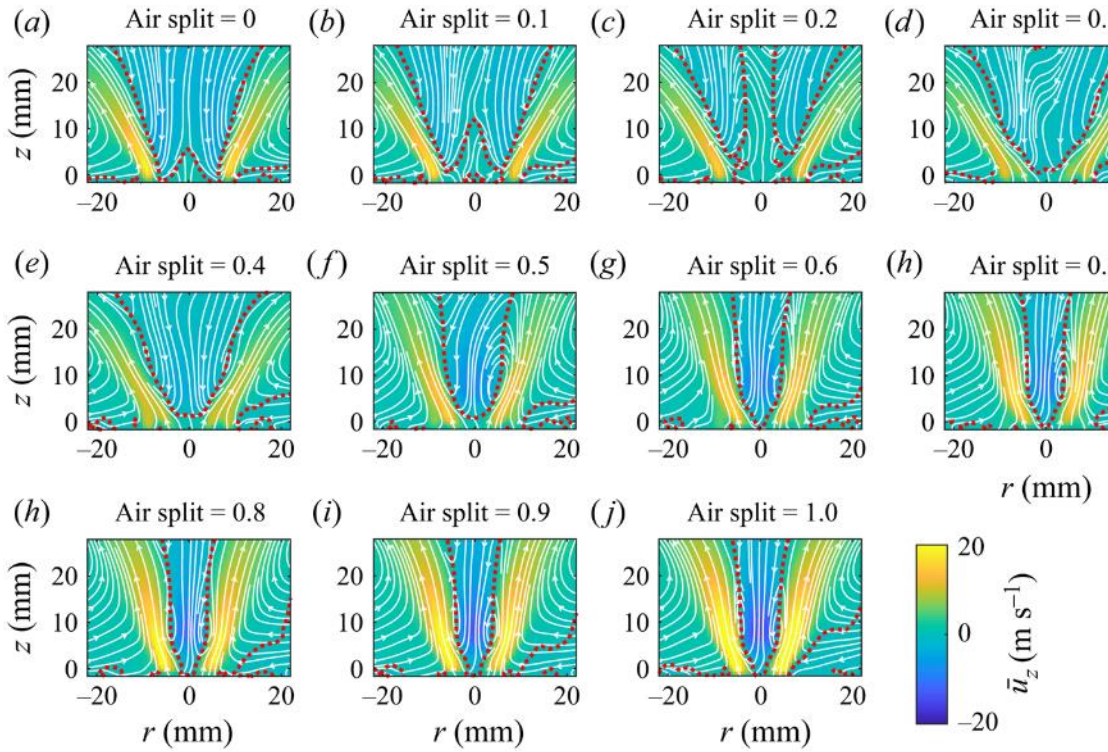
Parameter	Range
Combustion mode	Non-reacting Reacting
Equivalence ratio	0.67
Air split	0:0.1:1 (NR) 0.2:0.05:0.5 (R)
Reynolds number	19,490
Pressure	5 atm

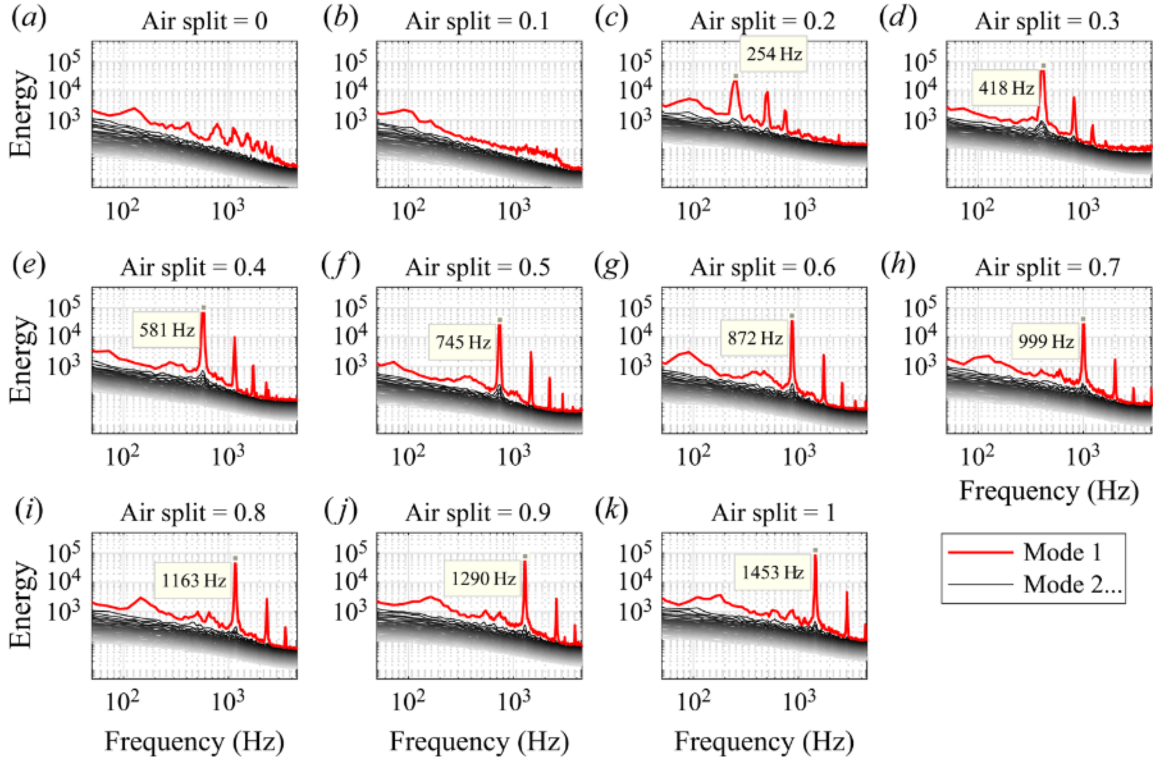
Karmarkar, A., Gupta, S., Boxx, I., Hemchandra, S., O'Connor, J. (2022) "Impact of precessing vortex core dynamics on the thermoacoustic instabilities in a swirl stabilized combustor." *Journal of Fluid Mechanics,* **946**: p. A36

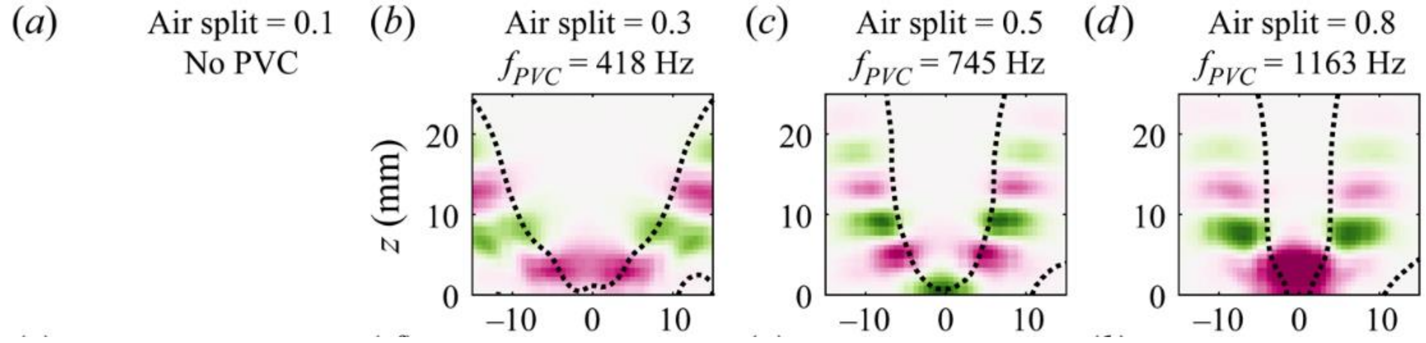
At the non-reacting conditions, increasing the air split moves the flow to the center, results in stronger central recirculation/higher backflow velocity



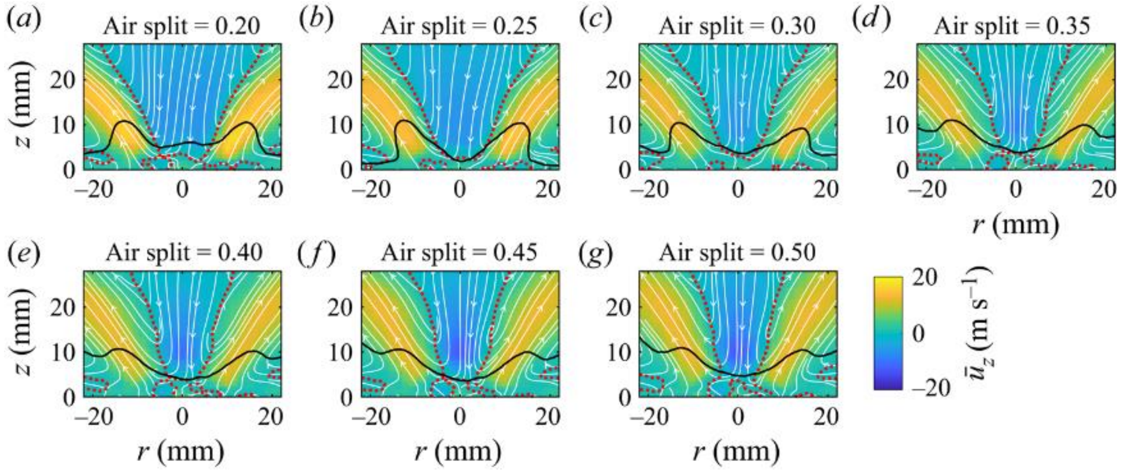
At the non-reacting conditions, we see a PVC develop as the air split increases and the strength of vortex breakdown increases



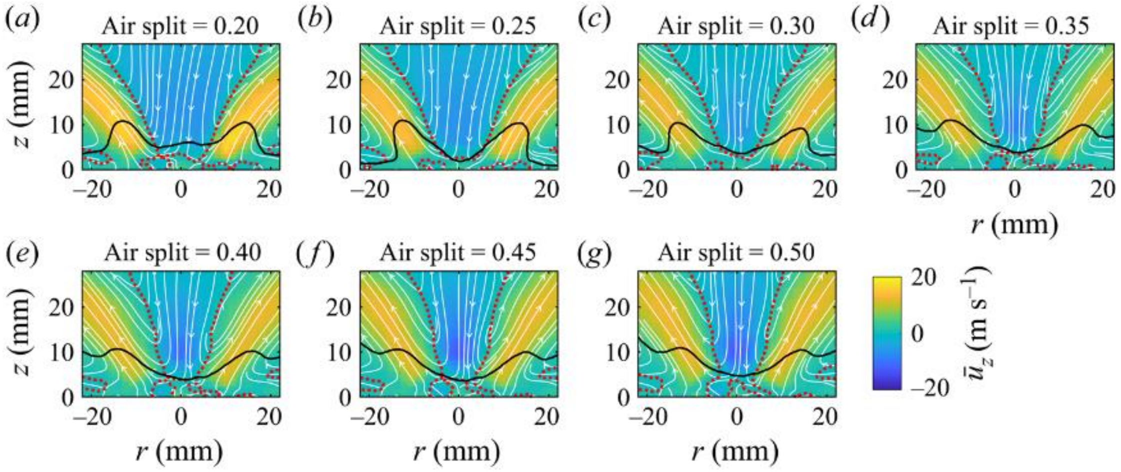


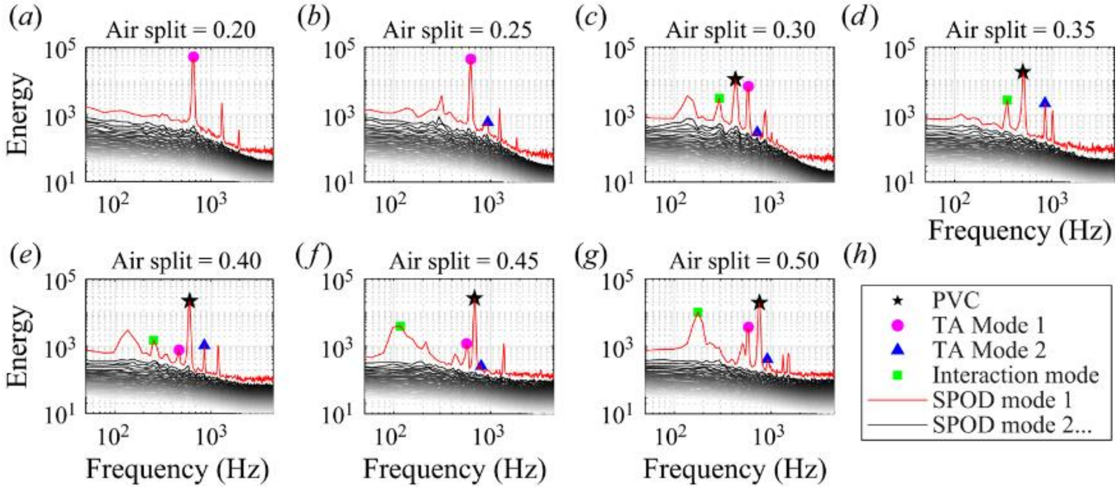


In the reacting conditions, there are three separate oscillations – two thermoacoustic modes and the PVC mode

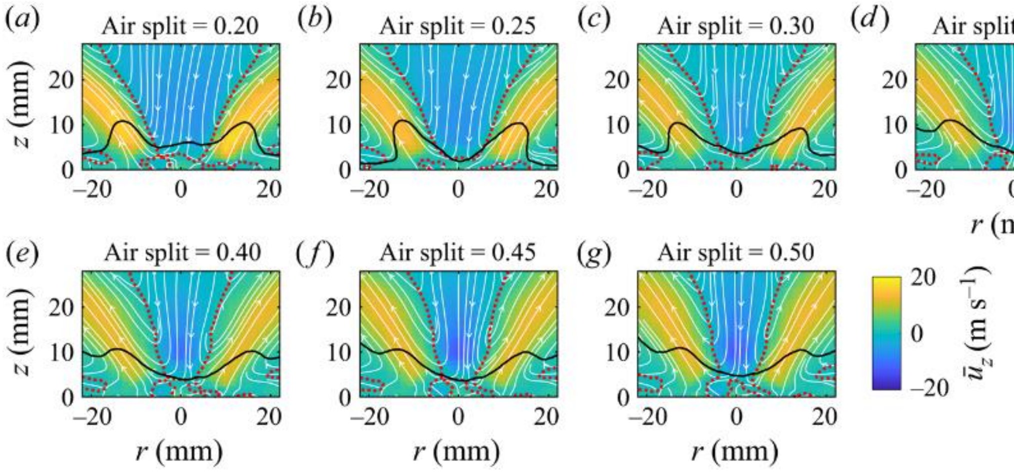


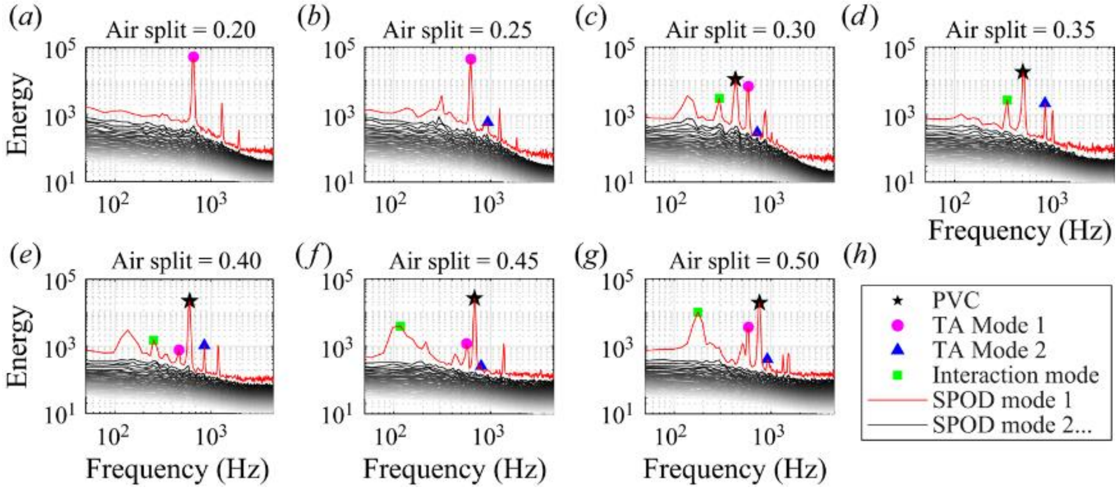
In cases like these, SPOD is significantly better than POD because the POD would not necessarily separate the different frequency oscillations

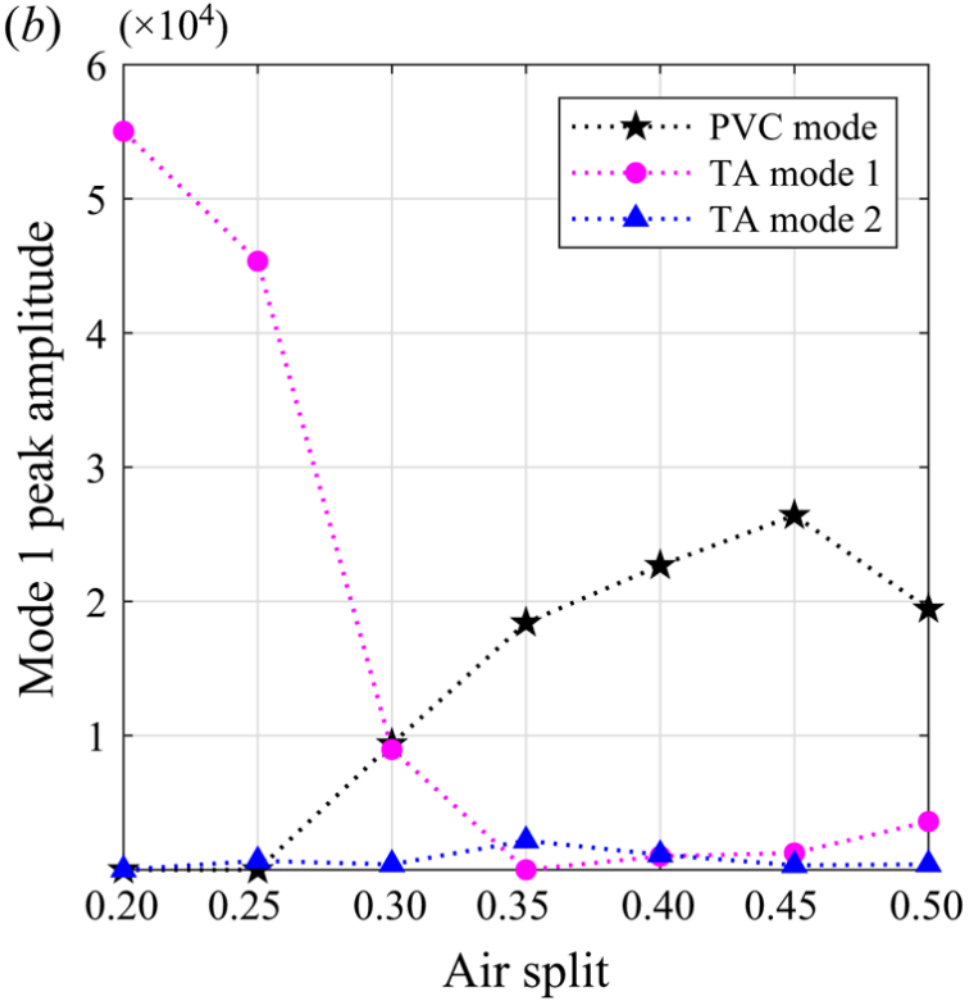




By tracking the mode amplitudes, it appeared as though the PVC mode was suppressing the thermoacoustic oscillation – is this possible?







Yes! It's possible! Nonlinear theory showed the mechanism by which the precessing vortex core could suppress shear layer response

The Navier Stokes Equations (for real):

$$\left(\mathscr{B} + \mathscr{B}^S\{\tilde{\boldsymbol{q}}\}\right)\frac{\partial \tilde{\boldsymbol{q}}}{\partial t} = -\mathscr{N}\{\tilde{\boldsymbol{q}}\}\tilde{\boldsymbol{q}} - S\mathscr{N}^S\{\tilde{\boldsymbol{q}}\}\tilde{\boldsymbol{q}} - S^2\mathscr{N}^{SS}\{\tilde{\boldsymbol{q}}\}\tilde{\boldsymbol{q}}$$

$$+\mathcal{L}_T\tilde{\boldsymbol{q}} + S\mathcal{L}_T^S\tilde{\boldsymbol{q}} + \epsilon^3\hat{\boldsymbol{q}}_{\boldsymbol{a}}(r,z)\cos(\omega_a t)$$

Solution expansion:

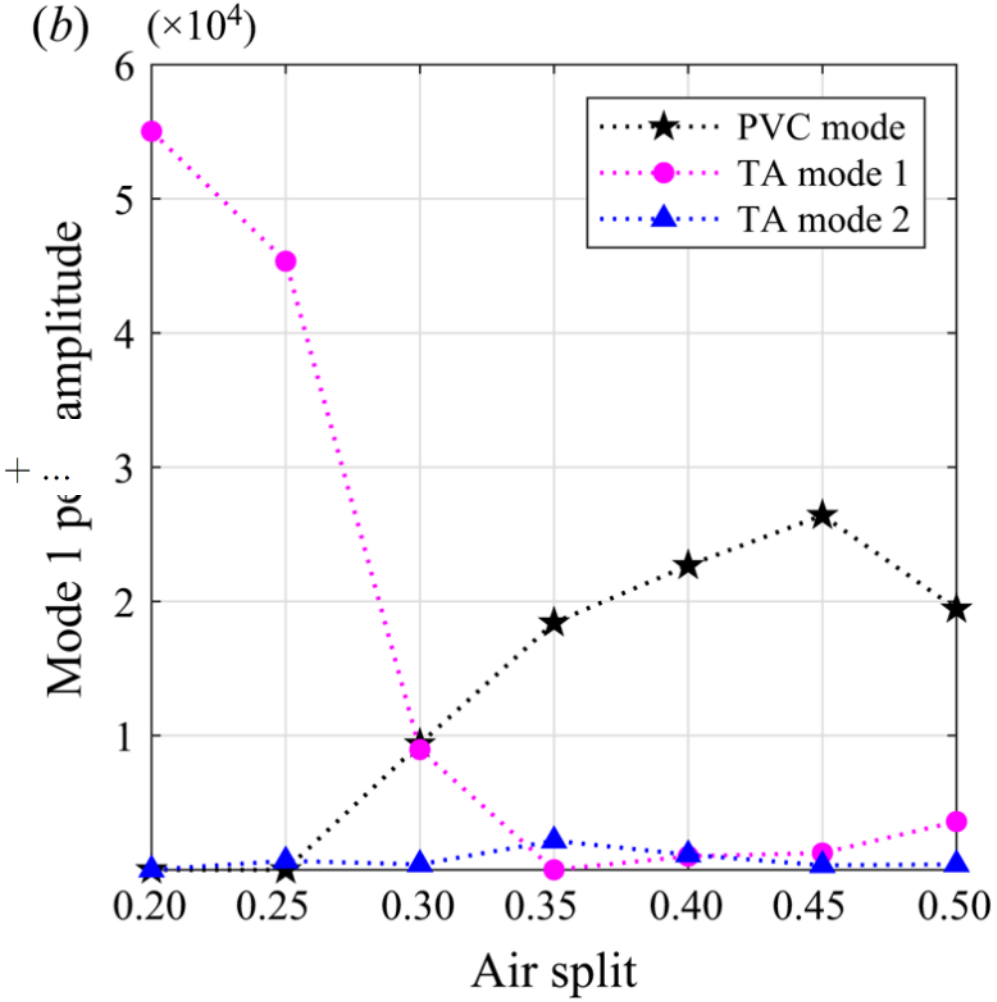
$$\tilde{\boldsymbol{q}}(r,\theta,z,t_1,t_2) = \boldsymbol{q_0}(r,z) + \epsilon\boldsymbol{q_1}(r,\theta,z,t_1,t_2) + \epsilon^2\boldsymbol{q_2}(r,\theta,z,t_1,t_2) + \epsilon^3\boldsymbol{q_3}(r,\theta,z,t_1,t_2) + \ldots$$

$$\boldsymbol{q_1}(r,\theta,z,t_1,t_2) = A_1(t_2)\hat{\boldsymbol{q}}_{\boldsymbol{1}}(r,z)e^{i(\theta-\omega_1 t_1)} + A_0(t_2)\hat{\boldsymbol{q}}_{\boldsymbol{0}}(r,z)e^{-i\omega_0 t_1} + c.c.$$

Oscillation amplitude:

$$A_{TH} = \frac{|\beta_{A_0 f}|}{\left|i\left[\omega_a - \omega_0 - i(S - S_c)\alpha_{A_0}\right] - \beta_{A_0 A_1}A_{PVC}^2\right|}$$

$$\beta_{A_0 f} = \frac{\langle\hat{\boldsymbol{q}}_{\boldsymbol{0}}^{\dagger}, \hat{\boldsymbol{q}}_{\boldsymbol{a}}\rangle}{2\langle\hat{\boldsymbol{q}}_{\boldsymbol{0}}^{\dagger}, \mathscr{B}_0\hat{\boldsymbol{q}}_{\boldsymbol{0}}\rangle}$$



Karmarkar, A., Gupta, S., Boxx, I., Hemchandra, S., O'Connor, J. (2022) "Impact of precessing vortex core dynamics on the thermoacoustic instabilities in a swirl stabilized combustor." *Journal of Fluid Mechanics*, **946**: p. A36

What have we learned so far?

1. Identify the technologies in which combustion instability is an issue
2. Describe the basic thermoacoustic coupling process
3. Explain the kinematics of flame response to harmonic inputs
4. Assess a flame transfer function to identify the potential coupling processes
5. Use your understanding of flame response and thermoacoustic coupling to explain instability control techniques

Course schedule

Monday:
— 1400-1500: Introduction and combustion instability in technologies
— 1500-1530: Thermoacoustic feedback
— 1530-1545: Break
— 1545-1730: Flame kinematics (with crafts!) and flame response

Tuesday:
— 1400-1530: Flame response and flame transfer functions
— 1530-1545: Break
— 1545-1630: Instability control
— 1630-1730: Advanced topics

Advanced Topics

— Transverse modes

— Nonlinear instability

— Entropy coupling

— What do you want to hear about?

Advanced Topics

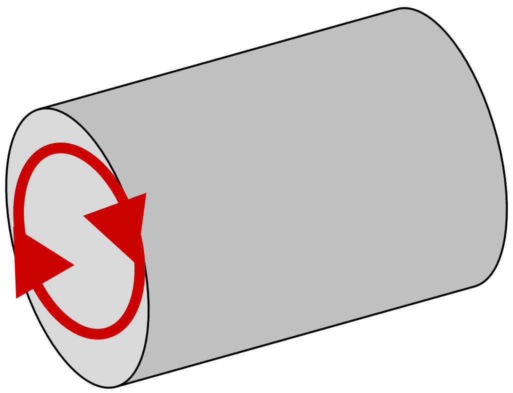
— Transverse modes (this was the subject of my PhD, so buckle up...)

— Nonlinear instability

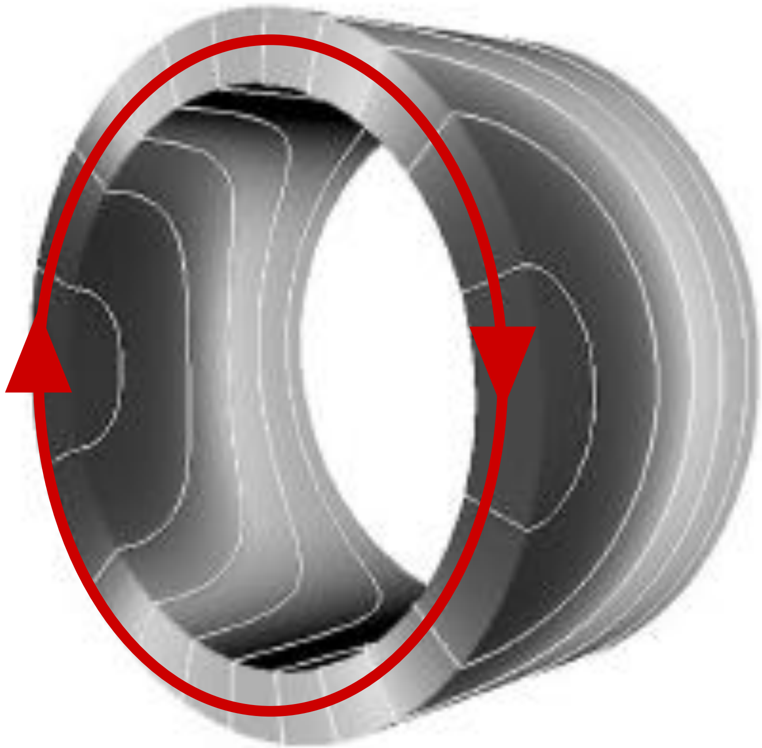
— Entropy coupling

— What do you want to hear about?

Transverse modes most often occur in annular combustor configurations where the circumferential and longitudinal length scales are similar



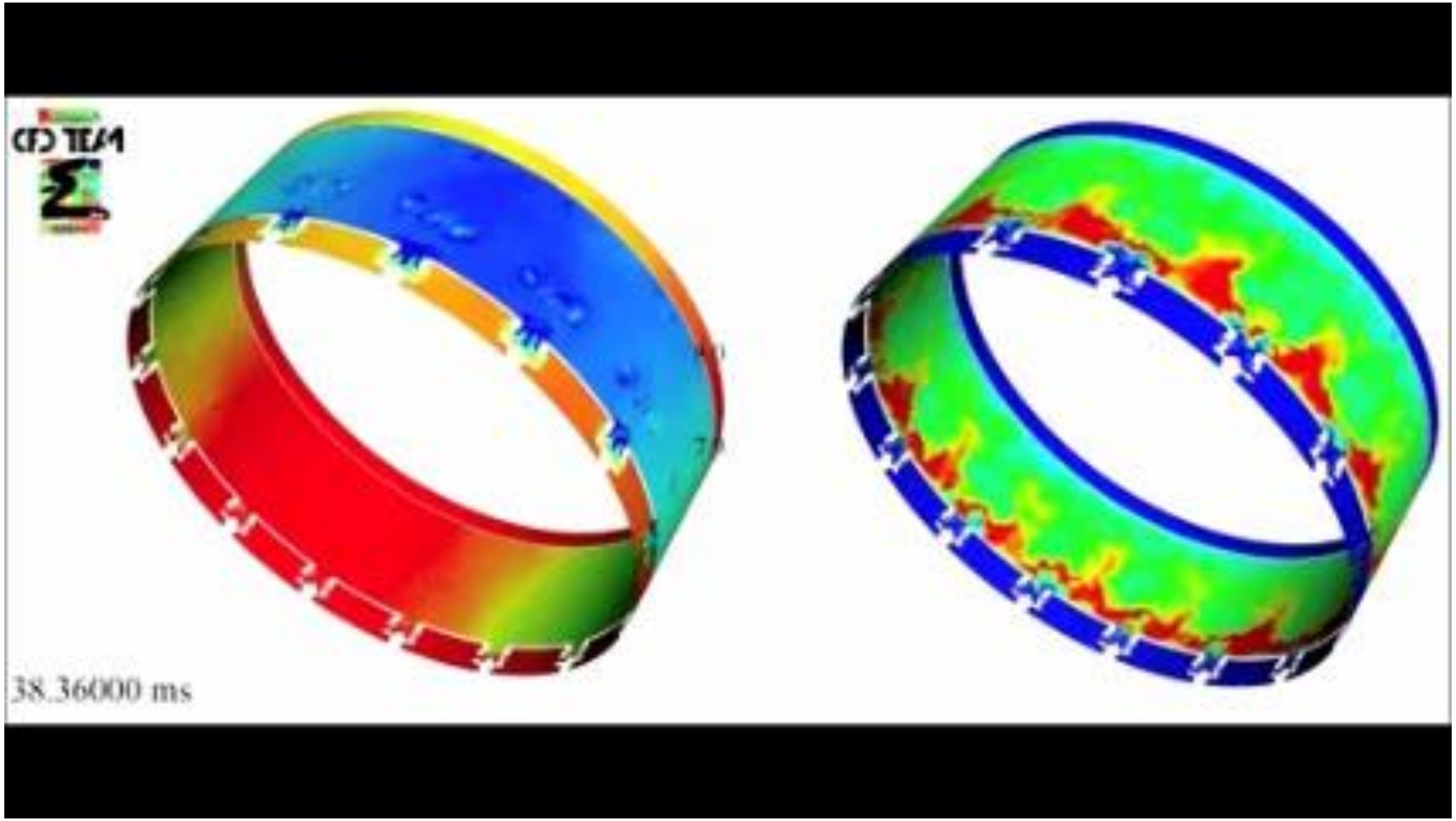
Can combustor geometry



Annular combustor geometry

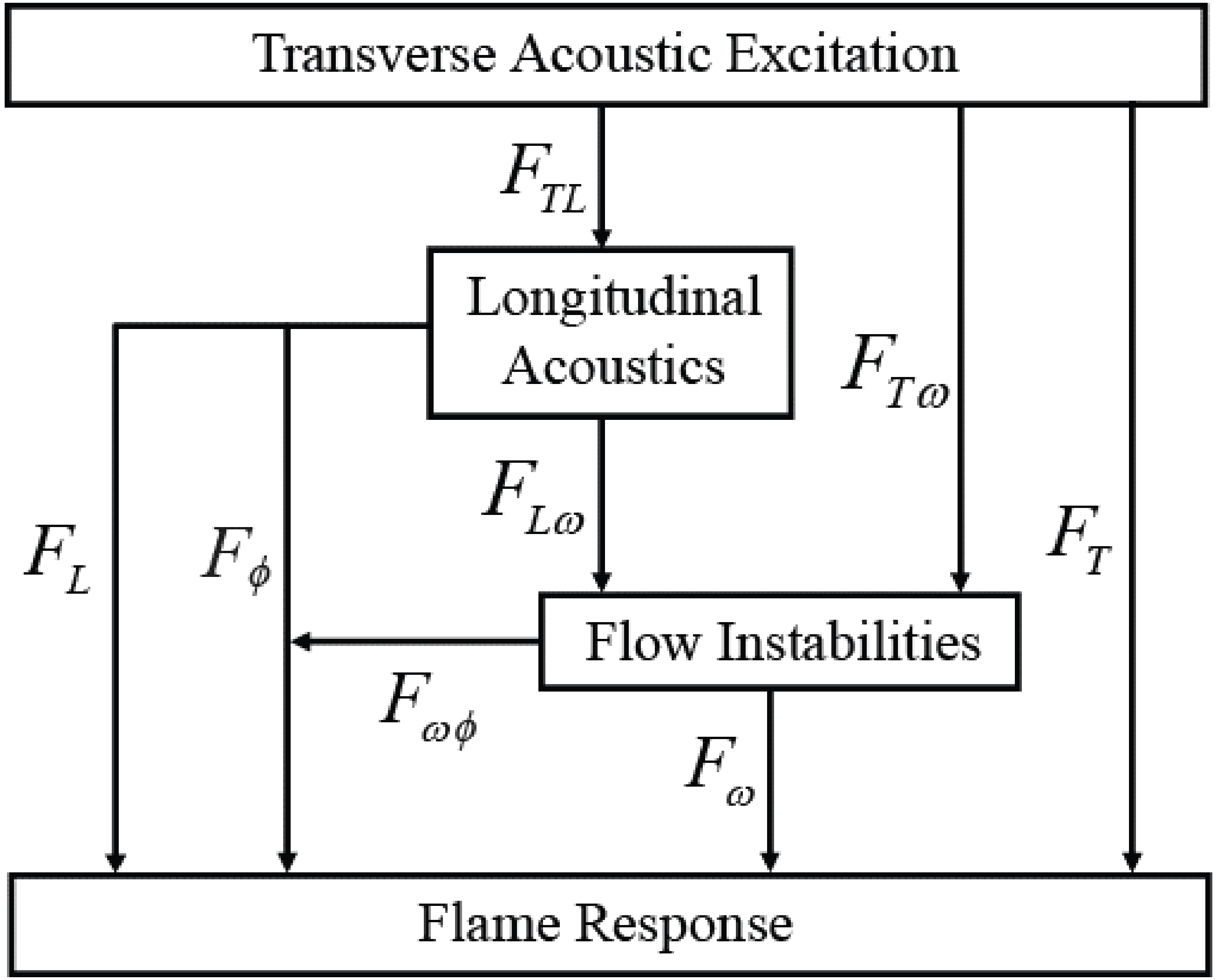
Sensiau, C., Nicoud, F., & Poinsot, T. (2009). A tool to study azimuthal standing and spinning modes in annular combustors. *International Journal of Aeroacoustics*, *8*(1), 57-67.

This instability mode can cause anti-symmetric and time-varying disturbance fields for the nozzles as they oscillate, which is more complex than a longitudinal mode



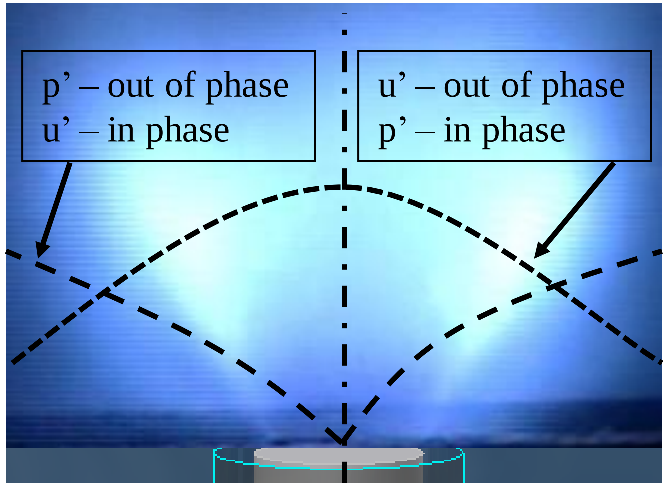
Wolf, P., Staffelbach, G., Gicquel, L. Y., Müller, J. D., & Poinsot, T. (2012). Acoustic and large eddy simulation studies of azimuthal modes in annular combustion chambers. *Combustion and Flame*, *159*(11), 3398-3413.

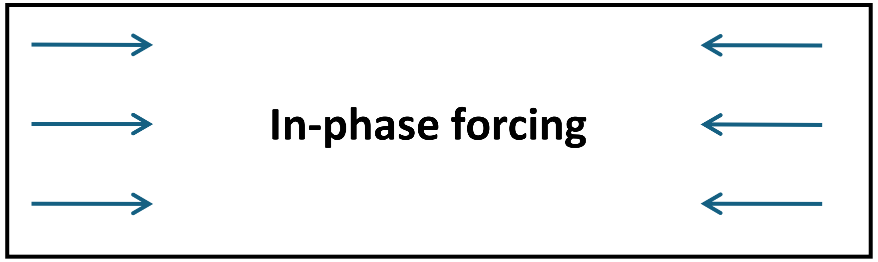
The flame response pathways are more complex in the transverse mode than in the longitudinal mode and is more geometry dependent

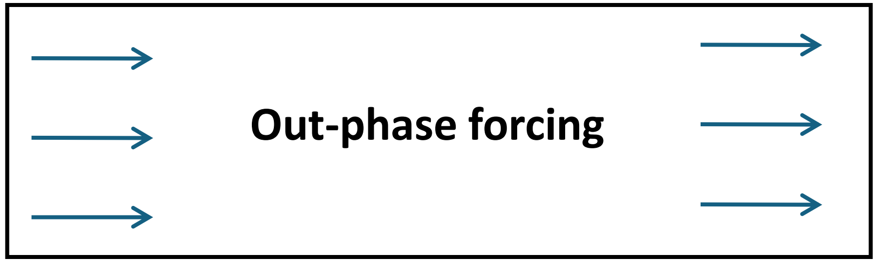


O'Connor, J., and Lieuwen, T. (2012). "Further Characterization of the Disturbance Field in a Transversely Excited Swirl-Stabilized Flame." J. Eng. Gas Turbines Power. 134(1): 011501.

We designed an experiment that allowed us to probe the extremes of the acoustic modes – the pressure node and pressure anti-node – and resultant flame response



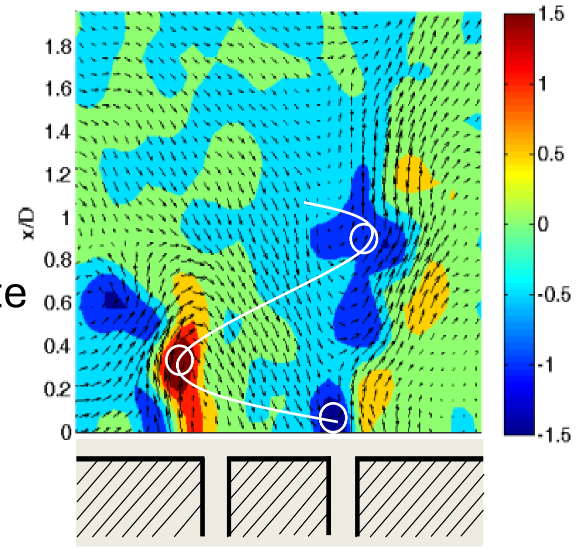


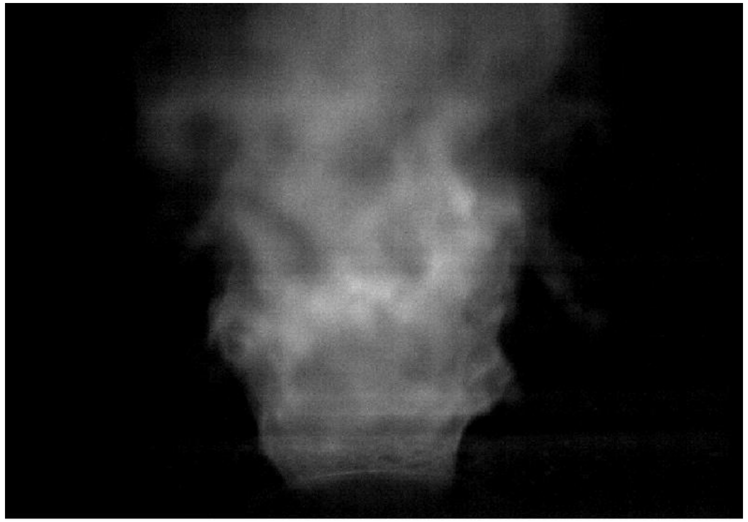


O'Connor, J., and Lieuwen, T. (2012). "Further Characterization of the Disturbance Field in a Transversely Excited Swirl-Stabilized Flame." J. Eng. Gas Turbines Power. 134(1): 011501.

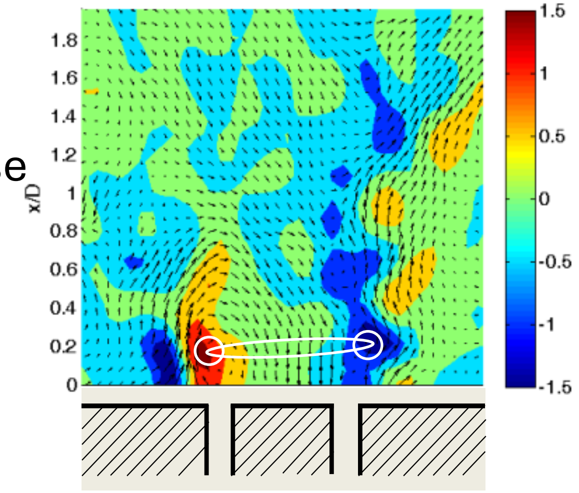
The flame response pathways are more complex in the transverse mode than in the longitudinal mode and is more geometry dependent

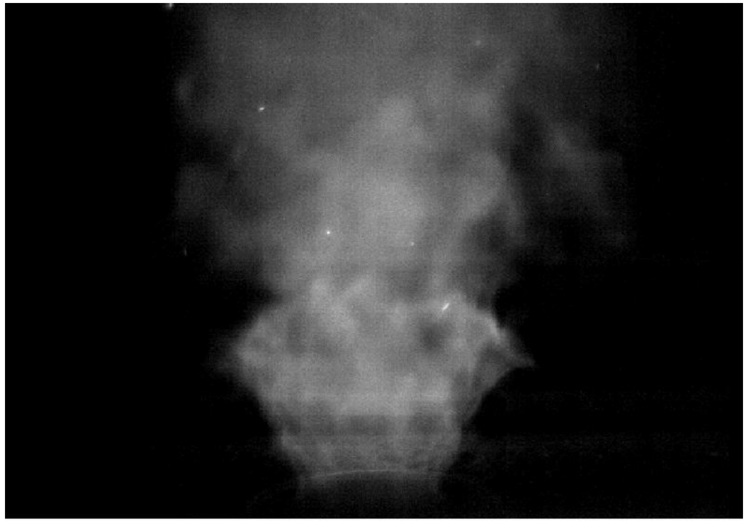
- Pressure node
- Velocity anti-node
- Anti-symmetric flame response
- No global heat release rate oscillation





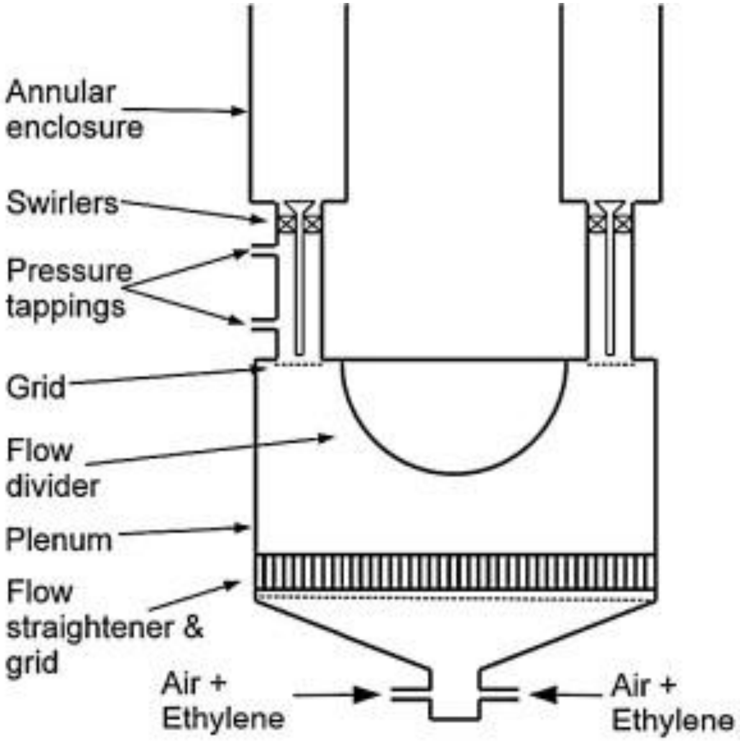
- Pressure anti-node
- Velocity node
- Symmetric flame response
- Finite global heat release rate oscillation

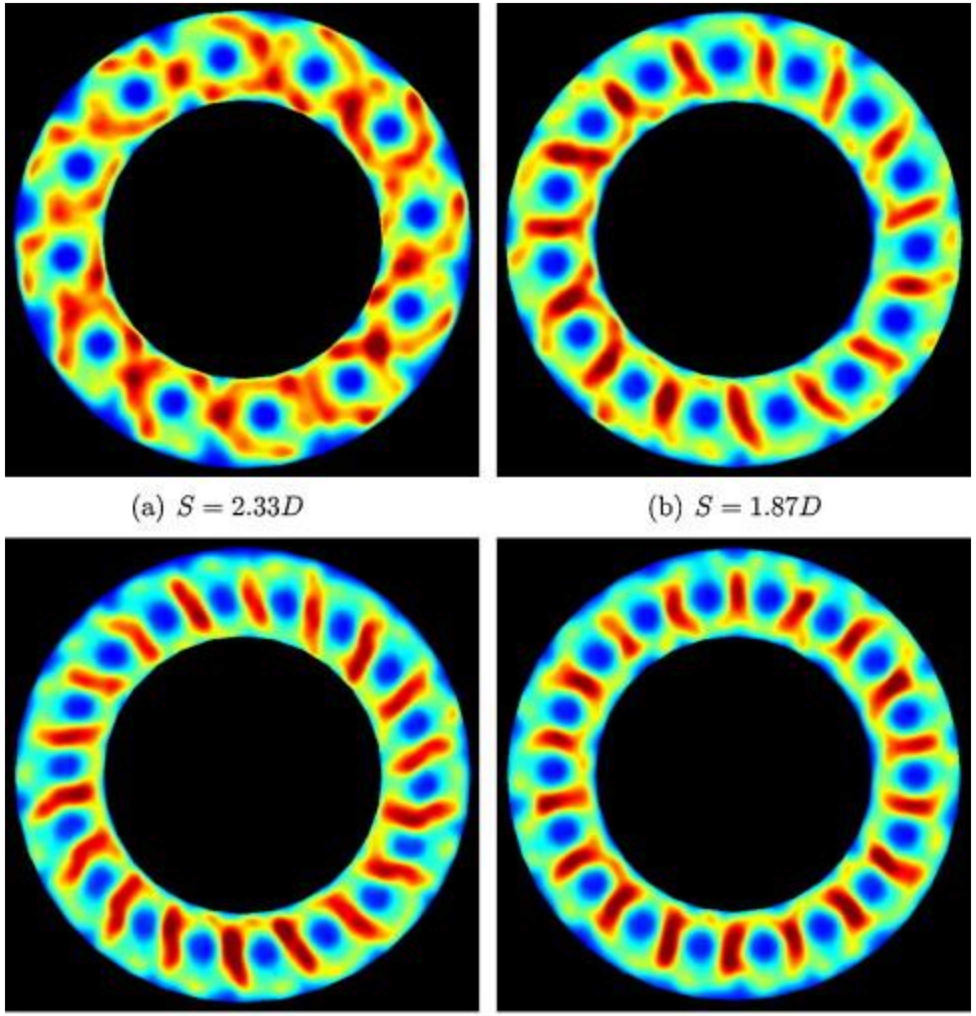




O'Connor, J., and Lieuwen, T. (2012). "Further Characterization of the Disturbance Field in a Transversely Excited Swirl-Stabilized Flame." J. Eng. Gas Turbines Power. 134(1): 011501.

Experiments in full annular combustors allow for investigation of spinning vs. standing modes and how flow couples with mode structure





(a) $S = 2.33D$ (b) $S = 1.87D$

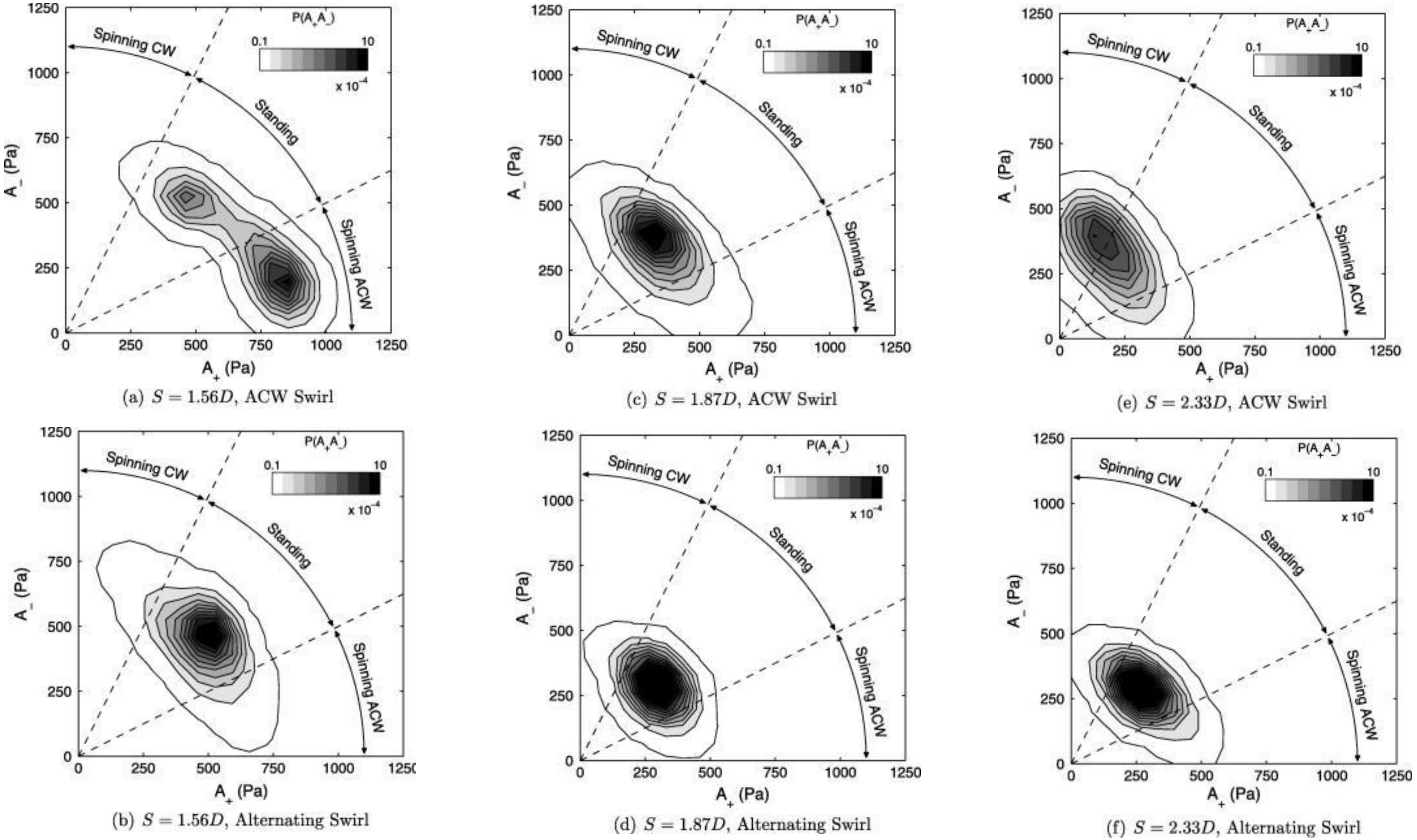
(c) $S = 1.56D$ (d) $S = 1.56D$

Worth, N. A., & Dawson, J. R. (2013). Modal dynamics of self-excited azimuthal instabilities in an annular combustion chamber. Combustion and Flame, 160(11), 2476-2489.

Experiments in full annular combustors allow for investigation of spinning vs. standing modes and how flow couples with mode structure

All nozzles co-swirling, three spacings (S)

Alternating swirl nozzles, three spacings (S)



(a) $S = 1.56D$, ACW Swirl
(c) $S = 1.87D$, ACW Swirl
(e) $S = 2.33D$, ACW Swirl
(b) $S = 1.56D$, Alternating Swirl
(d) $S = 1.87D$, Alternating Swirl
(f) $S = 2.33D$, Alternating Swirl

Worth, N. A., & Dawson, J. R. (2013). Modal dynamics of self-excited azimuthal instabilities in an annular combustion chamber. Combustion and Flame, 160(11), 2476-2489.

Advanced Topics

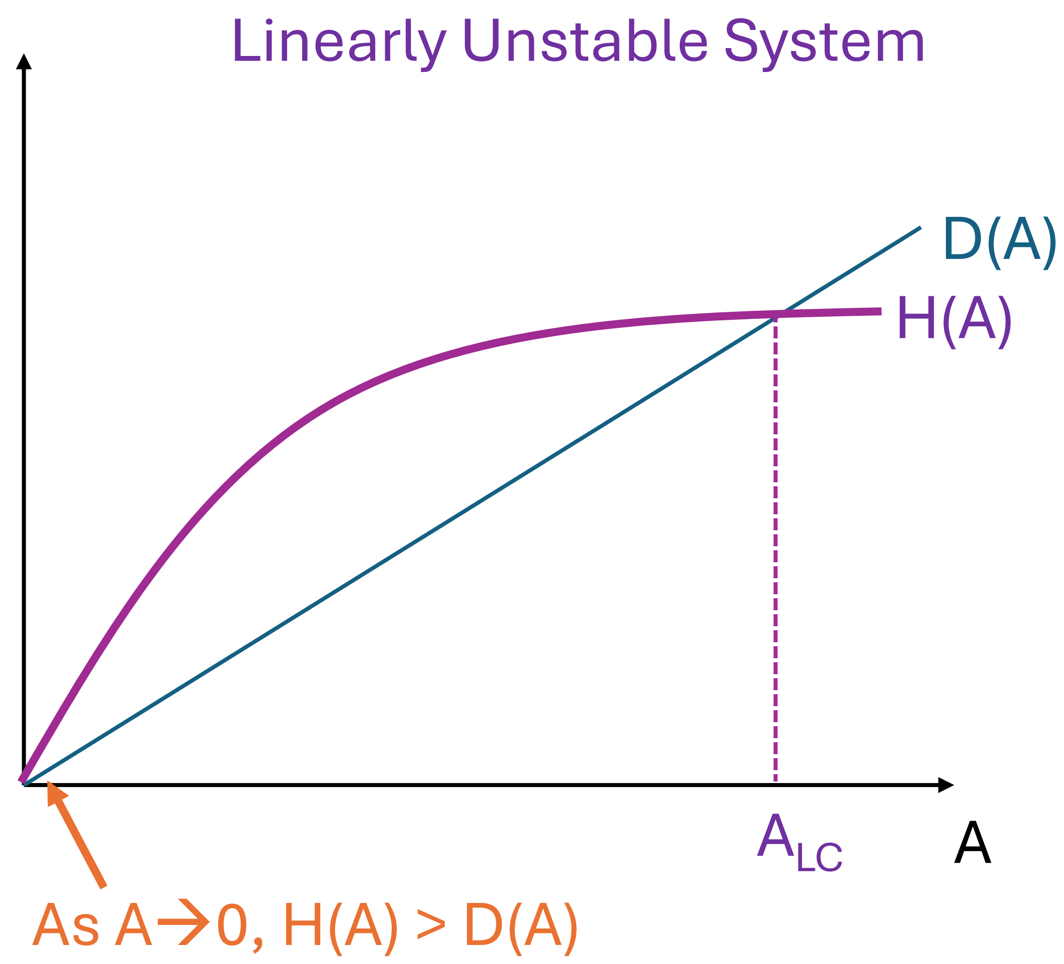
— Transverse modes

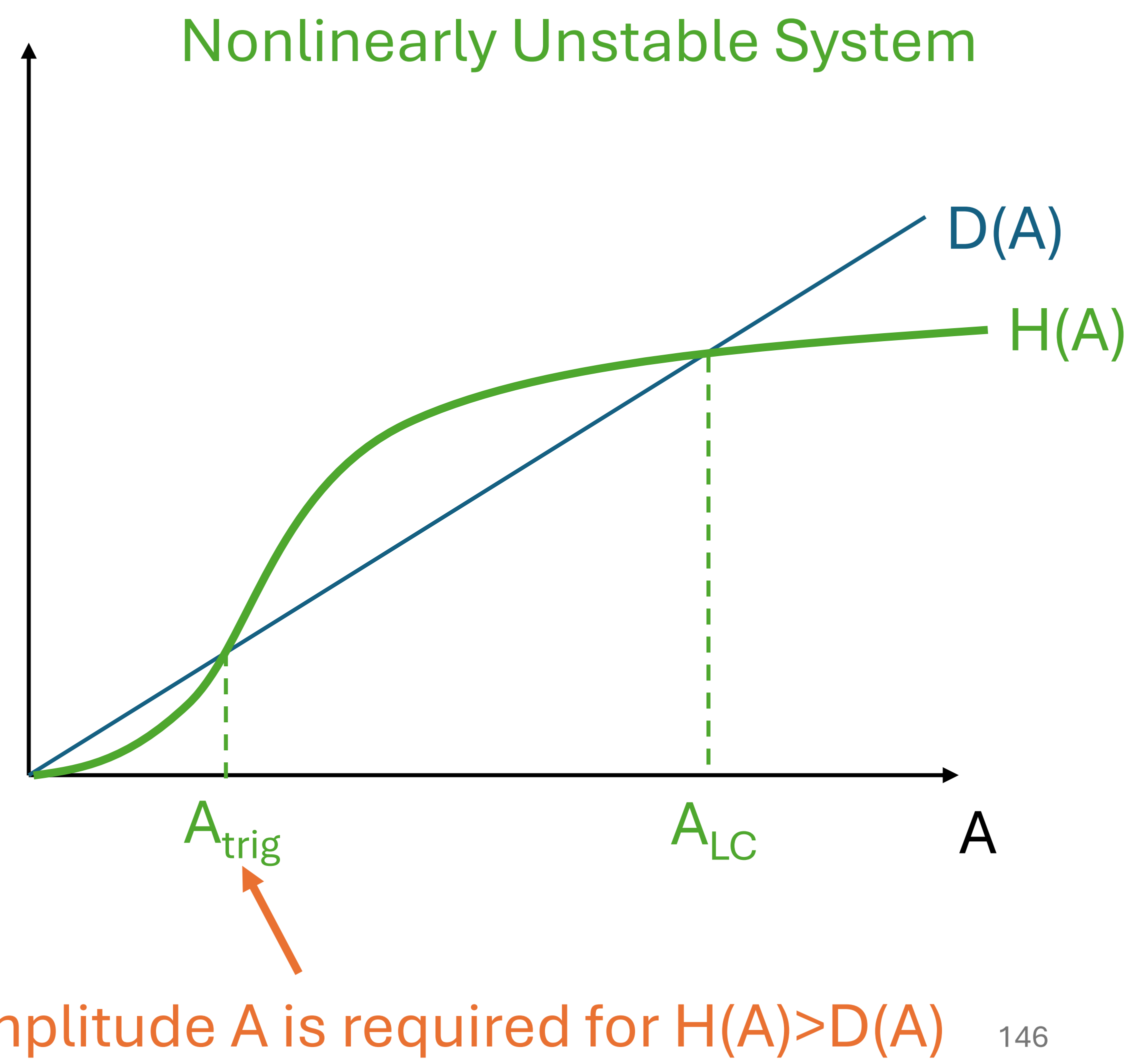
— Nonlinear instability

— Entropy coupling

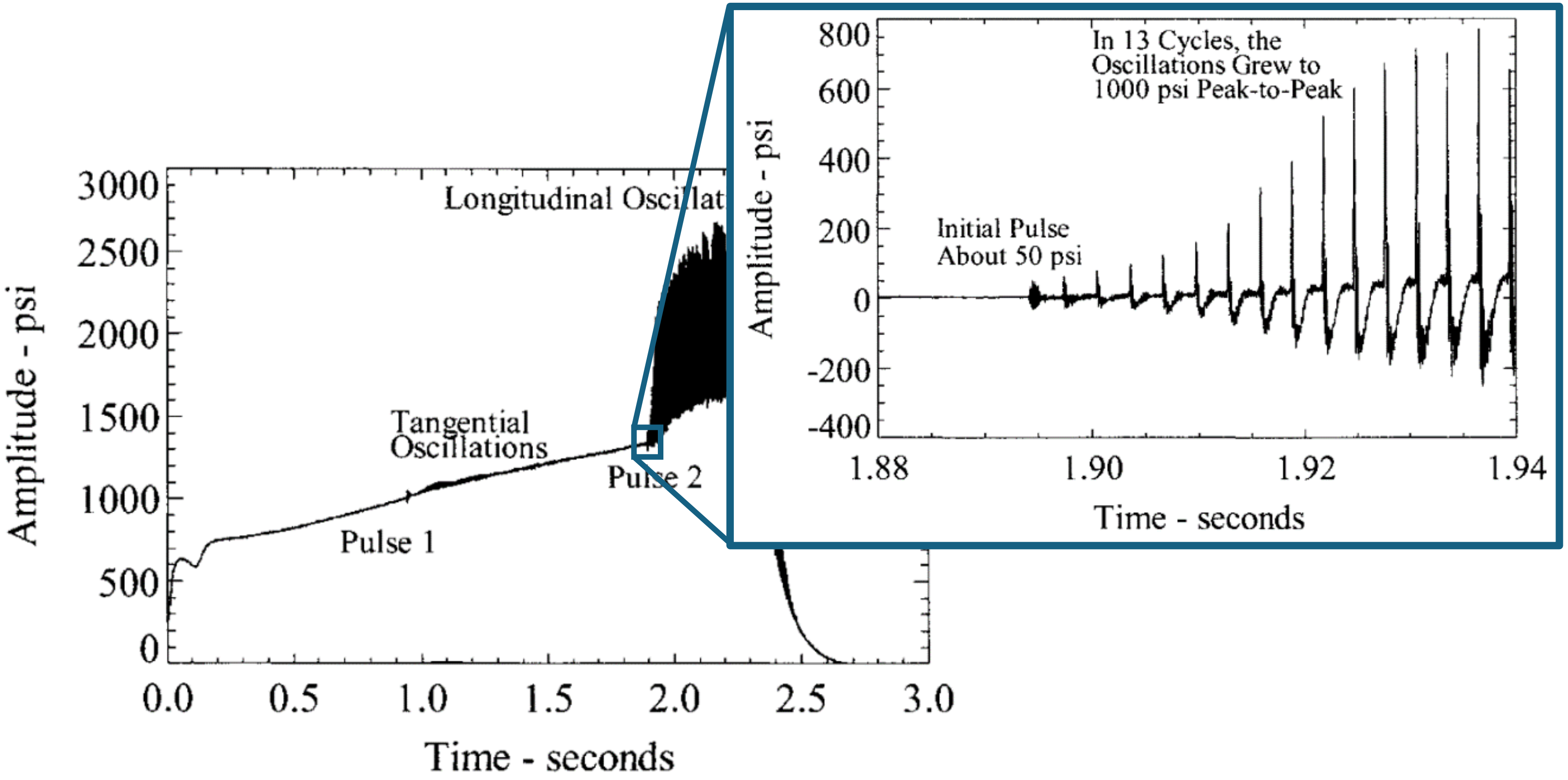
— What do you want to hear about?

Nonlinear instability refers to the type of instability and how it starts, not the response of the flame (which typically always becomes nonlinear)



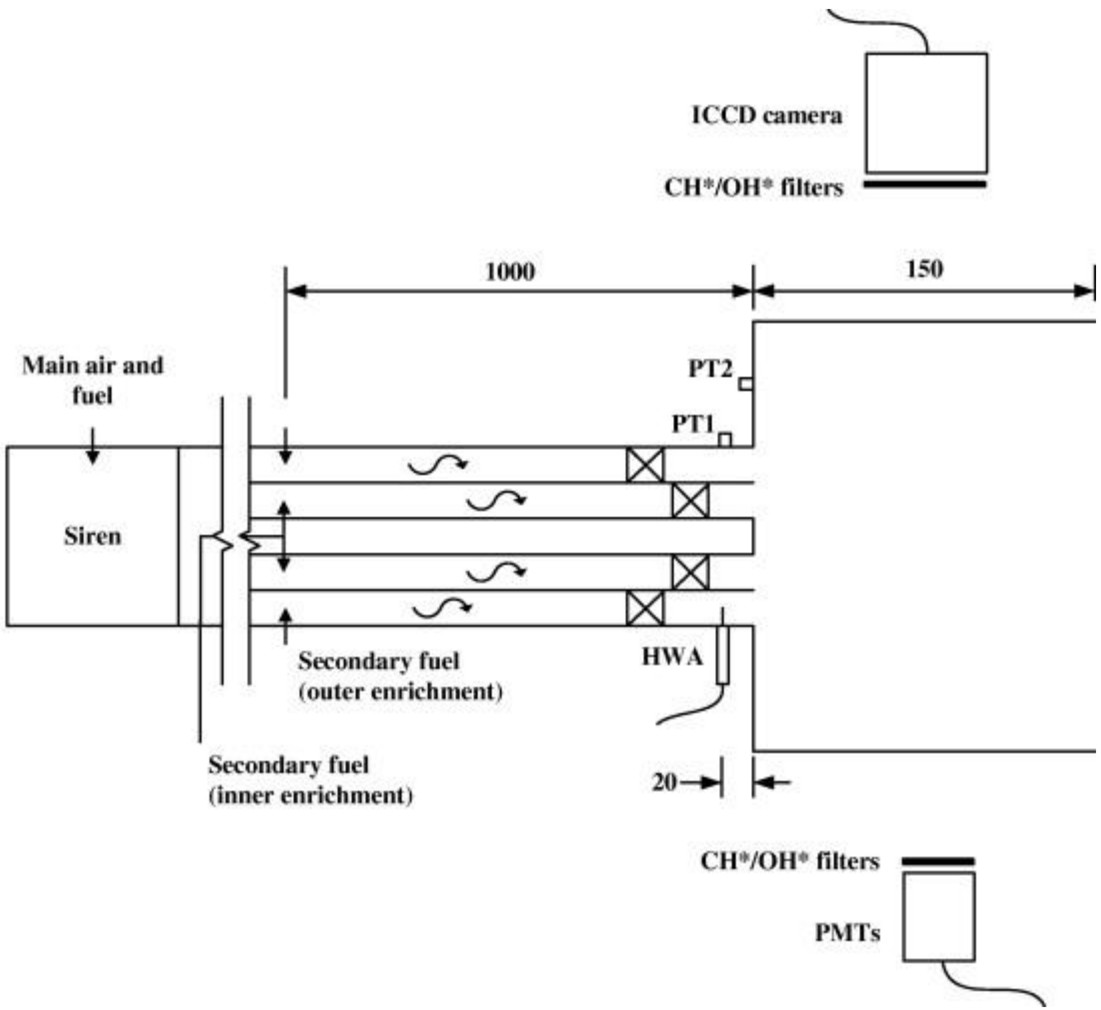


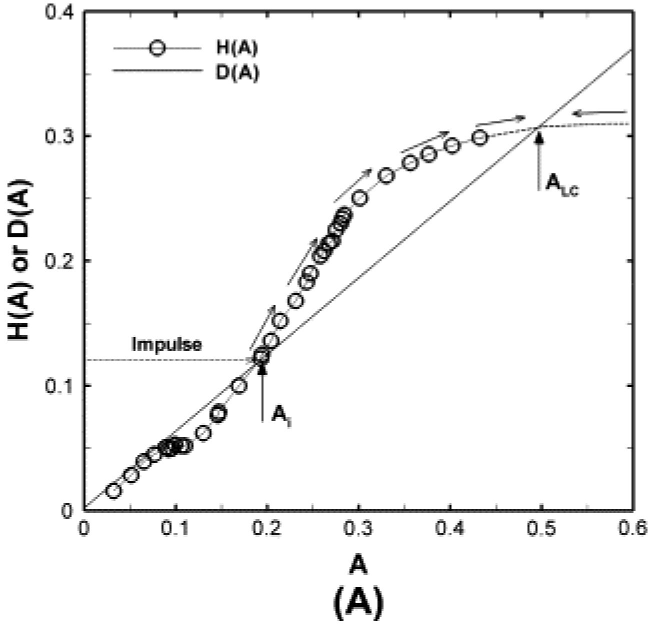
Example: Space Shuttle booster rockets

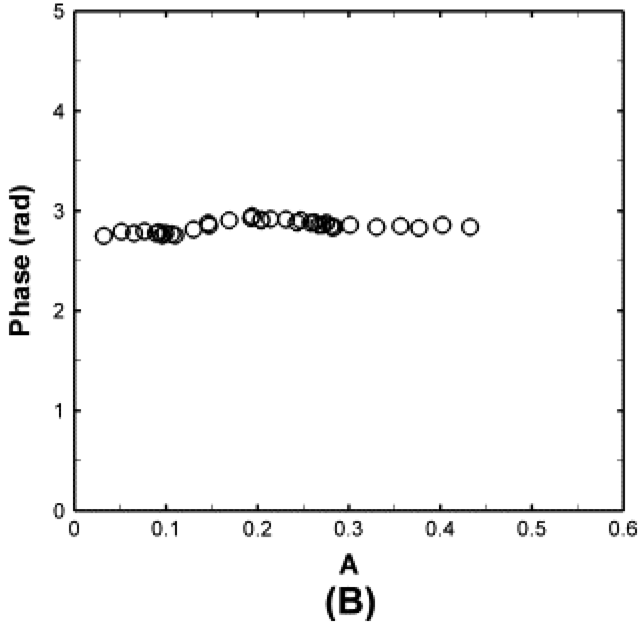


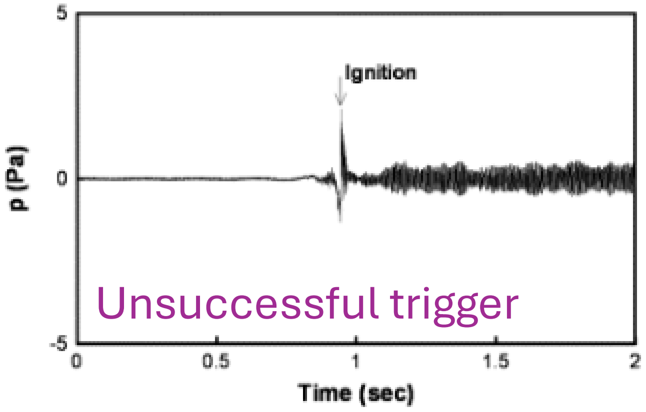
Blomshield, F. (2001). Historical perspective of combustion instability in motors-case studies. In 37th Joint Propulsion Conference and Exhibit.

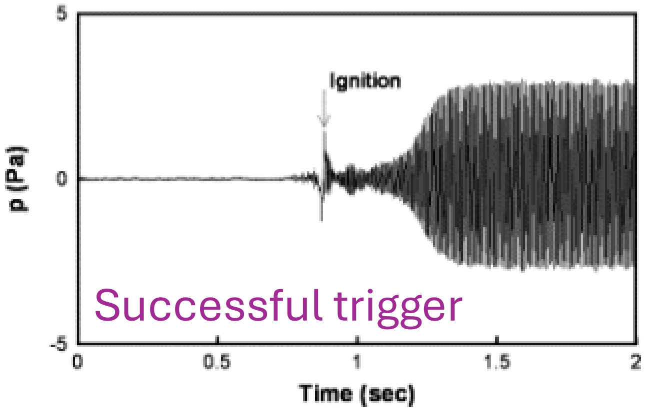
Example: gas turbine combustor from Kim and Hochgreb (it is rare to see this in gas turbines, so this is a cool paper)











Kim, K. T., & Hochgreb, S. (2012). Measurements of triggering and transient growth in a model lean-premixed gas turbine combustor. Combustion and Flame, 159(3), 1215-1227.

Advanced Topics

— Transverse modes

— Nonlinear instability

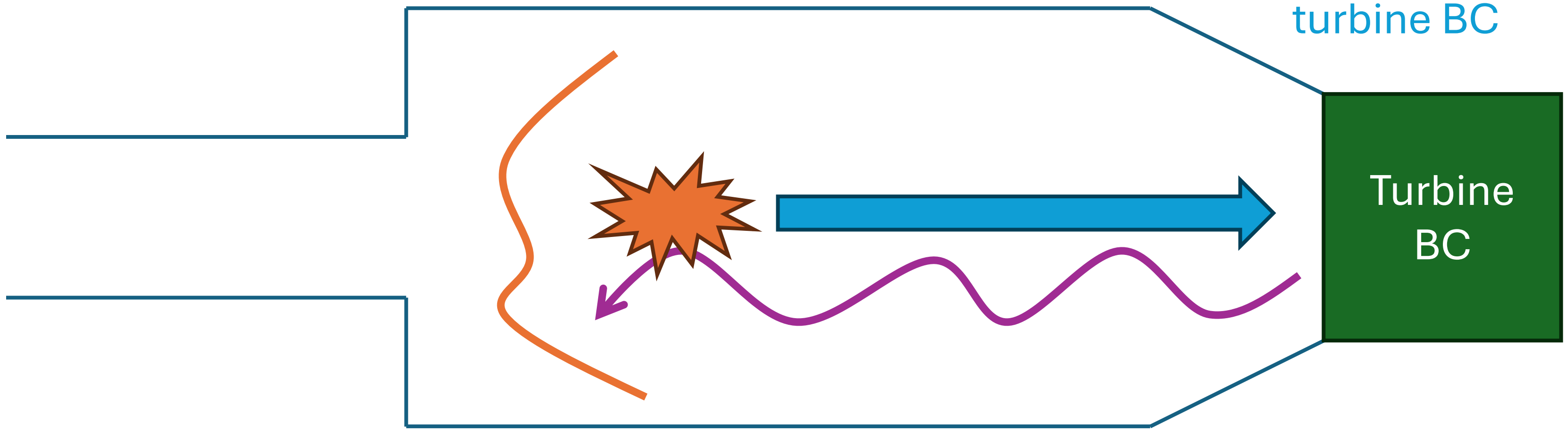
— Entropy coupling

— What do you want to hear about?

Entropy coupling is a unique instability mode as it relies on an acoustic/convective process and is heavily dependent on the combustor outlet boundary condition

Heat release rate perturbation leads to entropy oscillation

Entropy oscillation convects downstream and interacts with turbine BC

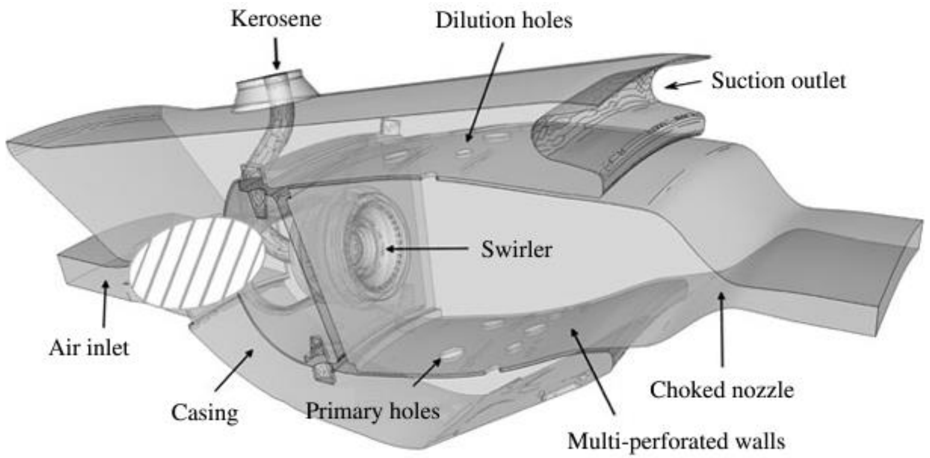


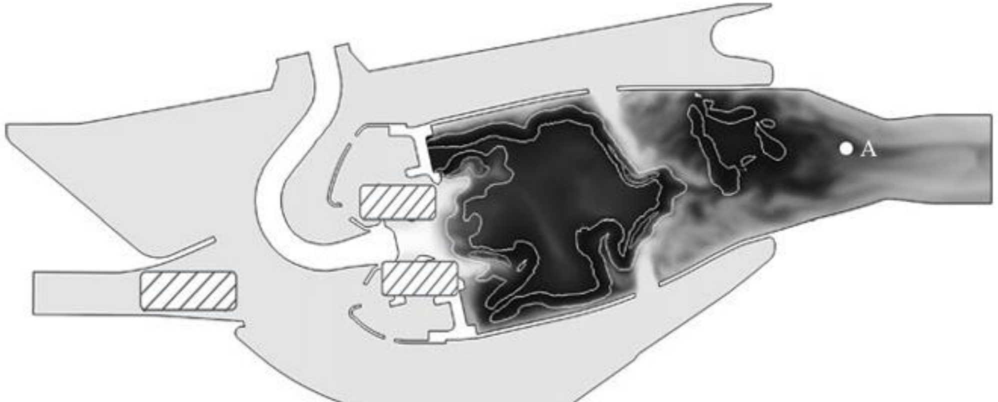
Acoustic disturbance drives an entropy disturbance at the flame and the cycle starts again

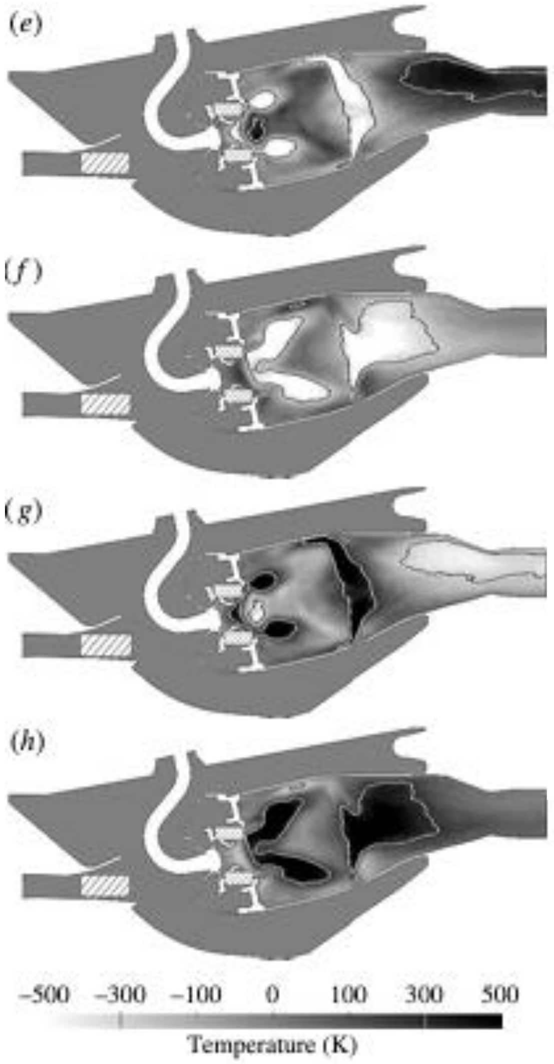
The interaction of an entropy disturbance with a high-impedance BC leads to an acoustic disturbance

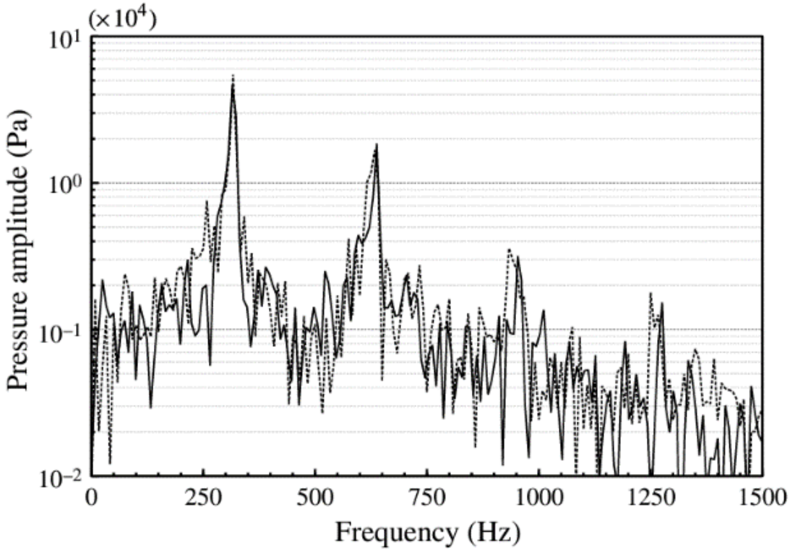
$$f = \frac{1}{\tau} = \frac{1}{\tau_{conv} + \tau_{acoustic}}$$ where $$\tau_{conv} = L_{comb}/u_{conv}$$ $$\tau_{acoustic} = L_{comb}/c$$

Because of the convective timescale, entropy modes tend to be low-frequency modes and typically occur when the engine is cold (often called “rumble”)









Motheau, E., Nicoud, F., & Poinsot, T. (2014). Mixed acoustic–entropy combustion instabilities in gas turbines. *Journal of Fluid Mechanics*, *749*, 542-576.

Advanced Topics

- Transverse modes
- Nonlinear instability
- Entropy coupling
- What do you want to hear about?

Thank you!

jxo22@psu.edu

https://sites.psu.edu/rfdl